

THE CHAUTAUQUAN

*A Monthly Magazine Devoted to
the Promotion of True Culture.*

*Organ of the
Chautauqua Literary and
Scientific Circle.*

Vol. V
March, 1885.
No. 6.

*Theodore L. Flood, D.D., Editor
The Chautauqua Press*

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THE CHAUTAUQUAN.

*A MONTHLY MAGAZINE DEVOTED TO THE PROMOTION OF TRUE
CULTURE. ORGAN OF THE CHAUTAUQUA LITERARY AND
SCIENTIFIC CIRCLE.*

VOL. V.

MARCH, 1885.

No. 6.

Officers of the Chautauqua Literary and Scientific Circle.

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REQUIRED READING FOR MARCH.

TEMPERANCE TEACHINGS OF SCIENCE; OR, THE POISON PROBLEM. PART VI.

BY FELIX L. OSWALD, M.D.

CHAPTER VI.—SUBJECTIVE REMEDIES.

“Deeprooted evils can not be abolished by striking at the branches.”—*Boerhave*.[\[1\]](#)

The history of the temperance movement has demonstrated the sad futility of palliative remedies. We have seen that the malady of the poison vice is not a self-limited, but a necessarily progressive evil. The half-way measures of “restrictive” legislation have resulted only in furnishing additional proof that prevention is better, because less impossible, than control.[\[A\]](#) The regulation of the poison traffic, the redress of the unavoidably resulting mischief, the cure and conversion of drunkards, in order to be effectual, would impose intolerable and never ending burdens on the resources even of the wealthiest communities, while the advocates of prohibition would forestall the evils both of the remedy and the disease.

But we should not overlook the truth that, in our own country at least, the poison plant of intemperance springs from a composite root. In southern Spain, under the dominion of the Saracens,[\[2\]](#) the poison vice was almost unknown during a series of centuries.[\[B\]](#) The moral code and the religion of the inhabitants discountenanced intemperance. The virtue of dietetic purity ranked with chastity and cleanliness. An abundance of harmless amusements diverted from vicious pastimes. Under such circumstances the absence of direct temptations constituted a sufficient safeguard against the vice of the poison habit; but in a country like ours the efficacy of prohibition depends on the following supplementary remedies:

1. INSTRUCTION.—In the struggle against the powers of darkness light often proves a more effective weapon than might or right. Even the limited light of human reason might help us to avoid mistakes that have undoubtedly retarded the triumph of our cause. We must enlighten, as well as admonish our children, if we would save them from the snares of the tempter; among the victims of intemperance, even among those who can speak from experience and can not deny that their poison has proved the curse of their lives, only a small portion is at all able to comprehend the necessary connection of cause and result. They ascribe their ruin to the spite of fortune, to the machinations of an uncharitable world, to abnormally untoward circumstances, rather than to the normal effects of the insidious poison. Intoxication they admit to be an evil, but defend the moderate use of a liquor as infallibly injurious in the smallest as in the largest dose; they underrate the progressive tendency of their vice and overrate their power of resistance; they cling to the tradition that alcohol, discreetly enjoyed, may prove a blessing instead of a curse. We must banish that fatal delusion. We must reveal the true significance of the poison habit before we can hope to suppress it as a life blighting vice. Our text-books should be found in every college and every village school from Florida to Oregon. Every normal school should graduate teachers of temperance. The law of the State of New York providing for the introduction of primers on the effects of alcoholic beverages was attacked by one of our leading scientific periodicals, with more learning than insight, on the ground that the physiological action of alcohol is as yet obscure even to our ablest pathologists, and therefore not a fit subject for a common school text-book. The same objection might be urged against every other branch of physiology and the natural history of the organic creation. “Every vital process is a miracle,” says Lorenz Oken, [3] “that is, in all essential respects an unexplained phenomenon.” A last question will always remain unanswered wherever the marvelous process of life is concerned, but our ignorance, as well as our knowledge, of that phenomenon has its limits, and in regard to the effects of alcoholic beverages it is precisely the most knowable and most fully demonstrated part of the truth which it behooves every child to know, but of which at present nine tenths of the adults, even in the most civilized countries, remain as ignorant as the natives of Kamtschatka who worship a divinity in the form of a poisonous toadstool. A boy may be brought to comprehend the folly of gambling even before he has mastered the abstruse methods of

combination and permutation employed in the calculus of probable loss and gain. We need not study Bentham^[4] to demonstrate that honesty is an essential basis of commerce and social intercourse. By the standard of usefulness, too, temperance primers might well take precedence of many other text-books. Our school boys hear all sorts of things about the perils encountered by the explorers of African deserts and Arctic seas, but next to nothing about the pitfalls in their own path—no room for the discussion of such subjects in a curriculum that devotes years to the study of dead languages. Is the difference between the archaic and pliocene form of a Greek verb so much more important than the difference between food and poison?

With such a text as the monster curse of intemperance and its impressive practical lessons, a slight commentary would suffice to turn thousands of young observers into zealous champions of our cause, just as in Germany a few years of gymnastic training have turned nearly every young man into an advocate of physical education. The work begun in the school room should be continued on the lecture platform, but we should not dissemble the truth that in a crowded hall ninety per cent. of the visitors have generally come to hear an *orator* rather than a teacher, and enjoy an eloquence that stirs up their barrenest emotions as much as if it had fertilized the soil of their intelligence, just as the unrepentant gamblers of a Swiss watering place used to applaud the sensational passages of a drama written expressly to set forth the evils of the gambling hell. Enthusiasm and impressiveness are valuable qualifications of a public speaker, but he should possess the talent of making those agencies the vehicles of instruction. The great mediæval reformers, as well as certain political agitators of a later age, owe their success to their natural or acquired skill in the act of stirring their hearers into an intellectual ferment that proved the leaven of a whole community—for that skill is a talent that can be developed on a basis of pure common sense and should be more assiduously cultivated for the purposes of our reform. A modern philanthropist could hardly confer a greater benefit on his fellow-citizens than by founding a professorship of temperance, or endowing a college with the special condition of a proviso for a weekly lecture on such topics as “The Stimulant Delusion,” “Alcoholism,” “The History of the Temperance Movement.”

Pamphlets, too, may subserve an important didactic purpose, and in the methods of their distribution we might learn a useful lesson from our adversaries, the manufacturers of alcoholic nostrums, who introduce their advertisements into every household, by publishing them combined with almanacs, comic illustrations, note-books, etc., *i. e.*, not only free, but winged with extra inducements to the recipient, and often by the special subvention of druggists and village postmasters—till quack annuals have almost superseded the old family calendars with their miscellanies of pious adages and useful recipes. Could we not retrieve the lost vantage ground by the publication of temperance year-books, compiled by a committee of our best tract societies and distributed by agents of the W. C. T. U.—with inspiring conviction to emulate the zeal stimulated by a bribe of gratuitous brandy bottles?

Popular books must above all be *interesting*, and with a large plurality of readers that word is still a synonym of entertaining. A German bookseller estimates that the romances of Louisa Mühlbach have done more to familiarize her countrymen with the history of their fatherland than all historical text books, annals and chronicles taken together, and we should not despise the aid of the novelist, if he should possess the gift of making fiction the hand-maid of truth, and the rarer talent of awakening the reflections as well as the emotions of his readers, for all such appeals should prepare the way for the products of the temperance press proper, by which we should never cease to invoke the conscience and the reason of our fellowmen.

2. PROSCRIPTION.—That union is strength is a truth which asserts itself even at the expense of public welfare, and in favor of those who combine to thwart the purposes of the law or prevent the progress of needed reforms. To the cabals of such adversaries, against whom the influence of moral suasion would be powerless, we should oppose weapons that would strike at the foundation of their strength, namely, the most effectual means to diminish the number of their allies. Many of those who are callous to the stings of conscience would hesitate to defy the stigma of public opinion; others who are proof against all other arguments would yield if we could make it their commercial interest to withdraw their aid from the enemies of mankind.

That the prescription of alcohol for remedial purposes will ultimately be abandoned, like bleeding, blue-pill dosing and other medical anachronisms, is as certain as that the Carpathian peasants will cease to exorcise devils by burning cow dung, and we can somewhat promote the advent of that time by patronizing reform physicians in preference to “brandy-doctors,” as Benjamin Rush[5] used to call them, and by classing alcoholic “bitters” with the prohibited beverages. It is mere mockery to prohibit the sale of small beer and permit quacks to sell their brandy as a “digestive tonic,” and obviate the inconveniences of the Sunday law by consigning their liquor to a drug-store. Does the new name or the admixture of a handful of herbs change the effects of the poison? We might as well prohibit gambling and permit musical lottery drawings under the name of sacred concerts. Till we can do better we should permit druggists to sell alcoholic bitters only on the certified prescription of a responsible physician, all such prescriptions to be duly registered and periodically reported to the Temperance Commissioner of a Board of Health. Nostrum-mongers[6] will probably continue to fleece the ignorant to the end of time, but they must cease to decoy their victims by pandering to the alcohol vice.

3. HEALTHIER PASTIMES.—There is no doubt that a lack of better pastimes often tends to promote intemperance. In thousands of our country towns, equidistant from rural sports and the amusements of the metropolis *ennui* rather than ignorance[C] or natural depravity leads our young men to the dram shop, and in recognizing that fact we should not delude ourselves with the hope that reading-rooms alone could remedy the evil.[D] The *craving after excitement*, in some form or other, is an instinct of human nature which may be perverted, but can never be wholly suppressed, and in view of the alternative we would find it cheaper—both morally and materially—to gratify that craving in the comparatively harmless way of the Languedoc[9] peasants (who devote the evening hours to singing contests, trials of skill, round dances, etc.), or after the still better plan of the ancient Greeks. Antiquity had its Olympic Games, Nemean and Capitoline arenas, *circenses*, and local festivals. The Middle Ages had their tournaments, May days, archery contests, church festivals and guild feasts. The Latin nations still find leisure for pastimes of that sort—though in modified, and not always improved, forms; but in Great Britain, Canada and the United States, with their six times twelve hours of monotonous factory work, and

Sunday laws against all kinds of recreations, the dreariness of existence has reached a degree which for millions of workingmen has made oblivion a blest refuge, and there is no doubt that many dram-drinkers use alcohol as an anodyne—the most available palliative against the misery of life-weariness. We would try in vain to convert such men by reproofs or ostracism. Before we can persuade them to renounce their excursions to the land of delirium the realities of life must be made less unendurable. They know the dangers of intemperance, but consider it a lesser evil.^[E] They know no other remedy. Hence their bitter hatred of those who would deprive them of that only solace. Shall we resign such madmen to their fate? I am afraid that their type is represented by a larger class than current conceptions might incline us to admit. Let those who would verify those conceptions visit a popular beer garden—not as emissaries of our propaganda, but as neutral observers. Let them use a suitable opportunity to turn the current of conversation upon a test topic: “Personal Liberty,” “The Sunday Question,” “Progress of the Prohibition Party.” Let the observer retain his mask of neutrality, and ascertain the views—the private views—of a few specimen toppers. Do they deny the physiological tendencies of their practice? The correlation of alcohol and crime? They avoid such topics. No, nine out of ten will prefer an unanswerable or unanswered argument; the iniquity of interfering with the amusements of the poor, with the only available recreations of the less privileged classes. Take that away and what can a man do who has no better pastimes, and can not always stay at home? What shall he do with sixteen hours of leisure?

The question then recurs: How shall we deal with such men? How reclaim them sufficiently even for the nobler purposes of the present life, not to speak of higher aims? How save them from the road that leads down to death? A change of heart may now and then work wonders, even the wonder of a permanent reform; but we have no right to rely on constant miracles, and for thousands in sorest need of help there is only one practical solution of the problem: Let us provide an opportunity of better pastimes—*not as a concession to our enemies, but as the most effectual method to counteract the attraction of their snares and deprive them of the only plausible argument against the tendencies of our reform.* We need not profane the Sabbath by bull-fights. We need not tempt the poor to spend their wages on railway excursions or the gambling tables of a popular

summer resort. But we should recognize the necessity of giving them once a week a chance for outdoor amusements, and unless we should prefer the Swedish compromise plan of devoting the evening of the Sabbath to earthly purposes, we should adopt the suggestion of the Chevalier Bunsen,[\[10\]](#) and amend the eight hour law by a provision for a *free Saturday afternoon*. Half a day a week, together with the evenings of the long summer days, would suffice where the means of recreation are near at hand. Even the smallest factory villages could afford a little pleasure ground of their own, a public garden with a free gymnasium, a footrace track, ball ground, a tennis-hall or nine-pin-alley, for the winter season, a free bath, and a few zoölogical attractions. In larger towns we might add free music, a restaurant managed on the plan of Susanna Dodds, M.D.,[\[E\]](#) and perhaps a museum of miscellaneous curiosities. Such pleasure resorts should be known as *Temperance Gardens*. They would redeem as many drunkards as all our prisons and inebriate asylums taken together; they would do more: they would *prevent* drunkenness. And above all, they would accustom the working classes to associate the name of Temperance with the conceptions of liberality, manliness, cheerfulness, and recreation, instead of—well, their present misconceptions. We might arrange monthly excursions, and the happiest yearly festival would be a Deliverance Feast; an anniversary of the day when the city or village decided to free itself from the curse of the poison traffic. Like some of the Turner halls[\[11\]](#) of the German gymnasts, temperance gardens could be made more than self-supporting by charging a small admission fee to the spectator-seats of the gymnasium, and selling special refreshments at a moderate advance on the cost price. The surplus might be invested in prizes to stimulate competition in such gymnastics as wrestling, running, and hammer throwing (“putting the club,” as the Scotch highlanders call it), with reserved days, or arenas, for juvenile competitors. In winter we might vary the program by archery, singing contests, and trials of skill in various domestic fashions, with an occasional “spelling bee”—at least for those who could be trusted to consider it a pastime, rather than a task, for the purpose of recreation should not be sacrificed even to considerations of utility. In regard to athletics, that apprehension would be superfluous; the enthusiasm of gymnastic emulation has exerted its power at all times and among all nations, and needs but little encouragement to revive in its old might. It would make the Temperance Garden what the Village Green was to the archers of Old England, what the palæstra was to

the youth of ancient Greece. It would supersede vicious pastimes; it would regenerate the manhood of the tempted classes, and thus react on their personal and social habits; they would satisfy their craving for excitement in the arena, they would learn to *prefer mechanical to chemical stimulants*. [G] Physical and moral vigor would go hand in hand.

The union of temperance and athletic education has, indeed, been the ideal of many social reformers, from Pythagoras to Jean Jacques Rousseau, [12] and the secret of their failure was a mistake that has defeated more than one philanthropic project. They failed to begin their reform at the basis of the social structure. He who fears the hardships of such a beginning lacks, after all, true faith in the destiny of his mission. Perseverance and uncompromising loyalty to the tenets of our covenant is to us a duty, as well as the best policy, for as a moral offense treason itself would not be more unpardonable than doubt in the ultimate triumph of a cause like ours. There is a secret which almost seems to have been better known to the philosophers and patriots of antiquity than to this unheroic age of our own, namely, that in the arena of moral contests a clearly undeserved defeat is a step toward victory. In that warfare the scales of fate are not biased by a preponderance of gold or iron. Tyrants have reached the term of their power if they have made deliverance more desirable than life; the persuasive power of Truth is increased by oppression; and if the interests of a cause have become an obvious obstacle in the road of progress and happiness the promoters of that cause have to contend with a law that governs the tendencies of the moral as well as the physical universe, and inexorably dooms the unfit to perish.[H] The unmasked enemies of mankind have no chance to prosper.

And even where their disguises still avail them amidst the ignorance of their victims we should remember the consolation of Jean Jacques Rousseau in his address to the Polish patriots: "They have swallowed you, but you can prevent them from assimilating you." Our enemies may prevent the recovery of their spoil; they may continue to devour the produce of our fields and of our labor, but we do not propose to let them enjoy their feast in peace; whatever their gastric capacity, it will be our own fault if we do not cause them an indigestion that will diminish their appetite. "All the vile elements of society are against us," writes one of our lecturers, "but I have no fear of the event if we do not cease to agitate the subject," and we

would, indeed, not deserve success if we should relax our efforts before we have secured the coöperation of every friend of justice and true freedom.

It is true, we invite our friends to a battle-field, but there are times when war is safer than peace, and leads to the truer peace of conscience. The highest development of *altruism* inspires a devotion to the welfare of mankind that rewards itself by a deliverance from the petty troubles and vexations of daily life; nay, all personal sorrows may thus be sunk out of sight, and those who seek release from grief for the inconstancy of fate, for the frustration of a cherished project, for the loss of a dear friend, may find a peace which fortune can neither give nor take away by devoting themselves to a cause of enduring promise, to the highest abiding interest of their fellowmen. At the dawn of history that highest aim would have been: security against the inroads of barbarism. In the night of the Middle Ages: salvation from the phantoms of superstition. To-day it should be: deliverance from the curse of the poison vice.

That deliverance will more than compensate all sacrifices. Parties, like individuals, are sometimes destined to conquer without a struggle; but the day of triumph is brighter if the powers of darkness have been forced to yield step for step, and we need not regret our labors, our troubles, nor even the disappointment of some minor hopes, for in spite of the long night we have not lost our way, and the waning of the stars often heralds the morning.

FOOTNOTES

[A] “All past legislation has proved ineffectual to restrain the habit of excess. Acts of Parliament intended to lessen, have notoriously augmented the evil, and we must seek a remedy in some new direction, if we are not prepared to abandon the contest or contentedly to watch with folded arms the gradual deterioration of the people. Restriction in the forms which it has hitherto assumed, of shorter hours, more stringent regulation of licensed houses and magisterial control of licenses, has been a conspicuous failure. For a short time after the passing of Lord Aberdare’s act, hopes were entertained of great results from the provisions for early closing, and many chief constables testified to the improved order of the streets under their charge; but it soon appeared that the limitation, while it lessened the labor of the police and advanced their duties an hour or so in the night, was not sufficient to reduce materially the quantity of liquor consumed, or the consequent amount of drunkenness.”—*Fortnightly Review*.

[B] “The western Saracens abstained not only from wine, but from all fermented and distilled drinks whatsoever, were as innocent of coffee as of tea and tobacco, knew opium only as a soporific medicine, and were inclined to abstemiousness in the use of animal food. Yet six millions of these truest sons of temperance held their own for seven centuries against great odds of heavy-armed Giaours, excelled all christendom in astronomy, medicine, agriculture, chemistry and linguistics, as well as in the abstract sciences, and could boast of a whole galaxy of philosophers and inspired poets.”—*International Review*, December, 1880.

[C] “Education is the cure of ignorance,” says Judge Pitman, “but ignorance is not the cause of intemperance. Men who drink generally know better than others that the practice is foolish and hurtful.” “It is not the most earnest and intelligent workers in the sphere of public education that make their overestimate of it as a specific for intemperance. While they are fully sensible of that measure of indirect aid which intellectual culture brings to all moral reforms, they feel how weak is this agency alone to measure its strength against the powerful appetite for drink.”

[D] “In a primitive state of society field sports afford abundant pastimes, our wealthy burghers find indoor amusements, and scholars have ideal hunting grounds of their own; but the large class of our fellow-citizens, to whom reading is a task rather than a pleasure, are reduced to the hard choice between their *circenses*[7] and their *panes*[8]. Even the slaves of ancient Rome had their saturnalia, when their masters indulged them in the enjoyment of their accumulated arrears of happiness; but our laborers toil like machines, whose best recreation is a temporary respite from work. Human hearts, however, will not renounce their birthright to happiness; and if joy has departed this life they pursue its shadow in the land of dreams, and try to spice the dry bread of daily drudgery with the sweets of delirium.”—*International Review*, December, 1880.

[E] “But beside their excitative influence, strong stimulants induce a lethargic reaction; and it is for the sake of this after effect that many unfortunates resort to intoxication. They drink in order to get drunk; they are not tempted by the poison-fiend in the guise of a good, familiar spirit, but deliberately invoke the enemy which steals away their brains.”—*International Review*, December, 1880.

[F] Author of “Health in the Household.”

[G] “I can not help thinking that most of our fashionable diseases might be cured *mechanically instead of chemically*, by climbing a bitter-wood tree, or chopping it down, if you like, rather than swallowing a decoction of its disgusting leaves.”—*Boerhave*.

[H] “The ultimate issue of the struggle is certain. If any one doubts the general preponderance of good over evil in human nature, he has only to study the history of moral crusades. The enthusiastic energy and self-devotion with which a great moral cause inspires its soldiers always have prevailed, and always will prevail, over any amount of self-interest or material power arrayed on the other side.”—*Goldwin Smith*. [13]

SUNDAY READINGS.

SELECTED BY CHANCELLOR J. H. VINCENT, D.D.

[*March 1.*]

Repose now in thy glory, noble founder. Thy work is finished; thy divinity is established. Fear no more to see the edifice of thy labors fall by any fault. Henceforth beyond the reach of frailty, thou shalt witness from the heights of divine peace, the infinite results of thy acts. At the price of a few hours of suffering, which did not even reach thy grand soul, thou hast bought the most complete immortality. Banner of our contests, thou shalt be the standard about which the hottest battle will be given. A thousand times more alive, a thousand times more beloved since thy death than during thy passage here below, thou shalt become the corner-stone of humanity so entirely, that to tear thy name from this world would be to rend it to its foundations. Between thee and God there will no longer be any distinction. Complete conqueror of death, take possession of thy kingdom, whither shall follow thee, by the royal road which thou hast traced, ages of worshipers.

The essential work of Jesus was the creation around him of a circle of disciples in whom he inspired a boundless attachment, and in whose breast he implanted the germ of his doctrine. To have made himself beloved “so much that after his death they did not cease to love him,” this was the crowning work of Jesus, and that which most impressed his contemporaries. His doctrine was so little dogmatical that he never thought of writing it or having it written. A man became his disciple, not by believing this or that, but by following him and loving him. A few sentences treasured up in the memory, and above all, his moral type, and the impression which he had produced, were all that remained of him. Jesus is not a founder of dogmas, a maker of symbols; he is the world’s initiator into a new spirit.... To adhere

to Jesus in view of the kingdom of God, was what it was originally to be a Christian.

Thus we comprehend how, by an exceptional destiny, pure Christianity still presents itself, at the end of eighteen centuries, with the character of a universal and eternal religion. It is because in fact the religion of Jesus is in some respects the final religion. The fruit of a perfectly spontaneous movement of souls, free at its birth from every dogmatic constraint, having struggled three hundred years for liberty of conscience, Christianity, in spite of the fall which followed, still gathers the fruits of this surpassing origin. To renew itself it has only to turn to the Gospel. The kingdom of God, as we conceive it, is widely different from the supernatural apparition which the first Christians expected to see burst forth from the clouds. But the sentiment which Jesus introduced into the world is really ours. His perfect idealism is the highest rule of unworldly and virtuous life. He has created that heaven of free souls, in which is found what we ask in vain on earth, the perfect nobility of the children of God, absolute purity, total abstraction from the contamination of this world, that freedom, in short, which material society shuts out as an impossibility, and which finds all its amplitude only in the domain of thought. The great master of those who take refuge in this ideal kingdom of God is Jesus still. He first proclaimed the kingliness of the spirit; he first said, at least by his acts, "My kingdom is not of this world." The foundation of the true religion is indeed his work. After him there is nothing more but to develop and fructify.

"Christianity" has thus become almost synonymous with "religion." All that may be done outside of this great and good Christian tradition will be sterile. Jesus founded religion on humanity, as Socrates founded philosophy, as Aristotle founded science. There had been philosophy before Socrates and science before Aristotle. Since Socrates and Aristotle, philosophy and science have made immense progress; but all has been built upon the foundation which they laid. And so, before Jesus, religious thought had passed through many revolutions; since Jesus it has made great conquests; nevertheless it has not departed, it will not depart from the essential condition which Jesus created; he has fixed for eternity the idea of pure worship. The religion of Jesus, in this sense, is not limited. The church has had its epochs and its phases; it has shut itself up in symbols which have

had or will have their day; Jesus founded the absolute religion, excluding nothing, determining nothing, save its essence....

Whatever may be the transformations of dogma, Jesus will remain in religion the creator of its pure sentiment; the Sermon on the Mount will never be surpassed. No resolution will lead us not to join in religion the grand intellectual and moral line at the head of which beams the name of Jesus.—*Renan*.^[1]

[March 8.]

Were you ever made to see and admire the all sufficiency of Christ's righteousness, and excited by the spirit of God to hunger and thirst after it? Could you ever say, my soul is athirst for Christ, yea, even for the righteousness of Christ? Oh, when shall I come to appear before the presence of my God in the righteousness of Christ; oh, nothing but Christ! nothing but Christ! Give me Christ, O God, and I am satisfied! My soul shall praise thee forever. Was this, I say, ever the language of your hearts? And after these inward conflicts, were you ever enabled to reach out the arm of faith and embrace the blessed Jesus in your souls, so that you could say, *My beloved is mine, and I am his*? If so, fear not, whoever you are—hail, all hail, you happy souls! The Lord, the Lord Christ, the everlasting God is your righteousness. Christ has justified you, who is he that condemneth you? Christ has died for you, nay, rather is risen again, and ever liveth to make intercession for you. Being now justified by his grace, you have peace with God, and shall ere long be with Jesus in glory, reaping everlasting and unspeakable redemption both in body and soul. For there is no condemnation to those that are really in Christ Jesus. Whether Paul or Apollos or life or death, all is yours if you are Christ's, for Christ is God's! ... Oh think of the love of Christ in dying for you! If the Lord be your righteousness, let the righteousness of your Lord be ever in your mouth.... Think of the greatness of the gift, as well as of the giver! Show to all the world in whom you have believed! Let all by your fruits know that the Lord is your righteousness, and that you are waiting for your Lord from heaven! Oh, study to be holy, even as he who has called you, and washed you in his own blood, is holy! Let not the righteousness of the Lord be evil spoken of through you. Let not Jesus be wounded in the house of his friends; but grow

in grace and in the knowledge of our Lord and Savior Jesus Christ day by day. Oh, think of his dying love! Let that love constrain you to obedience! Having much forgiven, love much.—*Whitefield*.^[2]

[*March 15.*]

But in proportion to the exaltation of the soul, and also in proportion to its purity and spirituality—the very opposite extreme or condition; in proportion to the impressibleness and moral sensibility of a man’s spiritual nature, he has direct communion with God, as friend with friend, face to face. “Blessed are the pure in heart, for they shall see God.” There are thousands of instances—they occur in every church where there are eminent Christians—of men and women who come to such a state of spiritual purity and spiritual openness that they talk with God as friend with friend. There is the direct operation of the Spirit of God upon their soul. Not that they less than any others are blessed by the spirit that applies the Word; not that they less than any others are subject to the indirect operations of nature and society; but there is, over and above these, also, for those that are able to take it, this direct inspiration of God’s soul. Whether it be by thought, I know not; or whether it be by moral feeling, I know not. “The wind bloweth where it listeth, and thou hearest the sound thereof, but canst not tell whence it cometh and whither it goeth; so is every one that is born of the spirit.” I do not know the mode of divine agency; but of the fact that the human soul in its higher spiritual relations is open; that there is nothing between it and God, as it were; that it palpitates, as it were, under the conscious presence of God, and is lifted up to a faith and a truth that are not possible to it in its lower realms—of that fact I have no more doubt than I have of my own existence.

There is such a thing yet as walking with God; there is such a thing yet as being under direct divine inspiration. I do not think there is such a thing yet as *authoritative* inspiration. Apostles are over and gone. Prophets have had their day. It is *individual* inspiration that exists now. It is authoritative only for the soul to which it comes, not lifting that soul up into authority, and enabling it to say “Thus saith the Lord” to any other soul. But I believe that still the divine Spirit works upon the individual heart, and teaches that individual heart as a father teaches a child.

Blessed are they that need no argument; and blessed are they whose memories take them back to the glowing hours of experience, in which they have seen the transfigured Christ; in which to them the heavens have been opened; in which to them the angels of God not only have descended upon the ladder, but have brought the divine and sacred presence with them. Many a couch of poverty has been more gorgeous than a prince's couch; many a hut and hovel has been scarcely less resplendent to the eye of angels than the very battlements of heaven. Many that the world has not known; who had no tongue to speak, and no hand to execute, but only a heart to love and to trust—many such ones have had the very firmament of God lifted above them, all radiant. There is this truth in the Spirit of God that works in the hearts of men directly, and in overpowering measure. Blessed be God, it is a living truth; and there are witnesses of it yet.—*Beecher*.

[March 22.]

Jesus Christ, in his dying discourse with his eleven disciples, in the 14th, 15th, and 16th chapters of John (which was, as it were, Christ's last will and testament to his disciples, and to his whole church), often declares his special and everlasting love to them, in the plainest and most positive terms, and promises them a future participation with him in his glory in the most absolute manner, and tells them, at the same time, that he does so to the end that their joy may be full. John xv:2: "These things have I spoken unto you that my joy might remain in you, and that your joy might be full." See also, at the conclusion of the whole discourse, chapter xvi:33: "These things have I spoken unto you, that in me ye might have peace. In the world ye shall have tribulation; but be of good cheer; I have overcome the world." Christ was not afraid of speaking too plainly and positively to them; he did not desire to hold them in the least suspense. And he concluded that last discourse of his with a prayer in their presence, wherein he speaks positively to his Father of those eleven disciples, as having all of them savingly known him, and believed in him, and received and kept his word; and that they were not of the world; and that for their sakes he sanctified himself; and that his will was that they should be with him in his glory; and tells his Father that he spake these things in his prayer, to the end that his joy might be fulfilled in them: verse 13. By these things it is evident that it is agreeable to Christ's designs, and the contrived ordering and disposition

Christ makes of things in his church, that there should be sufficient and abundant provision made, that his saints might have full assurance of their future glory.

The apostle Paul, through all his epistles, speaks in an assured strain; ever speaking positively of his special relation to Christ, his Lord, and Master, and Redeemer; and his interest in, and expectation of, the future reward. It would be useless to take notice of all places that might be enumerated. I shall mention but three or four. Gal., ii:20: "Christ liveth in me; and the life which I now live in the flesh, I live by the faith of the Son of God, who loved me, and gave himself for me." Phil., i:21: "For to me to live is Christ, and to die is gain." II. Tim., i:12: "I know whom I have believed, and I am persuaded that he is able to keep that which I have committed unto him against that day." II. Tim., iv:7,8: "I have fought a good fight, I have finished my course, I have kept the faith. Henceforth there is laid up for me a crown of righteousness, which the Lord, the righteous judge, will give me at that day." ... It further appears that assurance is not only attainable in some very extraordinary cases, but that all Christians are directed to use all diligence to make their calling and election sure; and are told how they may do it. II. Peter, i:5-8. And it is spoken of as a thing very unbecoming of Christians, and an argument of something very blamable in them, not to know whether Christ be in them or no. II. Cor., xiii:5: "Know ye not your own selves, how that Jesus Christ is in you except ye be reprobates?" And it is implied that it is an argument of a very blamable negligence in Christians, if they practice Christianity after such a manner as to remain uncertain of the rewards, in I. Cor., ix:26: "I therefore so run, not as uncertainly." And to add no more, it is manifest that for Christians to know their interests in the saving benefits of Christianity is a thing ordinarily attainable, because the apostles tell us by what means Christians (and not only apostles and martyrs) were wont to know this. I. Cor., ii:12: "Now we have received, not the spirit of the world, but the Spirit which is of God, that we might know the things which are fully given to us of God." And I. John, ii:3: "And hereby do we *know* that we know him if we keep his commandments." And verse 5: "Hereby *know* we that we are in him." Chapter iii:14: "We *know* that we have passed from death unto life." ... Verse 19: "Hereby we *know* that we are of the truth, and shall assure our hearts before him." Verse 24: "Hereby we *know* that he abideth

in us, by the spirit which he hath given us.” So chapter iv:13, and chapter v:29, and verse 19.—*President Edwards*.[\[3\]](#)

[March 29.]

Who has an understanding so exalted, so richly gifted, as to be able to say what love is! Should I say it is a dew, I merely describe its refreshing power. Should I say it is a star, I but describe its loveliness. Should I say it is a storm, I but describe the impossibility of restraining it. Should I say it is a ray of the sun, then I but describe its hidden source. Should I say it is produced in the utmost depths of the soul, when the breath of heaven unites with the heart’s blood of the new man, that it is the breath of the soul, still I should not have represented it, for I should but have said what it is in itself, not what it is to others. Should I say it is the light of the sun, that gives life and color to all creatures, still I should not have truly set it forth, for I should but have said what it is for others, not what it is in itself. Should I say it is a ray of the seven colors in a pure drop of water, still I should not have described it, for it is not so much a form as an odor, and a savor, in the depths of the human heart. Who has such a lofty understanding, such deep thoughts, as to be able to say what love truly is! The Scripture says—it is a flame of the Lord.[\[1\]](#) Yes it is a flame, steady, bright, and pure; a flame which lights up and warms, and shines through the heart into which it has entered, and then falls on other hearts, and the more light and warmth it gives to others, the brighter and stronger it burns in our breast.

But love, says the apostle, is greater than faith and hope, for beyond that limit where faith and hope depart, love still remains.... For as the door in this poor temporal life was but a little gate that did not always stand open, but was often shut by a strong gust of wind; in eternity the poor little gate will become a mighty portal, whose doors stand open night and day, which no storm-wind will ever close, through which the soul will freely pass into the heart of God and all his creatures. O, since in this life love has made us so rich, though but a little brook, which, when the sun shone fiercely, was almost dried up, how rich will it not make us when the little brook has become the stream, yea, the ocean, when it flows forth from the heart of God, in full spring-tide, and sin no more builds a barrier in the heart of the creature, and there will be a free and sacred giving and receiving between

heaven and earth, and among all that is in heaven and upon earth! O, who has so exalted an understanding that he can truly say what love is!—*Tholuck*.^[4]

FOOTNOTES

^[1] Canticles, viii:6, German version.

STUDIES IN KITCHEN SCIENCE AND ART.

VI. CABBAGES, TURNIPS, CARROTS, BEETS, AND ONIONS.

BY BYRON D. HALSTED, SC. D.

THE CABBAGE is a native of Europe, and grows wild along the sea coasts of England. The wild plant lives for two years, has fleshy leaves, and is so different from the cabbages of the garden as not to be recognized as their parent. Under cultivation this one species of plant (*Brassica oleracea*[\[1\]](#)) has produced the Savoy, Brussels sprouts, cauliflower, borecole,[\[2\]](#) etc. A more wonderful plant and a more useful one is seldom found in the whole range of the vegetable kingdom. The Romans did much to extend the culture of the cabbage. In Scotland it was not generally known until the time of Cromwell. Much improvement has been made in American sorts of cabbages within the past fifty years. In the wild state the cabbage has a hard, woody stalk, but the fine specimens in market have only a small stem, bearing a large, compact head, of closely folded leaves.

The first essential in the successful growing of cabbage is the right kind of soil. It should be a sandy loam, with a gravelly, and not a clayey subsoil. Soil that is naturally wet must be thoroughly underdrained before being devoted to cabbage growing. The importance of an abundance of well rotted manure can not be too fully impressed upon the mind of any person contemplating the production of excellent cabbages. Much that may be here said concerning the preparation of the soil for growing cabbages applies with equal force to the other vegetables treated in this article. Earliness is one of the leading points to be gained in raising most garden crops. It is the man with the first load of cabbages that gets the best price in the market. There is a great deal of stress to be placed upon the proper selection of seed, but seed is not all. The young plants of the earliest sorts must be fed, and

they require this food at an early stage in their growth, when chemical changes are only slowly going on in the soil. In other words, early crops need a far larger amount of manure for their satisfactory growth than crops started in midsummer, when the soil is rapidly yielding up its food elements. Early crops need to grow in cool spring weather, and therefore should be abundantly supplied with food in an available form. Mr. Gregory says in his excellent pamphlet on "How to Grow Cabbages," "If the farmer desires to make the utmost use of his manure for that season, it will be best to put most of it into the hill, particularly if his supply runs rather short; but if he desires to leave his land in good condition for next year's crop, he had better use part of it broadcast. My own practice is to use all my rich compost broadcast, and depend on guano, phosphates, or hen manure in the hill." This view of heavy manuring is confirmed by Mr. Henderson, in his "Farm and Garden Topics," when he says: "For the early cabbage crop it should always be spread on broadcast, and in quantity not less than one hundred cart loads or seventy-five tons to the acre.... After plowing in the manure, and before the ground is harrowed, our best growers in the vicinity of New York sow from four to five hundred pounds of guano, or bone dust, and then harrow it deeply in." The best sorts of cabbages for the early crop are: The Jersey Wakefield, which has a head of medium size, close, and of a deep green color; Early York, smaller, but quite early; Early Winningstadt, later, but an excellent sort. Among the best late kinds may be named: Large Flat Dutch, American Drumhead, Drumhead Savoy, and the Red Dutch. The last mentioned is largely used in pickling.

The young plants are obtained from seeds in various ways, determined by the numbers desired. When large quantities are needed for the early crop, the seed is sown in a hot-bed or green-house, about February 1st, for the latitude of New York City, and transplanted into other heated beds near March 1st. In this way fine plants may be obtained by the first of April. Many of the large cabbage growers prepare the soil, mark it in rows, and drop the seed in the hills where the plants are to grow. In this way much labor is saved, and there is the advantage of having several plants in each hill, to guard against losses from cut-worms. Cabbages quickly respond to good culture, and repay in large measure for every stirring of the soil, either with the hoe or the horse cultivator.

The most troublesome insect enemy is probably the Cabbage-worm, which in some localities has destroyed the whole crop. The mature insect deposits its eggs upon the under side of the cabbage leaves. These eggs soon hatch, and the green caterpillars begin their destructive work. No poisonous substances can be applied without endangering the lives of those who may afterward eat the cabbage. Hot water (160 degrees) has proved effective in killing the worms, while not doing injury to the plants. Flea-beetles have done some damage, as also the Cabbage-bug. After the crop is grown the cabbages may be kept by burying them in trenches, heads down. Three facts need to be kept in mind: Repeated freezing and thawing cause rot; excessive moisture also induces decay; and a dry air withers the head and destroys the flavor. About a foot of earth is usually a sufficient covering.

Cabbage in the many forms it is presented upon the table is a most wholesome and agreeable article of food. The farmer's garden is not complete without a full crop of cabbages. Any heads that are not needed for the family table can be fed with profit to the farm live stock. Poultry in particular, need some green food daily through the winter season, and a cabbage now and then satisfies this natural craving.

TURNIPS.—The garden turnips belong to the same genus (*Brassica*) with the cabbages, and are therefore closely related to them. The turnip is supposed to be a native of England and other parts of Europe. It is not known when this plant was first introduced into cultivation, and its wild state is unknown. At the present time it forms one of the prominent crops in all countries adapted to its growth.

The remarks made under the subject of cabbages concerning the free use of manure need not be repeated here. Turnips grow freely upon a rich and mellow soil, kept clean of all weeds. They do not require as fertile a soil as cabbages, and when the earth is very rich, there is sometimes an excessive growth of tops, without a corresponding development of the roots. It is not necessary to say that cabbages are grown for their many thick leaves, while turnips are raised for their roots. Plants as a whole have many places for the storing up of nourishment. Sometimes it is in the stems, as in the potato; in other cases the leaves or roots serve as a store-house of accumulated substance. The plant makes these deposits, to be drawn upon at some future

time, either for further growth of the same plant or for the early development of another. The root crops, for example, are naturally plants of two year's duration. The first season is spent in gathering and storing up substance in a large root. During the following year the starch, sugar, oil, etc., is withdrawn and used in the production of a flower-stalk, upon which the crop of seeds is finally borne, and after this the plant dies.

Turnips are mainly grown as a second crop, following early potatoes, etc. The soil should be made fine and rich before the seed is sown. Rutabagas may be sown from the 15th of June until the 15th of July. Yellow Stone, Aberdeen, White Cowhorn and Strap-leaved Red-top are sown in the order named, and from July 15th to the 1st or 10th of September. The seed is sown in drills, wide enough apart to admit of horse cultivation. The thinning of the plants in the row is of great importance. This work is best done with a hoe, the workman chopping out the turnips and leaving the plants about four to six inches apart in the row. In garden culture the rows need not be so far apart. It is very essential to keep the weeds down and the soil frequently stirred. The harvesting is simple. When growth is completed the roots are pulled, then the tops cut off and the turnips placed in root cellars or pits.

Turnips have an important place in a carefully planned system of farming. The root crop is a means of securing a large amount of most wholesome food for live stock, and at the same time it cleans the soil from weeds and prepares it for the growth of succeeding crops.

The leading insect enemy of young turnip plants is the Turnip-fly. If the seedlings can be protected until they get a good start in life there is no further trouble. Equal parts of wood ashes and land plaster scattered over the young turnip leaves is a good remedy. Air-slaked lime is also employed in the same manner.

THE CARROT.—The wild carrot, *Daucus Carota*,^[3] is a native of Europe and has become naturalized in this country to such an extent as to be ranked among the worst of weeds. The cultivated carrot was introduced into England by the Dutch, in the reign of Queen Elizabeth (last half of the sixteenth century), and has since been much improved and quite generally grown. In its native or wild state the root is small, woody, and of very little

value as an article of food. All of our so-called “root plants” in the wild state store up only sufficient food in the root to meet the wants of the plant the coming season. This tendency to accumulate has been developed under cultivation, and an excess is stored up, which is appropriated by man. The plant has enjoyed more favorable conditions for growth and been relieved in great part of the struggle for existence that is constantly going on among wild plants. All cultivated plants are living unnatural lives, being favored in various ways, and when they are left to shift for themselves either die or drift back, generation after generation, to the old original form from which the ancestors were forced to depart. No plant is a better illustration of this fact than the carrot. If left for only a few years, the fleshy rooted plants of the garden degenerate into the coarse, woody-rooted weeds of the pasture or hedge-row. We can not pass this point without endeavoring to enforce the importance of keeping up all the most favorable conditions of growth for garden vegetables, and carefully selecting seed of plants that show the least tendency to degenerate.

The plot for growing carrots should be nearly level, otherwise heavy rains may wash the seeds and young plants out of place. The soil should be deep, rich and mellow. Carrots are no exception to the rule that root crops flourish under high culture. When the barnyard fails to supply sufficient manure, it is well to use guano, superphosphates, and other quick acting fertilizers. If the soil is heavy, it is best to sow the seed in ridges made by a plow, thus enabling a horse-weeder to pass between the rows and not injure the young plants coming through the surface. Use seed not over one year old, and it is well to sow some radish seed with it, to come up first and show the rows, thus aiding in the early cultivation of the soil. It is of the greatest importance to keep the weeds down until the carrots get a good start. About six weeks after sowing, that is, the middle of July, thin the plants, leaving them four or five inches apart in the row. The carrots are dug and stored like most root crops. If grown in large quantities, most of the labor of getting the roots out of the soil is performed by horses. Carrots keep well in long piles, six feet wide at the bottom, and of any length. Ventilating holes need to be left at frequent intervals along the ridge of the covered heap. There are several varieties of carrots, some of them being earlier than others, while the size and general shape varies greatly. The Long Orange, Short Horn, Early Horn and White Belgian are among the

leading sorts. Market gardeners are now favoring the shorter sorts, the endeavor being to get them turnip-shaped, and thus save much labor in digging the roots.

BEETS.—The species *Beta vulgaris*,^[4] the parent of our common beets, is a native of Egypt, and grows wild along the Mediterranean Sea at the present day. The name is from the Celtic word *Bett*, meaning *red*, the prevailing color of most beets. This garden vegetable has been generally grown for six hundred years, and during that time has undergone many important changes. Long ago the beet arrived at a state of perfection beyond which it is not easy to pass. The Mangold-Wurzel^[5] and Sugar Beets are derived from another species. These are grown very extensively in Europe and are worthy of far more attention by American farmers. The Swiss Chard is another species of the genus *Beta*, largely grown in some countries for the leaves, which only are used. They are stripped off and used like spinach. The soil best adapted to the growing of beets is a rich, sandy loam, rather light than otherwise. It should be thoroughly pulverized by deep plowing, harrowing, etc., until a fine, mellow bed is prepared for the seeds. The seeds are sown in rows, and the soil should be pressed firmly upon them. For early beets the sowing may be done so soon as the ground can be worked. The late sorts may be sown in July. As soon as the plants are above ground a push-hoe should be passed close to the rows. A few days later the beets need to be thinned to five or six inches in the row. The removed plants make excellent greens. The remaining work until harvest time is keeping the soil free from weeds and loose by frequent hoeing. The rake is better than the hoe, if it is used frequently and no weeds get large. Beets should be harvested before frosts injure them. Handle carefully and store in a place where the temperature is uniformly a few degrees above freezing.

The Egyptian is among the best early sorts; it has a dark blood color, and much resembles a flat turnip in shape. The Long, Smooth, Blood Beet is considered as ranking first for general family and market uses.

The Mangold-Wurzels are coarse beets of large size, grown as a field crop for live stock. The White Sugar is a Mangold, free from much of the red coloring matter of the red sorts. These larger varieties of beets are very extensively grown in Europe for the manufacture of sugar, and it would add to our agricultural wealth if they were more frequently a part of a well

planned system of rotation of crops in America. It may not pay for us to make beet sugar, but the use of the roots as a wholesome winter food for stock is profitable.

ONIONS.—The onion (*Allium cepa*[\[6\]](#)) has been cultivated from early times, and its native country is unknown. As it is mentioned in sacred writings it is supposed that its home is in the far East. Onions thrive best on old ground, especially if it is a light, sandy loam. The onion field should be nearly level, clear of weeds, and liberally supplied with the best well-rotted manure; guano and superphosphates are excellent for onions. Deep plowing is not necessary. The amount of seed to be used depends upon the kind of onions desired. If they are to be pulled for early market, more seed is required than when they are to attain their full growth.

There are many varieties of onions grown from seeds. The Yellow Danvers, White Portugal and Weathersfield Red are well known sorts, representing the three prevailing colors. Onions are largely grown from sets, that is, bulbs that have ripened while quite small, and when set out grow and form large onions. The small size and early maturity are due to sowing the seed thick. From thirty to forty seeds are sown to each inch of the row. The sets are mature when the leaves begin to wither, and are then removed and dried. In planting the sets they are placed in rows about four inches apart.

The “Potato Onion” or “English Multiplier” is propagated by offsets. An onion of this class, if planted in the spring, will produce a cluster of small ones around it. These small onions will grow into large ones the next season. There are several sorts of onions that bear clusters of small bulbs upon the tops of the flower stalks, in place of seed pods. The “Tree,” “Top,” and “Egyptian” onions are of this class. These bulblets, when planted, produce large bulbs, and these latter, when set out the following season, throw up stalks bearing bulblets.

Onions are ready for harvesting as soon as the leaves droop and become dry. The bulbs should be well cured and placed in a dry, cool, storage room. The crop is sometimes badly injured by smut, especially when onions have been grown upon the same soil for many years. The onion maggot causes

some destruction. Guano and unleached ashes, when scattered over the bed, have both proved of value.

The above is only a brief consideration of five of the leading garden vegetables. The first four, namely: Cabbages, turnips, carrots and beets, are to a great extent farm crops, well suited for live stock. The composition of these is as follows:

	DRY MATTER.	ALBUMINOIDS.	FAT.	STARCH, SUGAR, ASH. ETC.	
Cabbage	14.3	2.5	0.7	7.1	1.6
Turnips	8.5	1.0	0.15	5.8	0.8
Carrots	14.1	1.3	0.25	9.6	1.0
Beets	18.5	1.0	0.1	9.1	0.8

The turnips contain the least dry substance, and the cabbages are far the richest in albuminoids. The carrot leads in starch, sugar, etc., followed closely by the beets. There is very little poetry in any of the five vegetables here briefly described, though they may enter into the daily food of those who think of lofty things and write in the most elegant style. They are the humble, unobtrusive toilers in the gardens of the world.

THE PREPARATION OF VEGETABLES.

There are two laws underlying the preparation of all vegetables for the table; the first is, cook until tender; the second is, do not cook until mushy and the juice extracted. By overlooking the first you are left with a rank, tough, indigestible dish; by overlooking the second with one watery, and—worst of all culinary adjectives—juiceless. A time-table regulating the exact number of minutes which each vegetable shall be cooked can not be perfectly exact. Not rules, but judgment must decide the limit of time. However a table of approximations may be of service to amateur cooks whose experience has not yet taught them that essential of successful cookery.

CABBAGE.—When young, requires an hour; winter cabbage, double that time.

TURNIPS.—When young, three quarters of an hour; winter turnips, two hours.

CARROTS.—When young, three quarters of an hour; winter carrots, two hours.

BEETS.—When young, three quarters of an hour; winter beets, four hours.

ONIONS.—When young, one hour; winter onions, two hours.

The temperature at which vegetables should be cooked is a point of great importance. A little reflection should easily settle the question, however. When young vegetables are tender, the juices are easily withdrawn, continued stewing or soaking extracts all the flavor and strength; when old they become tough, and only long stewing will make them tender and bring out the juices. By putting young vegetables into cold water we extract the juice before they begin to cook, and by the time they become tender they are tasteless; but by putting winter vegetables into cold water they are gradually softened, and by the time they are cooked tender the juice is fully developed; hence the reason for the rule which cooks have formulated: Put all young, green vegetables into salted boiling water; all dried and winter vegetables into cold water.

Add to your regard for these first principles a nice skill in draining all the water from your cabbage, turnips, carrots, beets and onions, and that most delicate of all cookery arts—the art of seasoning—and you can not fail of toothsome entrées^[1] and salads.

Cabbage Salad or Slaw.—Remove from a firm, fresh cabbage the outer leaves and slice fine. The simplest dressing is of sugar, salt and vinegar. Mayonnaise^[2] dressing may be prepared by taking the beaten yolks of six eggs and into them beating, drop by drop, two tablespoonfuls of salad oil; now alternate with every few drops of two tablespoonfuls of salad oil, small quantities of vinegar until two tablespoonfuls of vinegar have been used. Beat into this mixture, which should be very smooth, one saltspoonful of salt and half as much cayenne pepper, set in a cold place until wanted. A

cooked mayonnaise dressing is made by adding to each tablespoonful of boiling vinegar, the beaten yolk of an egg, and cooking until stiff. Remove the mixture and stir in an ounce of butter. When cool, season it with salt, pepper and mustard; then add sweet cream until it is of the desired consistency.

Hot Slaw is prepared by stewing chopped cabbage until tender, and then adding a dressing of vinegar, butter, salt and pepper.

Pickled Cabbage.—Chop, not too fine, a fresh cabbage, and season it with white mustard seed, salt and pepper. Now pack this firmly into a jar and add cold vinegar. Cloves should be sprinkled over the top to prevent mould. Or, pack a layer of chopped cabbage alternately with a layer of chopped onions, and having salted, allow it to stand for about twenty-four hours. A dressing of one pint of vinegar, one cup of sugar, and one teaspoonful each of ground mustard, black pepper, cinnamon, turmeric, mace, allspice, and celery seed is made for each head of cabbage and half dozen of onions, by scalding the vinegar and adding sugar and spices. Into this dressing pour the cabbage and onions. Allow them to simmer for half an hour, then put into jars.

Boiled Cabbage.—Quarter a cabbage from which the outer leaves have been removed, and which has been examined carefully for insects and slugs. Boil until tender. Drain well, being careful to press out the water. Boiled cabbage may be chopped, and a tablespoonful of butter, pepper and salt stirred in, or it may be served with white sauce or drawn butter. White sauce is made by cooking together one ounce of flour and two ounces of butter, and, after adding a pint of milk allowing the mixture to simmer slowly. Season with salt and pepper. Drawn butter differs from white sauce only in having water or broth in place of the milk. Cabbage may be boiled in water taken from the pot in which corned beef or pork is being cooked. This seasons it nicely.

Stewed Cabbage.—Chop cabbage fine and stew until tender. When “done” add sweet milk sufficient for a dressing and allow it to cook for ten minutes. Season with salt and pepper. Marion Harland gives this recipe for a stewed “stuffed cabbage:” “Choose for this purpose a large, firm cabbage. When perfectly cold bind a broad tape about it, or a strip of muslin, that it

may not fall apart when the stalk is taken out. Remove this with a thin, sharp knife, leaving a hole about as deep as your middle finger. Without widening the mouth of the aperture excavate the center. Chop the bits you have taken out very small; mix with some cold boiled pork or ham, or cooked sausage-meat, a very little onion, pepper, salt, a pinch of thyme, and some bread crumbs. Fill the cavity with this, bind a wide strip of muslin over the hole in the top, and lay the cabbage in a large sauce-pan with a pint of 'hot liquor' from boiled beef or ham. Stew gently until very tender. Take out the cabbage, unbind carefully, and lay in a dish. Keep hot while you add to the gravy, when you have strained it, pepper, a piece of butter rolled in flour, and two or three tablespoonfuls of rich milk or cream. Boil up and pour over the cabbage."

Baked Cabbage.—The cold boiled cabbage left over from dinner is very nice baked. Chop it fine and add a dressing made of beaten eggs and milk and seasoned with salt and pepper. Put it into a buttered baking dish, and having strewn the top with bread crumbs or rolled crackers, bake it brown.

Fried Cabbage.—Another excellent dish to be prepared from cold boiled cabbage is fried cabbage. Chop the cabbage fine and stir in a little melted butter, two beaten eggs, a little cream, pepper and salt, and cook until slightly brown.

Boiled Turnips.—Boil until tender and drain dry. After mashing them smooth, being careful to rub away all hard lumps, stir in a tablespoonful of butter and season with salt and pepper. If it is preferred to cut them in slices, they are nice served with white sauce or drawn butter as a dressing. A little vinegar added to the dressing is by many considered an improvement. Young turnips are nice served whole with either of these sauces.

Stewed Turnips.—An excellent way of warming over boiled turnips is to add sufficient milk to them to stew thoroughly, and then to season with pepper and salt.

Baked Turnips.—Cold boiled or sliced turnips may be "done over" by putting them into a baking-pan, covering with bread crumbs, moistening with milk, and then baking in the oven. Freshly boiled turnips, sliced thin, may be cooked in the same way.

Boiled Carrots.—If carrots are small and young they may be boiled whole, but if they are large they should be split into two or three pieces; when cooked they may be served with butter, salt and pepper, or with white sauce, like sliced boiled turnips.

Mashed Carrots.—Boiled carrots are very nice mashed with a large spoonful of butter, a little cream, and seasoning of pepper and salt worked into them. Serve as you would mashed potatoes.

Fried Carrots.—Cold boiled carrots, or those which have been parboiled, may be sliced and fried brown in butter. They must be seasoned, of course, with pepper and salt.

Stewed Carrots.—Parboil carrots for three quarters of an hour. Put them into a stew-pan and pour on them a teacupful of broth with seasoning of pepper, salt and butter, and stew until they are tender. A little cream and a lump of butter may be added and the whole allowed to boil up.

Boiled Beets.—In preparing beets for the kettle they should be washed, but not cut. When done, rub off the skin and slice. Butter, pepper and salt should be added for seasoning. If you like a dressing of vinegar put a tablespoonful of butter into half a cup of vinegar, add pepper and salt, and boil before turning upon the beets.

Baked Beets.—Slice your beets and place in a baking pan with butter, pepper and salt. Allow about twenty minutes longer for baking than boiling. This method preserves much of the juice of the vegetable which is lost in boiling.

Stewed Beets.—Parboil your beets until nearly done, rub off the skin and slice. Into your stew-pan pour enough milk to cover the beets, add a little butter, salt and pepper, and simmer slowly until they are done.

Boiled Onions.—Onions may be laid in cold water half an hour before cooking. Boil them in two waters until tender. When cooked, drain carefully and serve with butter, salt and pepper. Boiled onions are nice with a dressing of drawn butter.

Baked Onions.—Choose large onions for baking, and after peeling boil for an hour. Drain them thoroughly and about each wrap a piece of buttered

tissue paper, bake them until they are quite tender, then remove the paper and brown in the oven, basting with butter. Serve them with drawn butter.

Stewed Onions.—Onions which have been parboiled may be stewed in milk sufficient to cover. When done, a dressing of hot cream and butter, seasoned with salt and pepper, may be poured over them; or they may be chopped fine, and the cream, butter and seasonings be stirred in.

Fried Onions.—Slice into small strips and fry in butter, taking care to brown them evenly. Season with salt and pepper. Onions sliced thin and fried in hot fat are called *Saratoga onions*.

THE CIRCLE OF THE SCIENCES.

PHYSICS.

In the science of material things, mechanics takes account of forces that act on masses from without; physics, of those that act from within, or which, in some way, modify the condition of the bodies themselves. Both branches were, till recently, included in the vaguely comprehensive term “Natural Philosophy,” and the partial separation observed in modern treatises and text-books gives a little more distinctness to the facts presented. Under the former the earth is contemplated as a planet, obedient to the universal law of gravitation, and moving regularly in its orbit. The mechanism of the system is complete; the measure and adjustment of all the parts perfect.

GEOLOGY,

As a physical science, considers the earth apart from the solar system with which it is connected, and takes account of its materials and structure, and the forces that unite them. Its position in the group is about midway between mechanics and chemistry, being closely allied to other natural sciences, while its phenomena are occasionally varied by both mechanical and chemical agents.

PHYSIOGRAPHIC GEOLOGY

Treats of the earth's exterior physical features; of its form—an oblate spheroid—of its surface, oceans, continents, seas, lakes and rivers, hills, mountains, valleys and plains; of soils made from previously existing organic or inorganic substances, the detritus of rocks containing various minerals and small particles of decomposed vegetable matter. The materials of this outer covering of the earth are from many different sources, and variously constituted. From the finest grains of sand, clay, and loam, to

pebbles, boulders, and fragments of enormous dimensions, they are mingled apparently without any fixed order or proportions; sometimes but slightly covering the solid rock, at others piling it up in ridges and hills of considerable height. In this surface formation are included ancient sea-beaches, lake and river terraces, deltas, deposits of sand and clay, with vast beds of marls, peat and calcareous tufa,^[1] all the progressive accumulations since the present order of things began. In some of these deposits, more recent than the Drift^[2] period, fossils are abundant and very full of interest. In New Zealand the bones of a bird^[3] were found which exceed in bulk those of the largest horse, and are now in the museum of the College of Surgeons, London. The bird when alive was eleven or twelve feet high.

Less than a century ago what might have been a fossil elephant was found imbedded in ice on the coast of Siberia, and in such a perfect state of preservation that the people fed their dogs on its flesh. The animal was well covered with hair, and adapted to a cool climate, a representative of an extinct race. How it was imbedded, or how long it had been preserved in that condition, no one knows.

In Great Britain are found fossils of the rhinoceros and hippopotamus, of elephants, tigers, hyenas and giant elks, all of which are extinct species. The United States is especially prolific in the remains of huge mammals. The mastodon and megatherium were doubtless indigenous to this country. The latter had a thigh bone three times as large as the largest elephant, and the cavity through which it passed, indicates a spinal cord an inch in diameter. These largest skeletons were found in Georgia and South Carolina. Those of the mastodon are numerous, and found in many different places. Physiographic geology is a study intensely interesting, and of great practical importance, as it bears directly on many of the industries of life; but this general notice is sufficient.

LITHOGRAPHIC GEOLOGY.

The ultimate particles of material bodies, of which we know but little, exert such force or influence on each other as to decide the character of the mass; even if the atoms are identically the same in substance they may come together in a way to secure different results. The bulk of the solid part of the earth is rock, but all rock is not the same. We find several species of

granite, of limestone, and sandstone, a long list. But the whole may be divided into two classes, stratified and unstratified. Whatever the two classes seem to have in common, they are not of the same origin. The first occur in layers or strata, others are crystalline and massive. The loose materials, such as sand, clay and gravel, that have accumulated at the bottom of the pond or lake, are found arranged in beds or parallel layers. The streams carry the materials from the highlands, and they are at length deposited in the basin, and when hardened become stratified rocks. As this process is still going on, and recently formed strata are found approaching the consistency of stone, it is but reasonable to conclude that all rocks of this class, being formed in like manner under the water, are of aqueous origin. They are further classed according to certain peculiarities, either of material or formation.

Gneiss, abundant in all parts of New England, is a kind of stratified granite, of about the same materials, but splits readily into slabs that are used both for building purposes and flagging stones.

Mica slate resembles gneiss, has the same minerals, but more mica, and is of a more slaty structure, and the glistening particles of mica abound in it.

There are several other kinds of slate, named from the minerals that predominate in them, or the purposes for which they are mostly used. Roofing slate of excellent quality is extensively quarried in Maine, Vermont and Massachusetts.

Quartz rock consists mainly of quartz, but often has more or less mica. *Sandstone* is of kindred formation, the principal part of which is quartz, reduced to sand, and the grains more or less firmly united. In both the colors are various.

Conglomerate consists of water-worn pebbles of various kinds and sizes cemented together, and sometimes making a strong, compact rock.

The limestone formations are extensive in nearly all countries. In their structure some are very compact and break with a smooth surface. Those capable of a fine polish are called marble, the more common uses of which are well known. The purest crystalline limestone is used in sculpture; the

best quality being obtained from Carrara, Italy, and that called Parian from the island Paros.

Chalk, a useful formation, is a carbonate of lime. In some caves the dropping of calcareous water forms stalactites, which hang from the roof like immense icicles, and are often extended till they meet the accumulations below, called stalagmites, and form beautiful columns. Of the more than seven hundred crystals from this source alone, and of the many other varieties of minerals having much in common, and yet enough that is peculiar to distinguish them, no mention can be made. A careful reader and close observer will gather from familiar objects a fund of information of great value.

The parallel strata mentioned are not always horizontal, but sometimes nearly, if not quite perpendicular. Occasionally a ledge broken quite through separates, and the rock on one side of the fissure is either elevated or depressed, making what is called a fault.

The fissures crossing a bed of rock are often filled with a mineral entirely different from the rock itself. In some cases where the vein is small the foreign substance may have come in from above or laterally, deposited from water as in the case of stalactites. The larger fissures were evidently filled with the melted material thrust up from beneath.

The unstratified rocks are in masses, without fossils of animals or plants, and of igneous origin. Some of this class were probably formed later, and by the melting of secondary rocks, but most of them by the gradual cooling of the central mass containing the melted minerals embodied in them.

DYNAMIC GEOLOGY

Treats of the forces that move things on or beneath the earth's surface. The Drift shows not a little confusion. Things are evidently in an abnormal condition, and strangely mixed. Some of the disturbing causes are obvious. Currents of the atmosphere and ocean have done much, but are not sufficient to account for all the phenomena. Boulders brought from ledges north of the great western lakes, are found scattered over all the western states, some much battered on the passage, others bearing only marks of

long exposure to the elements. Deep furrows have been plowed in the rocks and hill tops over which they passed, at an elevation of thousands of feet above the level of the sea. Currents of water could never have lifted such huge masses from the lower to higher levels, or transported them any such distances. Icebergs or glaciers have evidently moved over the whole Drift region with fragments of rocks and pebbles frozen into their lower surface, that, like huge rasps, both cut away and polished the hardest rocks, at the same time bearing forward the boulders and whatever else chanced to be held in their cold embrace. There are other footprints of many and very great changes that have been wrought. Though many persons have erroneous impressions of the inequalities on the earth's surface, the height of the loftiest mountains being but little when compared with the earth's diameter, yet there is evidence that the normal condition has not been preserved. Large districts have, even within the historic period, been lifted far above their former level, and others sunk as much below. New islands have appeared in the midst of the sea, while others have sunk out of sight. Multitudes now live on what was once the bed of the sea, "in which were things innumerable, great and small beasts;" and ships sail over territory once covered with the habitations of living men. Rocks of immense thickness have been broken and the parts lifted into a vertical position, and many such great changes have taken place. What wrought them? It is safe to say that at least two forces have been operating, the one more gradual than the other. The cooling of the internal mass must cause contraction, which, in a globe of such dimensions, would be sufficient to break the strongest rocks constituting its shell. This force, when properly directed, might lift the rocks, and even throw them back on other strata of more recent formation. Then the expansive force of the gases within, when raised to their highest tension, is enough to cause earthquakes, and pour through the partially opened craters, or where the barriers are made less secure, floods of lava that are in time changed into rocks of that peculiar class. The vent will be found where the crust above the struggling giant is weakest, whether that be on the mountain top where the rocks had been shoved up into a vertical position, or at the bottom of the sea.

The dynamics of geology suggest problems of no ordinary interest, but our narrow limits forbid even a statement of them.

MINERALOGY

Is that branch of geology that treats of mineral substances, and teaches how to distinguish and classify them according to their properties. This is a wide field for investigation, and so fruitful that the temptation to linger in it is strong. Mining and work with the products of the mines engage the industry of so many that it would be especially pleasant to study with them a subject of such general interest. We relinquish that privilege, in order to state two or three things that seem thoroughly established by what is found written in the book of nature, and are in perfect accord with God's later scriptures, the Bible, when rightly interpreted.

1. The first fact is the great age of the earth. Processes are plainly indicated that must have required not only thousands, but millions of years for our planet, before man, made in the image of God, entered it as the theater of his responsible activities. The facts of the carboniferous^[4] period alone discredit, and utterly overthrow the theory which limits the days of creation to six of twenty-four hours each. The Bible gives the order of the successive creations, but does not fix the age of the things created. The word translated day often means an *age* or an indefinite number of years, as is seen by referring to the places where it is found. Give it this well established meaning in the first chapter of Genesis, and all is plain. There was time for millions of races of inferior creatures to live and die before the divine plans and works were consummated, and the earth became a suitable abode for the human race.

2. The second great fact is that all things were made on a plan, and in some connection. There are no isolated objects or superfluous parts in the physical world. The number may be countless, and the forms given them reveal an endless variety, but each has its connections, and all the parts are necessary to a perfect whole.

3. Another lesson is learned from the mute witnesses, which is that, while a long succession of races of animals, for which the earth, in its different stages of progress was a fit abode, existed, each higher in rank than its predecessor, the several races had distinctive characteristics, as the *radiates*, *mollusks*, *articulates*, and *vertebrates*. A lower species, when its purpose is served, becomes extinct, and is succeeded by a higher.

CHEMISTRY,

By analyzing compound and compounding simple substances, discovers their elementary properties, the forces that are resident in matter, and the laws that govern them. It demonstrates by experiments the affinity of ultimate particles, and of gases of unlike kinds for each other, an affinity which produces homogeneous compounds, often very unlike the elements that unite in forming them. The chemist has much to do with physical objects, but in handling them his appropriate business is to consider the changes produced by chemical attraction in all bodies, whether solid, liquid, or gaseous.

GEOGRAPHY

Is an ancient science, suitable for schools of all grades, and not for primary and intermediate departments alone. The child can treasure many of the facts that, if held in the memory, will be of use to him as he advances in years and knowledge, but his geography will benefit him little unless it is studied when his faculties are more mature. One who despises this study as beneath him, knows nothing yet of the important science as he ought.

PHYSICAL GEOGRAPHY

Has many things in common with both astronomy and geology, as it discusses the physical condition of the earth and its relations as a member of the solar system; describes its great natural divisions of land and water; and takes account of dynamic forces, such as aerial and oceanic currents, that are constantly causing important changes. The whole exterior structure of the earth, the phenomena of rain and dew, fog, frost, and snow, are geographical questions, to be discussed with special reference to the general laws or principles involved. It shows unity in the midst of diversity, and constancy of phenomena in the midst of apparent changes.

MATHEMATICAL GEOGRAPHY

Treats of the form and size of the earth, of the construction of globes to represent it; determines the latitude and longitude of places on its surface,

and all geographical problems pertaining to numbers, distances, and magnitudes.

POLITICAL GEOGRAPHY

Describes, in a general way, the countries and nations of men as they are politically divided, defines their boundaries, and to some extent characterizes their social and civil institutions. A great advance has been made in this branch during the present century. People respecting whom little was known, have come into the family of nations. The maps have been changed, and generally in a way that indicates the rapid progress of civilization. Asia has been so thoroughly explored that our general knowledge of the country may be regarded as nearly complete. No great *terra incognita* remains in that quarter, though fuller and more precise knowledge respecting the people in some parts is yet much to be desired. The interior of Africa is still but partially known, though the work of discovery has been pushed forward with considerable enterprise, and a host of explorers have struggled to penetrate the mystery that enveloped, for ages, that great division of the globe. The Upper Nile country has been explored far beyond the region assigned on the maps to the “Mountains of the Moon,” and all know the intense anxiety that is to-day felt for the safety of General Gordon and his little garrison, still shut up in Khartoum.

The study of geography, rightly pursued, is remunerative, full of inspiration, and as intensely interesting as any in the whole circle of physical sciences.

BIOLOGY

Is scientific discourse about life and vital forces. We give it a high position in the circle, since vitality is superior to either chemical or mechanical laws, suspending or modifying them for the production of organized structures of plants and animals. Even *vegetable biology* confronts us with that mystery of mysteries, life, which is quite inexplicable. We can only say it is a peculiar, indefinable something, necessary to the existence of such organisms, and without which they soon sink in ruinous decay.

The living germ is the determining power that shapes the organic body, and every germ will have its own body. Under no possible culture can the acorn develop into an animal. It will produce an oak, a tree of its own species, and nothing else can grow from it. So also of the animal germ. The form or kind is as determinate while the embryo is yet in the egg, as it will ever be. The life once begun in everything that lives and grows, there is a power that takes hold of the elements nature has in store for it, and, by a most wonderful transformation, works them up into its own body; and this power of assimilation must forever distinguish it from all lifeless inorganic matter.

The mystery deepens when we notice that living things exist in generations. The plant has seed in itself for the production of another plant. It has life in itself, and power to vitalize its successors. The products of the field and the forest grow and mature, then wither and decay; but they have successors of the same kind.

So human beings exist in successive generations. One generation passeth away, and another cometh, and so the race lives on. While alike in their power of assimilation and reproduction, there is a wide difference between the vegetable and the animal. They have not the same organs, and do not subsist on the same food. The plant is constantly consuming carbonic acid, and giving out oxygen, while animals consume the oxygen, and restore to the atmosphere carbonic acid. The difference of their physical structure, and their different relations to inorganic matter, suggest a wide difference in the “bios” or life, that animates them. Just what that difference is, no one can tell. It is a question for which science furnishes no answer. In his physical organization man differs but little from the lower animals. In this he is brother to the beasts that perish, having the same nature, needs, and liabilities. If he is “fearfully and wonderfully made,” so are they; in agility and strength many of them far surpass him. His peculiarities of form and structure do not secure, and, it may be safely said, were not intended to secure physical superiority, but rather to fit the organization for the indwelling of the rational soul, that is his distinguishing characteristic.

PHYSICAL ASTRONOMY

Has been made the subject of much diligent research and study. Some facts respecting the physical elements and structure of the sun and planets have been ascertained with reasonable certainty, but much is still in doubt. Assuming that the essential properties of matter are the same everywhere, we may tell with assurance of what the sun and stars are made, provided all solar and stellar phenomena are explained by physical laws that are understood, and in operation around us. This has been done in part, but not so as to harmonize the views of all astronomers. Since the use of the spectroscope[5] results have been more satisfactory, and on some questions of much interest, conjecture and theory have given place to certainty. By the decomposition of sunbeams or pencils of solar light, the refracted rays show the presence of several distinct chemical elements. Finding by a qualitative analysis that there is iron, copper, zinc, nickel, sodium, and other terrestrial substances in the solar and stellar spectra, we know that they enter into the composition of those celestial bodies. But in what proportions or combinations they exist is not known.

METEOROLOGY AND AEROLITES.

Who has not seen a shooting star? For a moment the bright objects dart through greater or less spaces in the heavens, and then disappear. Those of inferior size give but little light, and are seldom seen unless the eye is, at the time, directed toward the space they traverse. Occasionally one flames out with such brilliancy as to light up, for a moment, the whole heavens. These are called meteors—a name quite proper for both classes, and only the very ignorant suppose any of them to be real stars. They come singly, two or three in an hour, or in showers, such as were witnessed in 1833. When of such size that they strike the earth before being consumed by their intense heat, they are aerolites, or meteoric stones. Great masses of these are found in different places, and show such a peculiar combination of their chemical elements as to distinguish them from all other stones; and mineralogists generally conclude they were not formed on the earth. Whence they come is not certainly known. That they were formed by an aggregation of their materials in our atmosphere seems incredible. Nor were they thrown off by some great convulsion, from the moon, with force sufficient to carry them beyond the attraction of that body. Perhaps most astronomers now believe, on what they think sufficient evidence, that the celestial spaces are occupied

by innumerable small bodies moving round the sun, of whose nature and orbits nothing is certainly known. The earth, it is supposed, while making its annual circuit, must be constantly encountering them, and, as in passing rapidly through the upper region of the atmosphere they take fire and burn, the shooting star or meteor is simply the light of that flame. The mechanical production of heat, now well understood, shows why they burn. The rapid motion of the earth, especially if it be duplicated by that of the minute body striking through its atmosphere, would generate heat sufficient to quite consume the meteoroids; so that generally their solid substance is dissipated before they reach the ground. Sometimes the heated aerolite explodes when in such proximity to the earth that the fragments fall before they are consumed.

THE AURORA.

That most interesting atmospheric phenomena, the Aurora Borealis, though so familiar, has never been fully explained. It is rarely seen in equatorial latitudes, but increases in frequency and brightness as we go north, even to the arctic circle.

In this latitude all observers may at times notice two distinct forms of the aurora. The one, as we often see it, has a cloud-like appearance, with a soft radiance permeating it, and seems a vast, irregular patch of mellow light, ever changing, and at times showing a slightly reddish or purple tinge. It is more frequently seen near the northern horizon, having the form of a beautiful arch, the ends of the segment apparently resting on the horizon, and the middle, or crown, a few degrees above it. The other takes the form of streamers, reaching far up toward the zenith. Gently curved, like the celestial sphere on which they are projected, they are not stationary, but almost constantly in motion, but soon resuming their former position, spreading themselves out like immense flags, with their numerous silken folds, ever dancing, quivering, undulating, as if stirred by some gentle breeze, though all else seems in calm repose. To say that the phenomena are electrical, would, probably, not be the whole truth, though evidence is not wanting that the aurora is in some way connected with the electricity and magnetism of the earth and its atmosphere. Practical telegraphists testify that during a brilliant display of “northern lights” such strong, irregular

currents of electricity pass along the wires that it is difficult to send a dispatch; at other times the currents are so strong that they can communicate without the battery.

There is, perhaps, about as much against the theory of a purely electrical origin, as in its favor, and, on the whole, we conclude that the Aurora Borealis is one of the things respecting which modern observations have suggested more difficulties than modern science is yet able to explain.

HOME STUDIES IN CHEMISTRY AND PHYSICS.

BY PROF. J. T. EDWARDS, D. D.

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FIRE.—PHYSICAL PROPERTIES.

Clearness, accuracy, and brevity are the essentials of good definition. That it is no easy task to combine these, every teacher realizes.

Perhaps it is near the truth to say that fire is that operation in nature which at the same time evolves heat and light. The *operation* is, at the present time, supposed to be a certain vibration of ethereal or more solid substances. All matter is in motion. Whence this motion was first derived no philosopher can tell, unless he goes back to that primal source of both matter and motion, which in the beginning created the heavens and the earth, and said, "Let there be light, and there was light."

Prof. James Dwight Dana^[1] declares that the first act of creative power must have been heralded throughout the universe by a flash of light. Thus the geologist unites with the scriptural narrator, in the statement that light and heat belonged to the first day of creation, although scoffers for a long time ridiculed the idea that light could exist without the sun.

All space is supposed to be filled with a substance called ether, and that it permeates even solid material. When, for any reason, the natural motion of the molecules of matter is much increased, these molecules have the power of imparting their vibration to the ether in contact with them, and that in turn may produce vibrations in other substances, and if these vibrations come in contact with the nerves of touch, there follows the sensation of warmth or heat. If the vibrations of the ether are still more

rapid, when they fall upon the retina, we have the sensation of sight, and we call the agent light. Heat and light, then, are the same. In one instance the vibration is capable of affecting one set of nerves, and in the other, two sets of nerves. The heat-vibration can be discovered by the sense of touch alone, but the light-vibration may be detected both by the eye and the touch.

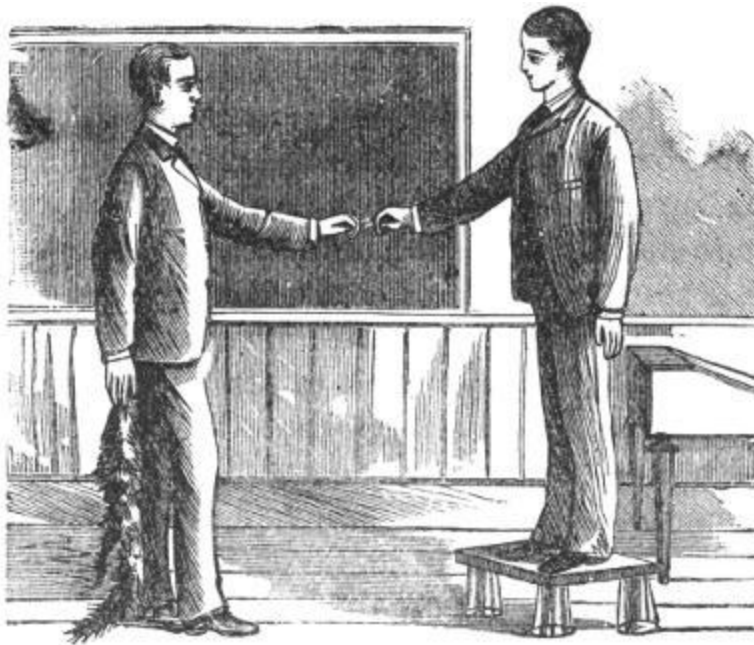
This variation in sensations, when produced by the same cause, may be illustrated as follows: Apply some salt to the tongue, and place some also in a wound, the two sensations are entirely unlike. Again, the vibrations of a body may be so slow that we can discover them by touch, as showing resistance, or so rapid that they are reported to the ear as a shrill sound, or they may be increased so intensely as to evolve heat, and if still more increased in rapidity, affect the eye as light. The spectrum affords us still another illustration of this truth. Pass through a prism a single ray of light, lo, it appears on the screen in all the colors of the rainbow. Nor is this all; *between* the bright colors, and *beyond* the violet and the red are invisible lines, and the various parts of the spectrum, although all are produced by the one ray, are capable of creating quite different results. If one should place a delicate thermopile below the red color, it at once reports heat, although the eye sees nothing there. The beautiful colors of the spectrum flash their light into the eye, raise the temperature of the thermometer and affect chemical transformations, while, still more wonderful, the dark lines above the violet, though unseen and not indicated by the thermopile, act upon the sensitized plate of the photographer with decided chemical force. Thus changes in vibrations as to rapidity, length and direction make changes in the resulting sensations.

Light-waves are always heat-waves, and heat-waves may, by increasing the rapidity of the vibrations, become light-waves. It will be observed that three of our senses are close akin. Hearing, feeling (as regards warmth) and seeing are all produced by vibrations. It is quite in accord with the doctrine of modern science to believe that the morning stars did “sing together,” for light is essentially rhythmic, and to senses adapted to the perception of their harmonies, the sunbeams would make music. The various colors of the spectrum differ solely in the wave-lengths of their vibrations. The red corresponds to low pitch in music and the violet to high pitch. As the vibrations of air striking upon the ear increase in rapidity, the sound rises in the scale. There is this difference between the ear and the eye—the former,

if trained, can detect all the tones in a chord of music, while the latter, however cultivated, can not discern the varied colors blended in white light.

There must be sixteen vibrations in a second to produce a continuous sound. When these vibrations reach thirty-eight thousand in a second they become inaudible.

Eisenlohr^[2] informs us that the red color in the spectrum has four hundred and fifty-eight trillion vibrations in a second, and extreme violet seven hundred and twenty-seven trillions. The former yields 37,640 waves in an inch, and the latter 59,750 waves in the same space. Now mark another beautiful analogy between sound and sight. In looking at the spectrum we can not discern the light or heat below the red color, because the waves are so slow. Ascending the gamut of color, the rapidity of the vibrations increases, until just *beyond* the violet it becomes so great that the eye can detect no color.



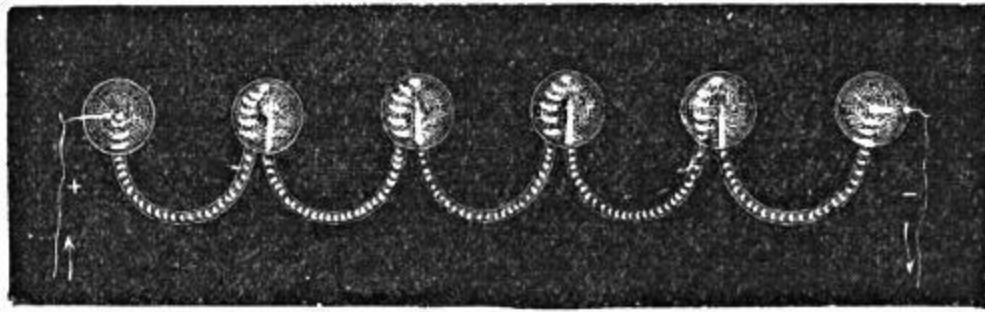
MECHANICAL ENERGY TRANSFORMED INTO ELECTRICITY.

Ex.—The boy on the insulated stool is repeatedly struck with some furry substance, like a tiger skin. He becomes highly electrical and capable of emitting sparks.

The same fact is discovered in the world of sound—beginning with vibrations which are too slow to be heard at all, we ascend the scale eleven octaves, when the vibrations become so rapid as to be inaudible. Complete darkness may be caused by either too slow or too rapid vibrations of light and heat, and utter silence by the same conditions in the sound waves.

SOURCES OF LIGHT AND HEAT.

These are five in number: The sun and stars, chemical action, percussion, friction and electricity. Stars are suns, but at a vast distance from our earth, the nearest being twenty trillions of miles away. To other systems they doubtless perform the offices of suns. Being so remote, however, although of myriad number, their influence upon our earth is hardly appreciable, and will not, therefore, be here considered.



GEISSLER'S TUBES.[\[3\]](#)

Ex.—This tube is filled with rarefied gases. Platinum wires convey the electric current through the tube, revealing curious striated sections of brilliant light, varying in shape and color, with the variety of gas and the degree of rarefaction.

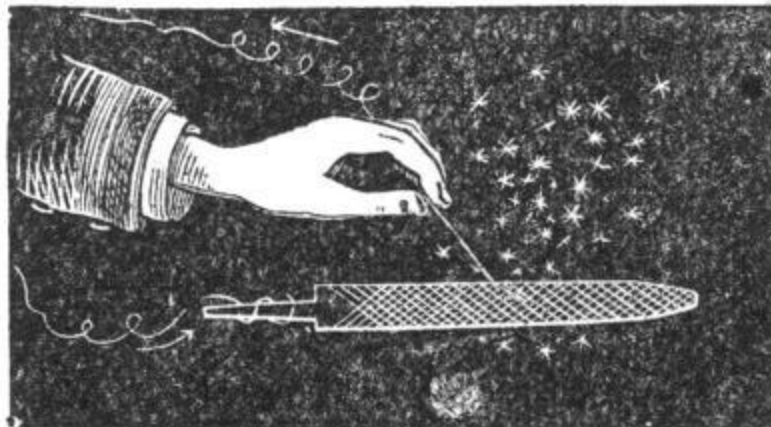
Our sun is an immense reservoir of energy. It is difficult to conceive its size. It would require twelve hundred thousand of our globes to equal it in volume. More than one hundred such worlds as ours might be strung upon the line forming its diameter. The sun has been for ages throwing off its vibrations of heat and light. Thousands of years before fires were kindled on hearthstones this form of energy, according to the modern doctrine of the correlation of forces, was locked up in the tropical vegetation of the coal periods, and in the great deposits of coal preserved for future use. The same anticipatory benevolence which projects on its journey the friendly ray of

the north star, forty-three years before the mariner's eye can see it, provided fuel for man thousands of years before it was needed.

This energy of the sunbeam reappears in the summer warmth of our dwellings in winter, in the expansion of steam, in the blow of the trip hammer, and throbs even in the pulsations of the human heart.

The cells of all plants need the force of the sun's rays to separate the carbon from the oxygen contained in the carbonic di-oxide absorbed by the rootlets and stomata of the leaves. Thus the great luminary builds the forests and clothes the earth with verdure. "All flesh is grass," and therefore to the forces of the sun's vibrations we must trace not a little of animal growth and strength. The sun gives out more heat than it would if six tons of coal were burnt on every square yard of its surface every hour. Sir John Herschel^[4] declares that its light is equal to that of one hundred and forty-six calcium lights, each one formed of a ball of lime equal to the sun in bulk; yet even a small calcium light is so dazzling that the eye can not look steadily at it.

The careless expression sometimes heard when the moon shines brightly, "It is as light as day," is a striking hyperbole, for it would require eight hundred full moons to equal the brightness of daylight.



ELECTRIC MOTION CONVERTED INTO SPARKS.

Ex.—A file is made part of the circuit, and as the wire conducting the electricity is rubbed along the file, the circuit is alternately formed and broken, and sparks follow each breaking of the circuit.

Of all forms of paganism, that of the Fire Worshipers^[5] seems least unreasonable, for the sun is even now, to us, the best symbol of beneficence

and unfailing energy. After thousands of years it shows no diminution of power, and although the imagination can conceive the possibility of its destruction, the most accurate scientific observations have not discovered the slightest indications of its lessening influence. "His going forth is from the end of the heaven, and his circuit unto the ends of it; there is nothing hid from the heat thereof."

CHEMICAL ACTION.

In a preceding article the chemistry of fire has been considered at some length. It only remains to mention briefly a few of the physical phenomena attending it. When elements unite by the force of affinity, it is supposed that their atoms rush together, and that their motion is converted into heat.

In the case of the galvanic battery the impetuous movement of the atoms toward the poles becomes electricity. We have constantly recurring instances in nature of that great truth that energy, though constantly disappearing is never lost, but reappears under new manifestations and a new name. It may for a time remain dormant, and anon become perceptible, as in the case of latent heat. For example, in mixing five pounds of water at a temperature of 212° Fahrenheit, and five pounds of ice, seven hundred and fifteen units of heat disappear in melting the ice, and the aggregate temperature of the mass is proportionally lower than that of the substances united. But upon their returning to their former state, this latent heat reappears as sensible heat.

In chemical action producing fire, the uniting materials are usually converted, first, into a gaseous form, but there are some exceptions. The most interesting is the following: When a few flakes of iodine are placed upon a fragment of phosphorus, the atoms of the two elements rush together with great energy, producing spontaneous combustion, and liberating sufficient heat to burn the superfluous iodine, with the evolution of beautiful violet fumes.

The mechanical action in flame is full of interest. Its brightness always seems to depend upon the incandescence of solid particles. This can easily be seen in an ordinary lamp. A piece of cold porcelain inserted in a flame will cool the incandescent carbon, and it will be deposited as soot.

The Bunsen[6] burner clearly proves that the brilliancy of our lights depends upon the incandescence of the carbon. This is a contrivance for passing jets of air through a flame, so that the intimate mixing of the oxygen of the air with the carbon will cause the immediate combustion of the latter. This results in converting it instantly to invisible gas (CO_2) before incandescence, and consequently the Bunsen flame, while it is intensely hot, emits but a feeble light.

Any *physical* change that facilitates the movement of atoms seems to increase the intensity of chemical action.



SHOWING THE PRODUCTION OF ELECTRIC LIGHT FROM CARBON POINTS.

Ex.—The rods are first placed near together, then as the circuit is formed they are drawn apart, and the electric light is formed between them.

An instructive experiment illustrating the characteristics of different kinds of flame may be performed as follows: Place near each other a small alcohol lamp and a piece of paraffine candle; when lighted observe the two flames. The three cones in each can be easily discerned, the *candle* burns with a much brighter light, showing it to be richer in incandescent carbon. Insert in each flame a piece of fine wire or narrow strip of glass, either of these will be much more quickly heated by the alcohol lamp, because its flame is richer in hydrogen. If a glass jar which is cold be placed over each, a film of vapor (H_2O) will gather on that covering the alcohol lamp with greater rapidity than on the other. If the jars remain over the flames until they are extinguished by the lack of oxygen, more carbonic anhydride (CO_2) will be formed from the combustion of the alcohol.

PERCUSSION.

When a blow is arrested by an object, the *motion* is converted into heat. The ancient flint-lock gun and the percussion-cap fire-arm both illustrate this fact. In the former, the descending flint struck out the spark, and in the latter the cap is exploded by the arrested hammer. The stroke of a cannon ball is attended with a flash. If the world were suddenly stopped in its course, heat enough would be generated to set it on fire. Nitro-glycerine and dynamite are exploded by percussion. Familiar illustrations of this scientific truth meet us in everyday life. It has even passed into a proverb with a moral application, that “hard cracks make the sparks fly.” A novel effect of percussion may have been noticed when a fall upon the ice has resulted in a mechanical disturbance of the optic nerve which revealed whole constellations of stars never yet catalogued.

FRICTION.

It is a spirited sight to watch the operation of sharpening tools upon a grindstone or emery wheel run by steam. Showers of sparks are produced by the friction. We often observe the same phenomenon when the brakes are applied to rapidly revolving car wheels. Rails are heated by the friction of the passing train. You may have had the misfortune, while riding, to have one of your carriage wheels become set, caused by the box of the hub, and the axle becoming so heated by friction as to “unite” their surfaces. All machinery requires constant watching and lubrication to prevent undue friction and serious wearing.

Mills have not unfrequently been set on fire by rapidly revolving belts coming in contact with the woodwork. When the whale, frantic with the pain of the harpoon, darts away with lightning speed, the sailors are compelled to dash water over the spinning wheel on which the rope is wound.

In all these instances motion is transformed into heat.

ELECTRICITY.

Galvanic, frictional, magnetic, thermal and animal electricity are all capable of producing heat. The first also produces an intensely brilliant light. We have long been acquainted with the “Voltaic arc”^[7] of the galvanic

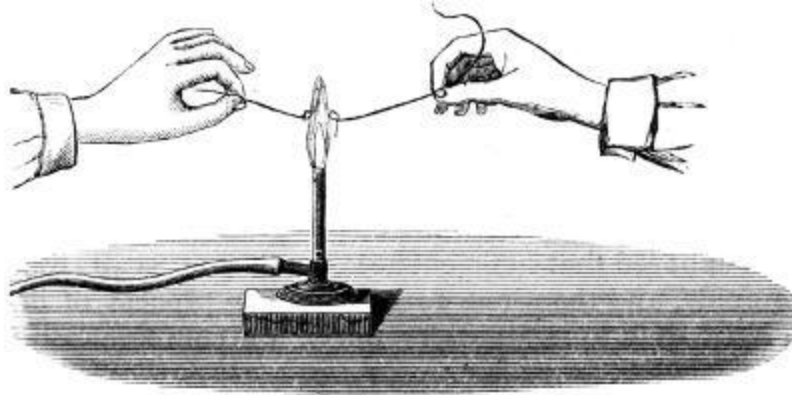
battery, but less familiar are the magnificent manifestations of frictional electricity. Dynamo-electric machines are of comparatively recent construction, and their object is to convert mechanical energy into that of electric currents, and vice versa.

A striking application of galvanic electricity is frequently seen in the discharge of gunpowder and other explosives, by making the electric current pass through a small platinum wire which is in close contact with them.

Electric energy is propagated in waves, and this wire, being so small, is incapable of transmitting them all at once, so they beat upon it until their repeated blows cause it to become red hot, and the material in contact is thus ignited.

Perhaps the grandest illustration of this action was seen in blowing up the rocks of Hell Gate^[8] in the East River, and thus opening a safe passage for the commerce of the world. The tiny finger of a little child, the daughter of the engineer, at a given signal, pressed the key that closed the circuit, and, like Æolus,^[9] when he struck the rock, set free the mighty elements of destruction.

This same principle, viz.: that resisted motion becomes heat and light, is seen in both the Brush and the Edison electric lights. In the former, electric currents pass along wires to carbon points, shaped like a crayon, and covered by a film of copper, and separated by a distance of about one half inch. The air between is a non-conductor, and here the flame is formed. In the Edison light, however, the two conducting wires enter a glass globe, from which the air is excluded. Here they are connected with a spiral wire about as large as a knitting needle, and three-quarters of an inch in length. When the electricity is turned on, this spiral glows with an intensely brilliant white light.



SOLIDS DIFFER AS TO CONDUCTING POWER.

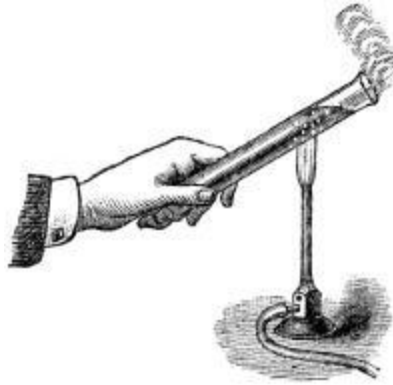
Ex.—If we hold a pipe stem or rod of glass in one hand and a copper wire in the other, and apply the ends of these to a flame, the wire will convey the sensation much more quickly to the hand than the other. This shows that solids differ as to conducting power.

A marvelous illustration of the relation between electric and sound vibrations is found in the telephone and microphone. The former is becoming a household necessity; the latter, though not so well known, is not less wonderful. It brings to our ear the tick of a watch miles away, and through it the walking of a fly sounds like the tramp of a horse.

DISTRIBUTION OF HEAT.

Heat is distributed by radiation, conduction, and convection. By the first we mean that heated bodies have the power of projecting from themselves, by means of the ether, their own vibrations. Thus the sun is constantly distributing its light and heat in all directions. Conduction takes place where the molecules of a substance nearest a fire first become heated and then impart their motion to the remainder of the mass, somewhat as in a row of suspended ivory balls, the first of which, when struck, transmits its motion from ball to ball, the last one flying off.

Convection takes place in liquids and gases. Here the particles in contact with the heated body becoming lighter by expansion, rise, and are followed by others, thus forming a current.



WATER A POOR CONDUCTOR.

Ex.—Fill a tube nearly full of water, applying a flame to the upper part of the tube. The water at this point will readily boil, while that in the lower part of the tube remains cool, showing that water is a poor conductor, and that liquids must be heated by convection.

The process of warming a room illustrates the three methods of heat distribution. The heat passes through the stove by conduction, away from it by radiation, and to the remote parts of the room by convection.

EFFECTS OF HEAT.

They are four in number. Rise of temperature, expansion, liquefaction, evaporation. The first indication of the presence of heat is discovered by an elevation in temperature. Though man is not a reliable thermometer, he would be able, ordinarily, even if blind, to chronicle the progress of the sun, from horizon to horizon, by the increasing and decreasing warmth. The little thermometer placed beneath the tongue of the invalid gives reliable report of the combustion going on within his system. We see a thousand illustrations of the expansive effects of heat, many of which are familiar to all. The exceptions are more interesting than the rule, and less known, the ordinary rule being that heat expands and cold (absence of heat) contracts. Water *contracts* by cold until it reaches the temperature of 39° , and then *expands* with great violence until congelation is completed, at 32° . A British officer in Quebec filled a twelve inch shell with water, and closed the fuse hole with a wooden plug securely driven in with a mallet. Upon being exposed to intense cold the plug was projected a distance of several

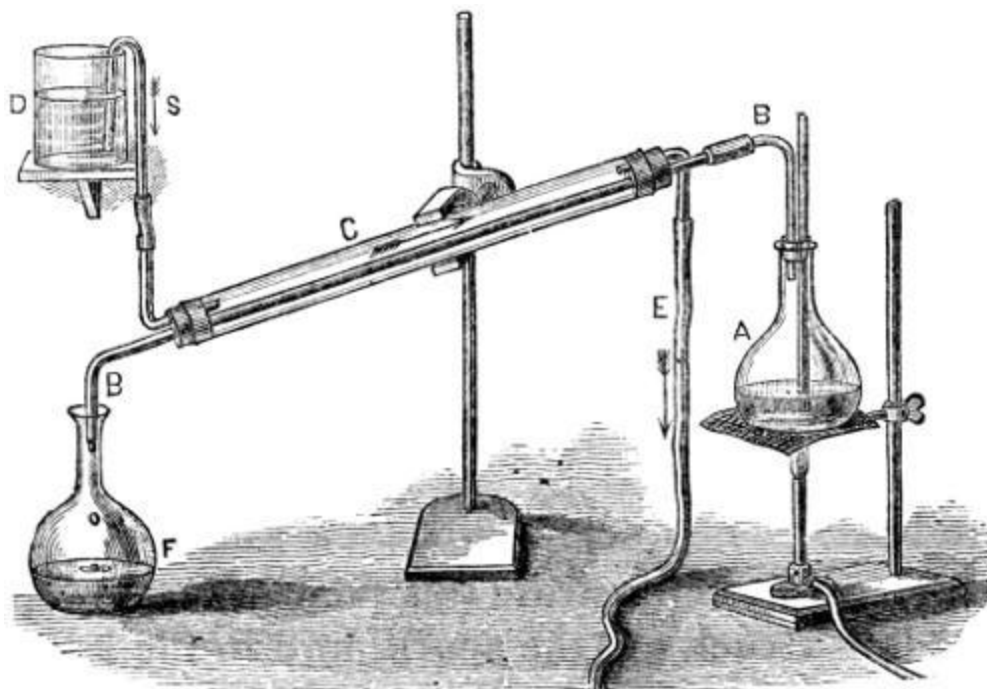
hundred feet, and a long tongue of ice was found protruding from the opening.

It is supposed that sufficient heat would convert all solids first into liquids, and then into gases. In the process of distillation, if we wish to retain its products, we combine both heating and cooling.

The knowledge of the melting and vaporizing point of substances is of immense value. We are enabled thus to drive off and secure the various ingredients entering into many complex substances. A notable instance is seen in the means used to secure the rich and varied products of petroleum.

THERMOMETERS.

These are not the only measurers of heat. We have the pyrometers, used for ascertaining the temperature of extremely hot bodies, and the thermo-electric pile, an apparatus which constitutes the most delicate test for heat which has been devised. It will detect heat in the body of a fly walking near it.



SHOWING DISTILLATION.

Ex.—Place a small amount of water, colored with ink, in a flask, and apply heat. The water will be vaporized, and in passing through the tube, which is surrounded by another tube containing cold water, it is condensed as a colorless liquid.

Thermometers are of three kinds, as to the materials used. They are air, alcohol, and mercurial. In each case the contraction and expansion of these respective substances are made to register variations of heat and cold. They are of three kinds, as to their system of grading—Réaumur's, the Centigrade, and Fahrenheit's. The first two make zero the freezing point; the last makes 32°. The boiling point of Réaumur's is 80°, the Centigrade 100°, and Fahrenheit's 212°. Once more changing the basis of classification, we find thermometers divided into three classes, with reference to the purposes they serve. The ordinary thermometer records the degree of heat or cold at the moment of observation. The differential thermometers can be made of two ordinary thermometers, by wrapping a piece of cloth around the bulb of one; these would show at any given moment whether it was growing warmer or colder. If it is growing warm, the column of mercury in the thermometer with the covered bulb will stand lower than the other, as the cloth prevents the heat reaching the quicksilver as readily as in the other. If it is higher than in the other, the weather is growing colder, as the cover prevents the heat from going off as rapidly as from the other. The third class, the registering thermometer, is so called because it marks the extremes of temperature. Without going into detail, it is perhaps sufficient to say that a minute bar of steel is placed on top of the column of mercury, and remains at any point to which it is pushed, thus recording the greatest degree of heat during any given interval of time. Somewhat similar in arrangement is the alcohol thermometer, marking the greatest degree of cold. It will, of course, be understood that almost all apparatus is greatly varied to serve special purposes. The limits of our article will preclude further discussion of fire in relation to light, although the subject of both physical and physiological topics is full of fascination and value.

End of Required Reading for March.

The most important question for the good student and reader is not, amidst this multitude of books which no man can number, how much he shall read. The really important questions are, first, what is the quality of what he does read; and, second, what is his manner of reading it. There is an

analogy which is more than accidental between physical and mental assimilation and digestion; and, homely as the illustration may seem, it is the most forcible I can use. Let two sit down to a table spread with food; one possessed of a healthy appetite, and knowing something of the nutritious qualities of the various dishes before him; the other cursed with a pampered and capricious appetite, and knowing nothing of the results of chemical and physiological investigation. One shall make a better meal, and go away stronger and better fed, on a dish of oatmeal, than the other on a dinner that has half emptied his pockets. Shall we study physiological chemistry and know all about what is food for the body, and neglect mental chemistry, and be utterly careless as to what nutriment is contained in the food we give our minds? Who can over-estimate the value of good books, those ships of thought, as Bacon so finely calls them, voyaging through the sea of time, and carrying their precious freight so safely!—*Prof. W. P. Atkinson.*

THE MOHAMMEDAN UNIVERSITY OF CAIRO.

BY BISHOP JOHN F. HURST, D. D., LL.D.

Years ago I had taken pains to gain all accessible information concerning the most celebrated, and certainly also the largest, university in the entire Mohammedan world. In 1871 when in Cairo a number of days, through lack of a proper guide and full knowledge of this important institution, I left the city without seeing it. I was determined this time, therefore, to make sure of a visit to it, and to see carefully, with my own eyes, this marvel of the Mohammedan faith. The University is located in a mosque, and is, in fact, the one chief business of the mosque itself. Religion—such as it is—is the fundamental feature of all Moslem education. Not a science is taught in any school of Mohammedanism which does not begin with the Koran, and again come back to it. Whether law or medicine or geometry—in fact, whatever is communicated to the young, the first and ever predominant lesson imparted with it and through it is, that the Koran is the fountain of all science. Very naturally, then, the school is a part of the service of the mosque. This idea is not new. It is an oriental habit. We find proofs even in the Scriptures that the church was God's first school. In ancient Egypt the temple, the palace, and the school were the perfected trinity in every city, and often the temple and the school were so closely enclosed that no careful observer could tell where one began and the other ended. The same idea reappears in the arrangements which Charlemagne made for the higher education of the Frankish empire. The school was often located under the palace and in close connection with the chapel roof, and was called *scholia palatina*, or the school of the palace. At first the object seems to have been that the emperor's children and other children of the court might have the best opportunity for learning; but very soon the limits became broader, and all who wanted to learn could have every advantage, within close distance of both church and palace.

The approach to the University of Cairo is a narrow street, with open booths on either side, where the artisans ply their crafts in full view of every passer-by. Three industries take the lead of all others—book-selling, book-binding, and hair-shaving. The nearest street to the University bears the name of the Street of the B, and such it may well be called. The Mohammedan has always a shaven head. He wears a great turban, of white or some other color. Green is the most infrequent shade, for that indicates that the wearer is a descendant of the prophet Mohammed. Not one hair is allowed under that turban. When it gets a little long the barber must shave the pate as clean as an ostrich egg. All along a part of the street leading to the University the barbers sit on the floors their shops, and shave the heads of their customers. The one to be shaved does not sit in a chair, but simply stretches out full length on the floor and puts his head in the lap of the barber, who also sits on the floor, with his feet doubled up under him. Then begins the process of shaving. It is a most lowly operation. No paper is used during the process, the barber getting rid of the shaved hair and soap by wiping the razor on his customer's face until the entire tonsorial feat is finished and an ablution of cranium and face is in order. In addition to the barber shops there are probably not less than twenty-five book shops, as many binderies, and a good number of stationery stalls. These are all of modest dimensions, but are well stocked with everything that a student needs that is to say, a student of the Mohammedan order.

Between the point where the street ends and the University enclosure proper, there is a large fore-court. Here one sees such a medley of all forms of life and strange habits, in connection with study, that he can never forget it. It is the place where no serious study goes on, but where the news is discussed and conversation enjoyed. Even the barbers have spilled over into this court, for I saw a number of them busily shaving the heads of outstretched students. One of them, seeing a Frank scanning his work, stopped a moment, and holding up his razor from the pate which he had nearly made bald again, asked me if I did not want to be shaved too. I thanked him—but had not time. Imagine a half-dozen students lying about in Mead Hall, in Drew Seminary, near the doors of Drs. Butts, or Strong, or Miley, or Crooks, or Upham, and having their heads shaved by busy barbers, who sit flat on the marble floor and relieve the crania of their theological patrons of their last capillary endowment! Then think of

students munching at a crust of dark bread or a pomegranate, or some edible, good or poor, according to his resources. Some students have families, and here the children come and play about them, at times when their fathers are not busy with their books. So far as I could see, there was no formal studying in this great fore-court. Perhaps there were a hundred persons in it, lying, sitting, walking. Some alone with their meditations, others entertaining a group of eager listeners, and gesticulating with oriental realism. Only one class had the appearance of any work, a group of boys. One of the number displeased his teacher, whereupon the latter beat him smartly with his fist until the little fellow's eyes swam in tears; my blood fairly boiled at the teacher's cruelty. I thought I was already in the University proper, but this was a serious error. The institution was yet to come; I was only approaching the great establishment.

I had no sooner touched the threshold of the great central hall than a man met me, and, with a most polite salaam, informed me that I must now put on slippers. He was a magnificent specimen of a well developed Egyptian—tall, muscular, grave, yet pleasant, and only answering such questions as were put to him. Unlike the European guides in blue and brass, those of Africa have no stereotype speeches which they hurl at you, as they have done at the thousands before you. In a moment four pairs of soft slippers, of yellow sheepskin, were brought to my companions and myself, and the wary hands which brought them slipped them on over our boots and tied them on with red strings. We were now to enter upon the holy stone floor of the great hall of Mohammedan learning, and only holy dust must fall upon that tessellated floor, and then only with softest touch. Here was a scene which baffles all description. The hall was about two hundred and fifty feet long and two hundred wide. All the classes were reciting, engaged in work, or listening to the professor. Every one who recited did it loudly. I stood beside one of the theological professors and watched his method. His class numbered forty students, whose various physiognomies showed that they had come from every part of the broad Mohammedan world. The professor sat squat on the floor, with his bare feet doubled up about him. There is no craze as yet among Mohammedans for only young teachers. This man, like many others, had long since passed beyond middle life. His heavy gray beard and very dark face were lighted up by as keen a pair of black eyes as ever became diamonds, when they saw in his young days the prophet's

torch in Mecca, or in vision beheld the curtain drawn aside which hides the Moslem paradise from human sight. The forty students sat about him in a circle, yet in such way that all were before him at once. He was one of the circle, in fact, and as he taught he swayed to and fro, and looked off into the distance as if in reverie, and then again at his class, and, with an intensity that only an Arab possesses, he burned his ideas into the very brain of the students. He sat at the foot of a stone pillar, and leaned against it at intervals, when his weary form needed a little rest.

This theological professor had the method of all. He held a thin book in his hand which seemed to be his own brief, and, after reading snatches from it, he gave a comment or explanation of it, and then had one student and then another repeat what he had said. Our American infant class method of teaching verses, and having them committed to memory while the class are together, and then repeating them, so that the teacher can see that the work is well and surely done, is precisely the method of both elementary and advanced education in this greatest university of the Mohammedan world. The brief of this theological professor was merely his collection of definitions, and these were committed to memory on the spot. Some of the students had sheets of tin, something smaller than the sheets of roofing tin with which we are familiar in the United States. On these they wrote in ink, with reed styles, and with such dexterity that a whole page was filled in a very short time. What was written on these tin slates was taken away, and designed to be committed to memory, when that process was not finished during the session of the class.

Now the entire floor of this immense hall was covered with classes at work. No teacher or student sat in a chair. There was not even a footstool in the entire University. The professors and students formed little or large groups all over the immense space, no class interfering with another, and each going on with its work as if alone, and yet not a partition or a curtain dividing the groups at study. I saw only a little eating here, an occasional student silyly making a lunch of new dates, the fruit with “gold dust” on it, now just in from the country.

I could not help noticing the various ages of the students. Some were really very advanced in years. They were waked up very late in life. Something had broken loose under their twenty-five yards of cotton cloth

which they call a turban, and they had come down the Nile with the rise, or had been wafted from the Darfur sands, and were going to study. They could do more, and be more, when they went back again. Here, too, was the old-time idea. The notion that a university is a thing for the young alone is a modern affair. The old conception was, it was everybody's place—the *universum* of men as well as studies. In Mohammedanism, as in Christianity, when once the passion for learning strikes one, the years count nothing. The person in the fifties or even in sixties is just as apt to be overwhelmed, swept on, by the learning frenzy as though he were only eighteen and smitten by other inspirations.

The entire number in attendance at this greatest University of the Mohammedans is about thirteen thousand. Some calculations place it at fifteen thousand. They come from every part of the world where the cimeter of Mohammed and his successors has drawn blood, and where the crescent now floats. Each part of the large hall has its nation, where the students are grouped territorially. Here, in one place, are the Benguelese, from southwestern Africa; in another place are the Algerines, from the sound of the Mediterranean surf. Yonder are only Thracians, from south of the Balkans. This group, as black as your hat, consists entirely of Nubians. Another is made up solely of natives of Zanzibar. These divisions reach into nearly all the Asiatic and African lands. There are Afghanistaneze and others from still farther east, from the very heart of India, and even from the far Pacific islands. One has only to see these collections of students, massed around a teacher of their own language and nationality, to become convinced of the broad field of Mohammedanism and the mightiness of the effort needful to uproot it.

Poverty! That is no name for the condition of the students. They come to Cairo from the far-off regions, impelled by some passion bordering on that for learning, living on a little crust and fruit, having no sleeping place at night save the space of the sacred mosque which serves as a university, never paying a piastre for all the instruction of years, and looking forward with earnest longing to the time when they can leave again and impart to their native villages, or the very desert wastes, the wisdom which they have gained in the shades of the great hall of learning in the Cairo of the caliphs. There is a dash of self-seeking in their coming hither. When the tocsin of war is sounded, there is no exemption from conscription save learning. He

who has once entered the doorway is safe from the conscription list. Were an attack made on the very citadel where Mohammed Ali put to death every plotting Mameluke—except one, who leaped upon his faithful Arab steed and plunged safely into the depths below—nothing could touch him. He has come to the fountain of knowledge, and Mars has no claim upon him. At the present time the number of students is not so large as usual, for there is no fear of a war, except such as the English are fighting and holding themselves responsible for. I looked carefully at the kind of food which these students ate, and in all cases it was of the simplest quality. Some were taking their solid dinner, and it was nothing more than a rude bowl of lentil soup or a flat cake of pounded grain. The clothing in most cases betokened the same poverty. The slippers were of rude construction, such as fifteen cents would buy, and even these are to be worn at the general prayer, which begins the day for all the students, only to be laid aside during the later hours. The habit is a loose black, or other colored robe, which has become threadbare by long usage. I am sure I saw many students, and professors as well, whose entire dress could not have cost five francs apiece. This dress they have on, moreover, is the whole scope of their wardrobe. When they get another suit it will probably be when they reach home again, and enter upon their calling for life.

The professors get no salary. They have passed through various stages of learning, and when once they have committed every word of the Koran, and perhaps some of the more noted commentaries on it to memory, and have given other proofs of aptness at teaching, they are declared able to instruct. But they get no pay for teaching. Neither the University treasury pays them, nor does the student do it. Their instruction is positively gratuitous. Now, if by copying the Koran or other book, or by private teaching in families, or by doing some outside manual work, they can be supported, well and good. But for sitting squat on the sacred marble floor and teaching students the holy laws, and all the holy sciences that come from them, there must be no itching palm. This is the one place, and only one, so far as I can recall, where I have been where there has been no call for backsheesh.

How, then, is this immense establishment supported? I answer, that many students are sustained, and so permitted to remain at the University, by the funds of the institution. The treasury, instead of taking care of the professor, goes rather to keeping the student from starvation. There are

many endowments which have fallen into the hands of the state which constitute a large part of this treasury. Education has always been an attractive investment, and many Mohammedans have left sums of money for this purpose, and so the University of Cairo owes a good part of its wealth to this source. Again, when funds fall from certain causes, into the treasury of the state—perhaps property for which there are no heirs—it is devoted to this purpose. The building and all its belongings, and all really needy students are thus provided for. Out of the three hundred professors and other teachers, only one is paid a salary. He is the general director, or rector, and his salary amounts to 10,000 piastres, or about five hundred dollars of our money.

Of one thing I was very careful to make inquiry. I mean as to the bearing of this institution on the propagation of Mohammedan ideas. In all descriptions I had become familiar with concerning the great purpose of the students, the thought was made predominant that the students went away with a missionary zeal, and became intense propagators of the faith throughout their lives. The Rev. Mr. Harvey, of that noble cause and magnificent institution for Egypt, the United Presbyterian Mission, from the United States, was a very kind escort during my visit. He has been many years a resident of Cairo, and is very familiar with every form of Mohammedan life, and he informs me that this zeal for the Moslem faith does not exist, that the students do not go away with it, and never exhibit it, except in rare cases, in later life. Their stay in the University may be long. They may be three or four or five years, and if no way to work opens they may spend most of their life there, but whenever they do leave, sooner or later, they go off not simply as teachers of theology, but as jurists, mathematicians, or professional men of other callings, and religion is less in mind than secular work. Even when they go out as imams, or priests, that profession carries with it certain functions which belong both to the town clerk or the district judge, and hence the priesthood is absorbed in certain legal and administrative functions which eclipse the sacred office altogether. As to a burning zeal to disseminate Mohammedanism, it does not exist. It has no unquenchable love for itself, and is only continuing its own means of propagation because of something better. That something better is at its doors, and is beginning to thread the labyrinths of the Dark

Continent. In due time Christianity will do for Africa what it has done for Europe, and is this day doing for the half of Asia.

The darkest feature of my visit to the University was the absence of women. Alas! you never see the Mohammedan woman in these oriental lands, save with veiled face and hesitant step. Only yesterday I saw a handsome carriage being driven along one of the principal Cairene streets, preceded by a gaily dressed herald, who cried, "Make way, make way," as is the fashion here still. The silken curtains were drawn, but the occupants were two ladies. They must live in the dark. In the mosque they must sit in the lofty spaces, far back behind the wooden screen work, and even then be veiled. The very small girls, who trip about with little rattling and tinkling bells around their ankles, are hardly old enough to learn the way to the next street before the veil is drawn over their face, and only their little eyes are permitted to look out. In the multitudes which I saw at the University, both as students and teachers, there was but one woman. She was probably the wife of a professor, and had come merely to bring the learned man his dinner, and then slip back again to the dark rear room of the house misnamed a home, and await his coming, and be the menial still to prepare his evening meal. Mohammedanism has no place for woman in its educational system. Its best interpretation of her office is that she is simply man's slave. But the better day is coming, and may it soon be here, when the right of all women, in all these oriental countries, to the highest and the largest knowledge, shall be recognized as equal to that of any men beneath the shining sun.

AS SEEING THE INVISIBLE.

BY MRS. EMILY J. BUGBEE.

To stand at the post of duty
Whether we rise or fall,
If this be a place of beauty,
Or the homeliest lot of all.

To walk with a soul undaunted
In the God appointed way,
Whether with praise enchanted,
Or in shadow land it lay.

The good of the world's bestowing
Is vanishing as the air,
And its loftiest honors throwing
A burden of ceaseless care.

But to live as always seeing
The invisible source of things
Is the blesseddest state of being,
In the quietude it brings.

For in all of the strife and clamor,
And the evil that is done,
We know that the Lord will finish
The good that he hath begun.

And we need not grope in blindness
Because of the dreadful days,
But sure of the Infinite kindness
May stand in the certain ways.

Oh! for a strong uplifting
And a courage that will stand
While the Judge of the earth is sifting
The peoples of every land.

Oh! Earth so full of the glory
Reflected from above,
~~We wait for your finished story~~

we wait for your finished story,
In the faith of a deathless love.

NATIONAL AID TO EDUCATION.

PART II.

BY GENERAL JOHN A. LOGAN,
U. S. Senator from Illinois.

Having ascertained the extraordinary fact, from a close analysis of tabulations of authoritative statistics furnished by the Census and Education Bureaus, that, assuming the cost of educating a child in Mississippi, Tennessee, and West Virginia to be equal to such cost in the New England states, every one hundred adults in the former pay more to educate the children in those states than is paid by the same number of adults in any one of the latter, let us explore a little further for the reasons underlying that fact.

It might naturally be asked: How can these calculations be correct, when, for example, we learn from the report of the Commissioner of Education that Massachusetts pays annually for each child enrolled in her schools \$15.44, while Mississippi pays but \$3.38?

There are several factors which aid in bringing about this result. Some of these can be exactly ascertained; others of them, for want of statistics, can not.

In the first place, the \$15.44 per scholar which Massachusetts pays amounts to but \$4.98 per capita of her adult population, while the \$3.38 per scholar that Mississippi pays amounts to \$2.12 per capita of her adult population. Hence the real difference, so far as the payers of the cost are concerned, is only \$2.86 per capita.

Another cause of this difference or inequality is the fact that Massachusetts pays her teachers, on an average, about \$49.06 per month, while Mississippi pays hers only \$30.07. While this doubtless affects the

efficiency and equality of the education, it does not necessarily indicate a less number of pupils.

Still another cause lies in the fact that while the length of the school year is in Massachusetts one hundred and seventy-seven days, in Mississippi it is but seventy-seven days.

And still another may grow out of the larger proportion of teachers employed in Massachusetts than in Mississippi, for we find that while in the former, one teacher is employed for every 35.7 enrolled scholars, in the latter, one is employed for every 42.5.

These items enable us to understand why there are differences between the amounts paid in the two states, and what those differences are that exist under the present order of things.

We perceive, therefore, that while a strict scrutiny may bring to light the facts that the education in the one state or section is more efficient, the terms of school attendance longer, and the amount paid for school purposes more liberal than in the other, yet this in no wise tends to invalidate the statistics heretofore presented, nor to affect the argument based thereon. Although it may be true that Massachusetts spends more than \$15.00 per scholar while Mississippi spends less than \$3.50, it is also true that the latter has forty-eight pupils enrolled in school to every one hundred adults, while the Bay State has but thirty-three; and that while it costs the adults of the Northern state but \$4.98 each to pay this \$15.44, a similar service, similarly compensated, for its enrolled scholars would cost the adults of the Southern one \$9.70 each.

The fact, then, that this remarkable inequality in the cost of educating the children of the different localities in the Union does exist, can not be successfully controverted; and that there is no method of equalizing the burden save by government aid can not be truthfully denied.

The time has gone by when it could be said that Mr. A., who is poor in this world's goods, but surrounded by a full household of ruddy youths, must provide for their education from his own depleted pocket, just as Mr. B., who is rich, and has but a single child, provides for its instruction out of his plethoric pocket.

The principle is now fully acknowledged that it is the duty of the state or government—of the people, as a body-politic—to bear this burden, and thus to equalize it. This is the principle upon which our common school system is founded, which, notwithstanding the tax it imposes, is even now looked upon by the people as one of our most important institutions, second only to the republican basis on which our government is founded.

To bring this vital institution as near to perfection as is possible, to distribute its benefits as equally as possible, to render the tax as light as is consistent with efficiency, and to bring the burden to bear as equally as is practicable on all sections and localities, should be one great aim of our Federal legislation.

All the great nations of Europe are beginning to throb with the divine impulse which is first seen in the great, questioning eyes of the speechless babe. Some of them have lain for long centuries encrusted in the densest ignorance, and awake but sluggishly to a realization of the tremendous national power, which others have long since discovered, embedded in the education of the masses. Thus Russia, with her population of 78,500,000, although almost exhausting herself with wars for territorial aggrandizement, has awakened to the necessity of granting to her schools \$9,000,000 annually—a mere pittance for such a nation, yet containing the germ of higher promise. So also Austria, with her population of 22,144,244, is slowly stirring. Education there is now made obligatory, and in 1881 she supplemented prior national aid to it by a grant of \$6,500,000. Italy, in 1882, with a population of 28,000,000, gave like aid to the extent of \$6,200,000, beside providing school buildings and other necessary desiderata—previous aid having borne good fruit in a marked decrease of illiteracy. Prussia, with a population of 27,251,067, is fortunate in the possession of endowed schools with regular incomes. Yet she gave national aid to education to the extent of \$10,000,000 in 1881, and \$11,458,856 in 1882. France, with a population of some 37,000,000—independent of the millions of dollars expended for a like purpose annually by her departments and communes—gave in 1881-2 to the extent of \$22,717,880 for the education of her masses. Little Belgium, with a population of but 5,403,006—about one twelfth of ours—in 1882 gave national aid to education to the extent of about \$4,000,000; for she perceives, as a direct consequence of periodical aid of this character, that Belgian illiteracy is surely and rapidly

decreasing, while in like ratio her prosperity is increasing. Great Britain is similarly alive to the necessity for government aid to elementary schools. Such aid was given by her in 1882 to those in England and Wales, whose united population is 25,968,286—less than half the number we boast—to the extent of £2,749,863, or—roughly calculating at five dollars to the pound—nearly \$14,000,000. This, too, in a land that is also rich in well endowed universities, colleges, grammar schools, and other institutions of learning. Such aid was also given in 1882-3 to elementary schools in Scotland, whose population is but 3,734,370, to the extent of £468,512, or, say \$2,342,560; and to Ireland, with a population of 5,159,839, to the extent of £729,868, or, say \$3,648,340. Thus, in addition to the great educational advantages arising from the numerous well founded and amply endowed educational institutions for the various grades and classes of the British people that have long existed in England and Wales, Scotland and Ireland, we find the government of the United Kingdom aiding elementary instruction to the extent of about \$20,000,000 in one year—the combined population being but 34,862,495 souls; the United States, with a larger population, is without the advantages either of such national aid or such endowed schools as those countries possess. Even the colonies of Great Britain are equally impressed with the importance and essential necessity of general public education. Taking as an example that one of her colonies with which our relations are most intimate—the province of Ontario. Its population comprises but 1,913,460 souls, yet the amount expended there upon education in 1880 reached \$3,414,267. A similar ratio of expenditure to the total population—counting the latter at 55,000,000—would call for nearly \$100,000,000 in the United States.

But while it may be of interest to note what other peoples and other governments are doing toward the advancement of general education within their borders, and while the contrast with that which is done, or fails to be done, in the same direction in the United States, furnishes food for instruction and ultimate benefit, yet it by no means follows that this nation, destined, as every one of its citizens proudly believes, to march in the van of the world's civilization, is to limit its aims, its labors, its appropriations in the furtherance of education—the prime factor in all civilization—by the standards of other nations. The rather should the comparison, while it may for the moment bring to our cheeks the blush of shame, act as a stimulus to

higher effort and larger expenditure, if necessary on our part to reach that preëminent position of prosperity, power, and enlightenment, of which the intellectual alertness of our people and the genius of our institutions give abundant promise.

In considering this subject we must not fail to remember that among the nations of the world ours stands alone in this: that here the sovereignty is in the people. An ignorant sovereignty is a tyrannical sovereignty, whether held by the many or the few. Its capabilities for good can alone be drawn out by education. That Liberty sits enthroned in this land is due solely to education and that proper spirit of freedom and independence in thought and action which is begotten of education. As has been well said by another: “We have gained all that we possess by reason of the education of the individual, and we hold it upon the same tenure. What we hold for ourselves we hold for mankind, and we hold it for both upon the same condition by which it was gained, and that is the continued and universal education and development of the people.”

Every child born in this great republic is born with the inherent right to be educated. He is born heir to that popular sovereignty which, upon coming of age, he is entitled to exercise. The coming responsibilities rest upon him from his very cradle up. He has an absolute right to such an education as will enable him to properly meet them. His parents who brought him into the world weighted with such responsibility, did it with the implied obligation on their part to give him that education without which his birth would be either a mockery or a crime. As with the parents, so with the state-local, and so with the state-national. If the parents fail in meeting this obligation it becomes a binding obligation upon the state-local, and if the state-local fails the obligation devolves upon the nation.

Again, the obligation of every parent in this republic to educate his children so as to enable them in due time to intelligently and wisely exercise the great power of the franchise, implies the obligation on his part to give them, up to that point, *equal* educational advantages. By a parity of reasoning it logically follows that in case of failure by parent or state-local—whether from inability or other cause—the obligation to secure to all children within its domain not only facilities, but equal facilities, for the attainment of a sufficient education to enable them to cast an intelligent

ballot, rests upon the nation. Nor does this obligation cease when such equal facilities are provided. It goes further. It extends, if necessary, to the compulsion of those children to avail themselves of the facilities which the nation provides for their education.

That it is the right, then, of every American child to have a rudimentary education, and that it should be equal to that of every other American child, seems clear; and that where, through any cause, that child fails to get such education, it is the duty of the national government to enable him to gain it, seems equally manifest.

To what extent, and from what resources, the nation should grant this educational aid to its children, and through what channels and upon what basis the distribution of that aid should be made, are subjects that will now command our attention.

The burden of educating the children of the nation is a heavy one—a fact perhaps not as fully realized by our rulers and legislators as it ought to be. From the report of the Commissioner of Education, 1882-3, it appears that the estimated real value of sites, buildings, and all other school property in all the states and territories, is \$216,562,197. That of course is the existing “school plant” as it may be termed; but to get such a “school plant”—utterly insufficient as it may be—has been more or less burdensome. From the same authority it appears that the amount imposed and expended for common school purposes, in all the states and territories for 1880, was \$91,158,039; a large sum, yet after all but little more than half the amount absolutely needed in order to provide adequate school facilities for all entitled thereto.

A careful and conservative estimate founded upon all attainable data will show that not less than \$160,000,000 annually must be provided to secure the education of all the children of our country of lawful age. Of this amount, provision, as we have seen, is already made in the various states and territories to the extent of over \$90,000,000 annually. Of the various measures relating to the subject of national aid to education that have been urged upon the attention of Congress, none has ventured to appropriate a larger annual sum [\[J\]](#) than \$50,000,000. Should Congress at any time make

an appropriation of that amount, there would still be an annual deficiency of some \$20,000,000.

It is not at all certain that our national legislators have considered the magnitude of the subject with which they are to deal, nor that they have all investigated it with that degree of care and seriousness which it plainly deserves and even demands at their hands.

Every one, without controversy, admits the importance of educating our children; and without doubt, every one of our legislators has not only a warm and friendly feeling for this work, but also a willingness to do something to afford it national aid. But with how many of them is this a willingness without a formed and definite purpose? It were almost better that the importance of such education should be a disputed point—that a storm of controversy should arise and shake them in its throes, forcing them to lay hold of the very horns of the sacred altar of education—rather than that the dead, arid level of inert concession should bring forth nothing save a deceptive mirage. It is time to wake up to the fact that government aid in the line of education means nothing unless it be in the form of an annual appropriation of sufficient amount to produce tangible results.

Do our legislators appreciate the significant fact that of the \$91,158,039 expended on the public schools in the thirty-eight states and nine territories and the District of Columbia during 1882, more than one quarter of that entire expense was borne by the three states of Illinois, Ohio, and Iowa? That nearly one third of that great expense was borne by and expended in the four states of New York, Ohio, Illinois and Pennsylvania? That more than one half of it was borne by and expended in the six states of New York, Ohio, Illinois, Pennsylvania, Iowa and California? That nearly two thirds of it all was borne by and expended in the nine states of California, Missouri, Indiana, Iowa, Massachusetts, Pennsylvania, Illinois, Ohio and New York?

Of what practical avail, then, is the bill [\[K\]](#) which passed the United States Senate on the 7th of April last, so far, at least, as the amount to be appropriated is concerned? It proposes to appropriate a total amount of \$77,000,000. That amount certainly sounds well and looks generous at first sight. But how is it appropriated? Let us see.

This \$77,000,000 that looks so large and adequate, is to be scattered over the whole country, and over a period of eight years, thus:

The first year, \$7,000,000—which is much less than Illinois alone gives in one year for her own children; the second year, \$10,000,000—which is much less than Massachusetts and Iowa together give in one year for their own children; the third year, \$15,000,000—which is much less than Ohio and Pennsylvania together give for one year's schooling of their own children; the fourth year, \$13,000,000—or about what Massachusetts, Indiana, and Wisconsin together give a year for such purposes; the fifth year, \$11,000,000—or less than New York alone gives in one year; the sixth year, \$9,000,000—very little more than Ohio alone gives; the seventh year, \$7,000,000—or only a trifle more than Missouri and California together give in a year; and the eighth year, \$5,000,000—or a trifle over what Indiana gives, and less than Iowa gives, in one year!

Now, if such appropriations as these are not absurdly inadequate, what are they?

They are limited to eight years, and during those eight years the mean average annual appropriation is less than \$10,000,000. Think of it for a moment. An amount (\$9,625,000) appropriated by Congress to cure the illiteracy of the whole nation—only \$1,057,325 more than Illinois now spends in a year for educational purposes; only \$1,361,755 more than Pennsylvania spends, and only \$804,086 more than is spent by Ohio; while it is \$1,797,593 less than the state of New York expends in a single year within her borders for like purposes!

Take the exact figures of the census returns, and the amount actually needed is easily ascertained for that year—though it must be remarked that the amount needed is not remaining the same, nor diminishing, but increasing every succeeding year. The school population in 1880 was 16,243,822. To educate that population required an assumed average annual expenditure of not less than \$10 each, or \$162,438,220. The real expenditure was but \$91,158,039. Hence there was in that year a necessity for an expenditure of at least \$72,085,783 more than was actually expended.

But let us examine the statistical facts a little more closely. It is true that the school population then was 16,243,822, but it is also true that of that number only 10,013,826 were enrolled in the public schools, and of these again only 6,118,331 took advantage of their opportunities for instruction by daily attendance at those schools. Here, then, we find that the \$91,158,039 was expended in educating the 6,118,331 children who daily attended school, and that the actual average cost per scholar, therefore, was \$14.90, and not \$10. We discover also, that while 6,118,331 children were in daily attendance at the public schools, 3,895,495 children on the rolls of such schools were not in daily attendance, and that 6,229,996 other children of school age had not even the opportunity or facilities for any such education! It is plain, therefore, that had the 10,125,491 children of school age in the two latter classes—those who failed to take advantage of the school opportunities offered them, and those who had no such opportunities at all—been compelled, as they should be (except in case of sickness or other very sufficient cause), to daily attend public schools, then instead of the \$91,158,039 actually expended in such schools that year, there should have been expended \$242,027,855 that year, in order to give all children of school age an equal educational chance. In other words, the expenditure, as compared with the necessities of the case, left a deficit for that one year of \$150,869,816.

Now it is to make up for the deficiencies in the school facilities already provided in the states and territories, that Congressional legislation and national aid is proposed. But it would puzzle the combined mathematicians of all countries and ages to demonstrate that an annual deficiency of \$150,000,000, or more, can be made up by an expenditure of \$77,000,000, dribbled out in annual sums varying from \$5,000,000 to \$15,000,000, during eight successive years.

While, however, to meet the necessities of the case fully and absolutely would call for enormous annual appropriations, yet as the utmost conservatism and moderation should govern all experimental legislation involving large appropriations, so in legislating upon this subject it were safer to adopt the basis and estimate of least requirement heretofore given, and adopt \$50,000,000 as the amount that should be annually appropriated for this important purpose.

It is to be kept in mind, also, that an annual appropriation to this extent need not add one dollar to the burden of taxation now borne by the people.

In this connection it is not necessary to discuss any of the questions relating to the methods of raising our national revenue. Whatever differences of opinion there may be touching those methods or means, it must be conceded that our nation, under the present system and laws, holds a high and even commanding position among the civilized governments of the world, and that our people are enjoying more than an average degree of prosperity. It is our duty to use every effort to advance to still higher prosperity. In the meantime, however, any bill appropriating national aid to education should be based upon our present condition. Our revenue now exceeds our expenditures per annum by fully the amount (\$50,000,000) sought to be appropriated by the bill referred to. Hence its enactment would not add one dollar to the taxes already imposed. It follows, then, that should Congress be asked to support a measure making annual appropriation of \$50,000,000, derived from the internal revenue taxes and the sale of public lands, for school purposes, opposition to such a measure on the pretext that it would impose additional burdens upon the people would be flimsy and without force, and only transparently veil an opposition to increased facilities for educating our children.

If our children are to be provided with adequate facilities for proper and necessary instruction, the burden must be imposed in some form; and none can be devised that will bear more equally upon all, and be felt as little as this.

It is an old truism that "every rose hath its thorn." The advance of civilization and knowledge has its drawbacks as well as its advantages. This is manifested very distinctly in one direction in our own country. The rapid invention and introduction of labor-saving machinery has had a very marked tendency to draw the laboring population from the rural districts, and congregate it at the manufacturing centers. This, although it may be attended with many important advantages, has some very serious disadvantages, and is, perhaps, in part the cause of the serious contests we have seen of late years between capital and labor. It increases the population of the cities, and proportionately decreases that of the rural districts, and, as a consequence, increases the cost of living, as it advances the price of

property in the cities. It also tends very largely to increase the power and influence of corporations, monopolies, and other associations of this kind. The single item of transportation is vastly enlarged by this fact, and thus is increased the necessity for, and the power of, the railroads of our country. The effect of bringing together at these manufacturing centers large bodies of employ  s is, that for self-protection, combinations of labor, as against the encroachments of capital, are formed. Irritation and contests follow.

It is from these facts that we are confronted with one of the most difficult problems forced upon any nation for solution—a problem which thus far seems to be beyond the reach of legislation.

To check the advance of scientific and inventive genius, or to stop the progress of knowledge, is neither desirable, practicable, nor possible.

The only possible solution of this perplexing problem would seem to lie in the education of the masses, and thus elevating the laboring population as nearly as may be to the educational level of the capitalists—the rural districts to the educational level of the cities. By adequate national and state legislation, very marked and important progress in this respect may be secured. Should the government adopt the policy of adequate national aid to education, its distribution according to the number of persons under twenty-one years of age would perhaps be the best basis for such distribution at the start, but future experience and more exact knowledge would, no doubt, enable the remedy to be applied, in due time, more exactly to our needs. At present the statistics of illiteracy are not sufficiently definite and thorough to take them as a reliable guide in determining the basis for the distribution of so large an amount of funds.

One means, however, of meeting the difficulty named—one possible step toward the solution of this puzzling problem—is certainly within our reach. Educate the masses, elevate the laboring and producing population, and bring them up as nearly as possible to the educational plane already reached by those who hold and wield the moneyed power.

Education increases our wants and demands; increase in demand brings increase in supply; and this of necessity increases the demand for labor.

Economy on the part of the nation as well as the individual is a correct principle, and holds good in all states and conditions of life, but we must not forget that it is a relative term. For the individual who can neither read nor write to expend money for books and writing materials is a useless expenditure; but would you count that an extravagance on the part of him who can do both, so long as he keeps within his wants and means? What constitutes the difference in the application of the principle to the two cases? *Education.*

The pioneer farmer may have spent a life of patient toil on his farm, satisfied to live in his log cabin, with possibly a single room, a puncheon floor, and a clapboard door, unable to read or write—an upright, honest man, and probably as nearly contented as it falls to the lot of mortals to be. But mark the change! His sons and daughters are growing up toward manhood and womanhood; the free school has invaded his neighborhood; and they attend it. How soon it affects the household arrangements, manners, dress, and everything about the family! What has wrought the change? Education. Their wants, and what are now their necessities, are greatly increased. What follows? The desire to meet and supply these wants brings increased effort and industry for the purpose. And every family thus advanced in its views of what is necessary to comfort and happiness increases to the same extent the demand upon the producer and manufacturer, and thus widens the field of labor. Hence the solution of this great and knotty problem is to be reached chiefly by the education of the masses—by raising them toward educational equality with the wealthy.

There are many who delight in picturing the days of primitive simplicity, when wants were few and easily supplied; but is there one of these moralizers who would willingly go back to them? “Strict economy as gauged by our means” is a correct maxim everywhere and at all times. But civilization and enlightenment are progressive, and no laws save such as would trample under foot the inalienable rights of the people to “life, liberty, and the pursuit of happiness” can check that progress. We must therefore either foster the comparatively few more fortunate and energetic of our people, or we must endeavor by appropriate and legitimate and adequate legislation to link together and advance the entire mass. The noblest work of man is the elevation of his fellowman, and the grandest work in which a government can engage is the enlightenment of its people.

But these can alone be accomplished by the aid of the great lever:
Education.

FOOTNOTES

[J] And that bill was introduced by myself.

[K] S. 398, 1st Session XLVIIIth Congress.

A TRIP TO THE LAND OF DREAMS.

BY ROBERT R. DOHERTY.

It is strange how soon we all turn into redoubtable adventurers, after the “soft dews of kindly sleep” have fallen. Not Marco Polo, fresh from the glories of the Cathayan court; nor Orellana, with his glittering lies about Dorado; nor Hans Pfaali, big-mouthed with the wonders of his voyage to the moon; not even Baron Munchausen himself, could tell more astonishing tales, than can the prosiest among us on his return from Dreamland.

Dreams were believed by the ancients to be vehicles of supernatural communication with mortals. Homer says that they come from Jove; Mohammed tells us that Allah sends them; and according to Job, “God speaketh in dreams.” Milton, on the other hand, pictures Satan, “squat like a toad close at the ear of Eve,” assaying by his devilish art to reach the organs of her fancy, and with them to forge phantoms and dreams. So deep-seated was the belief in the supernatural origin of night visions, that the law of ancient Rome required those who dreamed of public affairs to report to the augurs, so that an authoritative interpretation might be promptly given to the rulers. There was hardly a governor or general of antiquity, but had a number of professional augurs in his retinue; and the course of events was often modified by the meanings they attached to their patrons’ dreams. Professor Creasy has written a unique volume on the “Fifteen Decisive Battles of History,” and has suggested another, on the dozen or more “Decisive Love Affairs.” As many fateful dreams could easily be selected, around which, as on a pivot, the destiny of the world has seemed to turn. The most ludicrous, and in many cases wicked interpretations were given to dreams; and Cato—himself an augur—said it was strange how two augurs could meet without laughing in each other’s face.

Even at the present day belief in the prophetic character of dreams is widely prevalent, and many lists of “interpretations” are in circulation among the credulous. When Rory O’More assured us that dreams go by

contraries, he followed current superstition. Tears are supposed to indicate joy, and laughter, woe. Dream of the dead, and you may expect tidings of the living; dream of the living, and unlooked for danger—perhaps death—is imminent. Many of the interpretations printed in the “guide books” are, however, exceedingly natural, as, for instance, that visions of gold foretoken wealth, and orange blossoms, marriage.

Let us place in contrast with such fanciful absurdities a tabulation of some of the veritable indications of dreams, as made by a modern scientist. Lively dreams, according to Dr. Winslow, are a sign of the excitement of nervous action; soft dreams, of slight irritation of the brain, often in nervous fever announcing the approach of a favorable crisis; frightful dreams, of determination of blood to the head; dreams of blood and red objects, of inflammatory conditions. Visions of rain and water are often signs of diseased mucous membranes and dropsy; distorted forms frequently point to disorder of the liver. Dreams in which the patient sees any part especially suffering indicate diseases of that part. Dreams about death often precede apoplexy, which is so connected with determination of blood to the head. The nightmare, with great sensitiveness, is an indication of determination of blood to the chest.

To adequately define dreaming must ever be a difficult, if not an impossible task. Professor Wilson, of Edinburgh, has graphically outlined peculiarities which distinguish dreams from the imaginings of wakeful hours and from the hallucinations of madness. The current of thought that rushes through the sleeper’s mind is quite free from the control of his will. Dr. Rush has called a dream a transient paroxysm of delirium, and delirium a permanent dream; but the dreamer’s intellect is withdrawn from almost all relation to external objects; while the lunatic holds communication by all his senses with the world about him. But while sleep has thus closed “the five gateways of knowledge” to the dreamer, he still hears and sees and feels and smells and tastes. An imaginative person, on visiting Niagara Falls, can afterward reproduce it graphically in memory; but his most vivid mental picture seems pale and hopelessly inaccurate when the scene is revisited. The visions of our sleep, on the contrary, are among the most vivid of our life, and where the objects have been seen before, the most accurate. “The main difference,” says Dr. Smith, “between our sleeping and waking thoughts appears to lie in this, that in the former case the perceptive

faculties of the mind are active, while the reflective powers are generally asleep. Thus it is that the impressions of dreams are in themselves vivid, natural, and picturesque, occasionally gifted with an intuition beyond our ordinary powers, but strangely incongruous and often grotesque; the emotion of surprise or incredulity, or of unlikeness to the ordinary course of events, being in dreams a thing unknown.”

Of the vividness of impressions made in dreams, illustrations are plentiful. Dr. Abercrombie first told the often quoted story of the English army officer whose susceptibility was so remarkable that “his friends could produce any kind of dream they pleased by softly whispering in his ear.” On one occasion they led him, in this way, through a long quarrel, which threatened to end in bloodshed. Just as the dreamer was to meet his enemy a pistol was handed to him; he fired it off in his sleep, was awakened by the report, and repeated to his laughing friends the fancies they had whispered to him a moment before. A well authenticated case is on record of a young Englishman who, at the age of twenty-eight, through disease, lost the power of speech for four years. He dreamed that he fell into a cauldron of boiling beer, and in his agony and fright shrieked for help. Of course, he at once awoke, and from that moment the use of his tongue was fully recovered. A bottle filled with warm water, which touched the feet of Dr. James Gregory after he had fallen asleep, produced an awful vision of a bare-footed tramp over the hot crater of Mount Ætna, through clouds of sulphurous vapors, and amid spurtings of scalding lava. Because of a blister on the head of Dr. Reid, he “positively endured all the physical torture of being scalped, while dreaming that he had fallen into the hands of a party of red Indians.” A lady dreamed that a man entered her chamber, and tightly clasped her left hand in his without offering her further violence or uttering a word of explanation. She remonstrated with him in vain; she shrieked for help, but could not make herself heard; then began a desperate struggle with the imaginary stranger, which culminated in awaking the sleeper—but not in releasing her hand, which, to her great alarm, was still held as in a vise. Summoning all her will-power, she rose from her couch and crossed the room, and it was only when she attempted to light a lamp that she discovered that she was holding her own hand with the other, which had become numb by the tightness of the grasp.

Indefinite expansion of time—or, rather, a total ignoring of the limitations of time—is another peculiarity in dreaming. It has been demonstrated that a man can dream in detail the events of years, and consume in the act of dreaming only a small fraction of one minute. “I sometimes seemed to have lived for seventy or one hundred years in one night,” says De Quincy, the prince of dreamers: “nay, I sometimes had feelings representative of a millennium passed in that time.” Dr. Macnish, from whose delightful essays several of these illustrations have been taken, within an hour dreamed that he made a voyage, remained some days in Calcutta, returned home, then took ship for Egypt, visited the cataracts of the Nile, Cairo, and the Pyramids; “and, to crown the whole, had the honor of an interview with Mehemet Ali, Cleopatra, and Alexander the Great!” A gentleman dreamed that he had enlisted as a soldier, performed many military duties, deserted, been apprehended, carried back, tried, condemned to be shot, and at last led out for execution. His eyes were blindfolded; after an interval of awful agony he heard the rattle of the fatal musketry, and awoke—to find that “a noise in the adjoining room had at the same moment produced the dream and awakened him.” We have, perhaps, all, though in less degree, had similar experience of the rapidity of thought in dreaming.

There is hardly any limitation to the fancy of the dreamer; he may even lose his identity, and for the nonce personate Cæsar, or Cromwell, or the King of the Cannibal Islands. It is said, however, that no man or woman ever dreamed that he or she belonged to the other sex; although the strange notion that the dreamer is a fish, or beast, or bird, is not infrequent. Usually, however, “we are somewhat more than ourselves in our dreams.” The tired school girl cries herself to sleep over some difficult arithmetical task, dreams, perhaps, that her teacher assists her, and wakens with the correct “answer” in her mind. So Condorcet successfully pursued his most intricate calculations in his dreams; and Benjamin Franklin has acknowledged his indebtedness to his midnight visions for the solution of many grave political problems which had hopelessly taxed his reason during his waking hours. An austere philosopher, who ordinarily seemed to be destitute of risibility, tells us that in one dream he could compose a whole comedy, witness its performance, relish its jests, and laugh himself awake.

But to the marvels of Dreamland there is no end. “Strange it is,” says the poetical essayist, “when regal Mab rides forth, drawn by a team of little

atomies across men's noses as they lie asleep, galloping through lovers' brains, and over courtiers' knees, and lawyers' fingers, and soldiers' necks, and ladies' lips!" Strange, indeed, and blessed as strange. Let us thank God for our dreams. They are the great levelers of life. The cruel distinctions of wealth and blood are forgotten, and our personal disadvantages are set aside. The bashful stutterer talks with the grace and fire of Demosthenes, and the wasted invalid regains his pristine vigor. In dreams

"The child has found its mother,
And the mother finds her child,
And dear families are gathered,
That were scattered o'er the wild."

The poor drudge who toils wearily through twelve long hours for the mere necessities of life, can at night sit on a golden throne and dispense royal favors. The ambitious soldier can fight bloodless contests, and win empires, without staining his soul with the crimes of a Napoleon.

And if the dreams of the mass of mankind be so full of wonders, what must be those of the giants of intellect and passion? What exquisite sensuous delight must have thrilled the poet Coleridge during his vision of Xanadu of Kubla Khan, when the mere fragmentary strains that he then heard sung are so beautiful! How wild and spectral, how awfully magnificent, were the dreams of Albrecht Dürer, judged by the allegorical pictures in which he has attempted to reproduce them! If to read of the visions of a Bunyan or a De Quincey thrills us, what must it have been to experience them—to have floundered with Pliable in the Slough of Despond, and stood with Christian on the Delectable Mountains—to have been "grinned at, stared at, chattered at," by thousands of alligators such as the "Opium-eater" describes, or to have with him "sunk fathoms deep in Nilotic mud."

Physiologists have made many curious and valuable observations bearing on our subject. They have found that when a sleeper dreams, the brain swells greatly, and becomes red in color, while the brain of the dreamless sleeper is "pale, shrunken, and bloodless;" they have shown that, from physical causes, he that sleeps on his left side will have visions of

fantastic incongruities, while the dreams of the slumberer who reclines on his right side will at least be logical and self-consistent; they have divided “the exciting causes of dream-images into peripheral and central stimulations”—that is, into those caused by muscular movements or positions, and by the hygienic condition of the various organs of the body, and those which originate somewhat mysteriously, in the nerve-centers.

After all, however, very little is known of the true philosophy of dreaming; and perhaps the quaint fancy of Sir Thomas Browne may not be as utterly absurd as at first it seems—that this life is but a dream, and that death will be an agreeable awaking to our real life, whose past is now forgotten only because we are now asleep.

THE HOMELIKE HOUSE.

BY SUSAN HAYES WARD.

CHAPTER III.—THE DINING ROOM.

Iss was gar ist,
Trink' was klar ist,
Sprich was wahr ist.

—*German Dining Room Motto.*

The central work-room of the house is the kitchen. There labor is continuous. There three times a day, year in and year out, the meals must be cooked, and the pots and pans washed. Slovenly work there tells all over the house. An ill-regulated kitchen involves poor cookery and waste, and cheapens the most artistically arranged dining room. But the importance of good, careful and intelligent cookery hardly comes within the limits of this article.

It behooves us, however, to insist upon it that the room where so much of the necessary work of home is carried on, should be airy, sunny, cheerful, well stocked with the implements essential to the lightening of kitchen labor, and adapted in every way to the comfort of its occupants.

A good farmer supplies himself with tools and machines for his farm work; but his wife often toils with cracked stove, green wood, and a scant supply of kettles and pans, when only a slight outlay would save her many weary steps and much worry of mind.

The kitchen should have painted walls that can be readily washed. Indeed, every surface in the room should be washable. There should be

plenty of closet room, a large sink, a large work-table, comfortable chairs, at least one easy chair, a shelf for books, and room in the window for a few plants if desired. A picture or two would not be out of place if protected by glass, nor an occasional motto—like the charge to the German cook:

“Köchin, denk’ an deine Pflicht,
Vergiss du heut’ das Salz ja nicht.”

Or the admirable rules for home living which Dr. Watts wrote for children:

“I’ll not willingly offend
Nor be easily offended;
What is ill I’ll strive to mend,
And endure what can’t be mended.”

There are many small houses where either kitchen or sitting-room has to serve also as dining room. Any sensible woman can make shift to get along comfortably in this way and eat her bread and honey with the queen in the kitchen when necessity compels, so long as she has neatness and despatch for hand-maidens. One large, light room is often far better than two small dark ones; but where a room does double duty there can hardly be unity in the arrangement and furnishing.

To my question, “What is of most importance in the dining room?” a man made answer, “the kitchen,” and a woman, “the outlook.” No doubt the provision of wholesome and abundant food for her family is the housewife’s first duty, but while fully endorsing the masculine paradox, we must not ignore the woman’s plea for a cheerful outlook.

If possible, the dining room should have as good a view as the house affords. Let it look out on the orchard, the sea shore, or the distant hills, rather than the stable or the clothes line. The view of a terraced, box-bordered garden, of a tulip bed and apple blooms, as seen from an old-fashioned country house dining room is one of the sweet memories which childhood has stored up for the enrichment of my coming years. Three

times a day the household gathers here to take the goods the gods provide them, and then, if ever, they should enjoy a little leisure, and be in the mood to appreciate the best of the out-of-door world that surrounds them. A good view is better than pictures or stained glass for a dining room; but when a good view is out of reach and an unsightly one is unavoidable, then stained glass comes to our aid. If that darkens the room too much, ground or cathedral glass panes can transmit the light, surrounded by a border of color. That would be over-leaping the obstacle; but it can be quietly set aside by means of a pretty sash or half-sash curtain of Madras muslin or any pretty, thin, colored curtain material. A curtain is a simpler, franker, and consequently better solution of this difficulty than any of the pasted-on, semi-translucent, paper cheats that simulate stained glass

“In faint disguises that could ne’er disguise.”

Let honest poverty hold up his head and hang up a width or two of ten penny Turkey-red calico by the aid of button rings and a brass wire, so that it can be drawn across the lower sash, and if the color be in keeping with the room, it will look better than anything more pretentious and less true. Good stained glass, such as Mr. Tiffany or Mr. La Farge devise, is very beautiful, but like Adolphus’s tea-pot, it has to be lived up to throughout the room, and so is more expensive than in its first cost. The fine view, however, involves no extra outlay, and beside adding good cheer to that which the housewife spreads upon her board, it is no inconsiderable factor in the table-talk of the year, helping not a little in the entertainment of guests.

The dining room should also be conveniently near the kitchen, either in point of fact, or made practically near in the case of a basement kitchen by a “lift” or dumb waiter. The kitchen should not open directly into the room, or all the kitchen odors will abide there. An intermediate pantry or entry way shuts off many of the smells of cooking, and a small slide through which dishes can be passed serves to the same end, as it obviates the necessity of keeping the door ajar while food is carried back and forth.

How to light the dining room is a question of some importance. There should be light enough to show the table to advantage, but it should be

possible to darken the room with shutters or blinds in the inevitable summer warfare with flies. A room looks better, artistically, where the light enters from but one side. Cross lights are the artist's abhorrence.

In city houses a conservatory built out on one side gives a pleasant suggestion of the woods and out-of-doors, and at the same time gives the right effect of light and shade to the room. Kerosene lamps are not ornaments to the dining or tea table. They are cumbersome, malodorous, and their room is better than their company. A chandelier over the table, burning gas or holding lamps, is the easiest and cheapest arrangement, but not the most picturesque. The prettiest light of all, and probably the most expensive, as the prettiest things are apt to be, is given by wax candles from tall candlesticks. Four of these judiciously arranged on the table will give an abundance of light for those seated about it, if additional light be provided for the rest of the room by a lamp on the sideboard or in side sconces.

A dining room should not be too warm. It is an old boarding house trick to so heat the dining room as to take away all appetite for food. The room should, in fact, be kept a little cooler than the rest of the house, partly because the lower temperature provokes appetite, and in part because on leaving the table it is natural to feel a sensation of chilliness, the blood of the body being called aside to the business of digestion, so that it is comfortable after eating to step into a room a few degrees warmer than that in which one has been seated.

The color of the dining room depends upon its size, exposure, and upon whether it must do double duty as sitting room, or library. Dark, rich furniture and wall-hangings have been the rule for dining rooms for many a year. The larger the room, the more elaborate, rich and dark can be the furnishing, but a dark room that hardly gets a glimpse of sun throughout the year must be made sunny by plenty of yellow in woodwork, or walls; bright, sunny pictures with gilt frames; and by the glitter of brass or of the pretty, yellow English ware with which the china shops are aglow this year. For rich wall effects Japanese or leather paper is good, Lincrusta better, the latter being a comparatively new material, in substance something like linoleum, washable and very durable (so the manufacturers assert), having figures raised upon it, and coming in good designs and colors. With elaborately decorated walls, plain curtains are called for. Where the walls

are to be furnished freely with oil paintings, let the pictures supply the decoration, and let the walls be as unobtrusive as possible—only ensuring that they are of a good back ground color; sage, olive-green, olive-brown, or dull red, in paint or paper.

Family portraits, if good, are not out of place in the dining room; but poor, old photographs in bungling, black walnut frames should be preserved in the private apartments of those who value them. They are never decorative; nor are pictures popularly known as dining room pictures much more pleasing. The effigy of a silver salver of leaden hue loaded with fruits of all climes, with a decanter of wine and a half empty glass, of fishes hanging by their gills, or dead ducks, each from one web-footed leg, is not nearly so attractive as a good portrait, landscape, *genre* picture or flower painting, however good practice the manufacture of such studies may be in the art schools. I know a dining room where, outrivaling some amateurish fruit painting, an engraving after Rosa Bonheur of a shepherdess with her sheep has been a daily delight for years, and another where the only picture space in the room, that directly over the mantel shelf (the walls being darkly paneled), is filled with a water color copy of Sir Joshua's "Angel Choir" that seems fairly to light and hallow the air around it, like the glories round the heads of saints.

An over-mantel is appropriate in the dining room if anywhere, as it affords shelf room for choice china or glass that ought to be seen. A little ingenuity can go a great way in dressing up a commonplace shelf so that it shall have dignity and importance. I have known one to have a very aristocratic air which was only an adaptation of part of a four-post bedstead, graceful, slender posts standing on either side of the fireplace, built up with shelves of varying width and length.

If books in every room are a prime necessity, as our model home-maker assures us, then there should be at least one book-shelf in each room.

"Pray what is that book-shelf for?" asks the visitor while seated at the dinner table in "The Poet's House," which Mr. Scudder has described for us. "Books of reference," said Stillwell, promptly. "It's extraordinary how many little questions come up for discussion at the table, questions of dates, of names, of quotations. So I keep a dictionary, a book of dates, a brief

biographical dictionary, a dictionary of poetical quotations, and one or two other such books at hand. It is the sideboard to our mental feast. We don't keep everything we possibly need on the table itself." And others beside poets would find such a shelf of great convenience.

There should be at least a square of carpet in the dining room under the table, not only for warmth and the look of comfort, but to prevent the noise of chairs scraping over the bare floor. In other rooms small rugs scattered here and there may suffice, though one large rug is always more restful to the eye, but a table around which a family gathers, either in dining or sitting room should for these obvious reasons always stand upon carpet. Where the floor is carpeted throughout and a crumb-cloth or drugget used, pains should be taken in the selection of the latter to get the colors in harmony with those of the carpet. Cheap druggets as a rule are so glaring and crude in color that any carpet that respects itself loses tone at once, and appears thoroughly commonplace when forced into association with the blowzy things. The patient seeker, however, may be rewarded in his search by finding a "Bocking" or drugget that shall be as thoroughly becoming to his carpet as is the tidy morning apron to the neat-handed Phyllis who wears it. Since the days of King Arthur the round table has been held the most delightful for social and hospitable purposes. With a small family there is a cosiness about a round table that is very charming, but when one is forced to enlarge the circle a small, round table can not be expanded to the required circumference. A solid table seven feet long and four feet wide will seat six comfortably, and eight without crowding, and is of delightful dimensions to sit about of an evening, when work or study is toward. If such a table be used (an Eastlake table, our furniture dealers would call it, since Mr. Eastlake inveighed so severely against what he styled the "telescope" table) there should be two side tables made four feet long which could be of service in the room, standing against the wall, but on occasion could be used to enlarge the dining table. H. J. Cooper, an authority on house-furnishing, after speaking of Mr. Eastlake's objections and suggestions, says:

"We do not find, however, any great revolution in the matter of expanding dining tables, and are inclined to think their great convenience will prove a barrier to any wide-spread reform."

A good table should be polished, not varnished, and protected when not in use by a substantial cover.

Side tables are serviceable when one's space or purse will not allow of a sideboard, but sideboards are useful articles of furniture, and look better when made of the same wood as table and chairs. They should be simple of construction, with straight rather than curved lines, smooth surfaces, and with drawers easily get-at-able and lightly pull-out-able.

The sideboard should hold the table linen for daily use, the daily silver and cutlery, teacups and saucers. If, in addition, it can find room for a convenient box of biscuits, pot of ginger, or any simple refreshment for the late worker who likes a bite before going to bed, so much the better will it serve its purpose as a sideboard. Our grandmothers kept here decanters and wine glasses, but in these days of the W. C. T. U. wine is seldom found standing out boldly in sight in the dining room. In addition to the sideboard a small table or a butler's tray should be ready to hand to hold dishes or food that must be used in the table service.

Cabinets for the display of china, closets let into the chimney for the safe keeping of nice glass, should feel themselves at home here, while plaques and old china plates seem to belong of right to the dining room, and can be arranged over doors by means of a tiny balustrade, or on the frieze, or as over-mantel decorations, while choice cups and saucers can fill the cabinet spaces. In the breakfast room, where I am now writing (not my own), I have just counted fifty-nine pieces of glass or pottery which hang on the wall or stand exposed on sideboard, mantel or shelves, besides the tiles of the fireplace, two small cabinets, each holding a half dozen rare and precious Japanese cups and saucers protected by glass, and two large cabinets filled to overflowing with specimen china, and yet the room does not seem at all overstocked with ceramic treasures.

Growing plants are charming dining room ornaments, but will only thrive where a minimum of gas and furnace heat and a maximum of sunshine and fresh air is supplied, with regular attention as to water and shower baths. They are sure, however, to reward painstaking care.

Decorative china and plants, however, are luxuries, though less or more within the reach of all. A screen, though usually looked upon as a luxury, is

in the dining room almost a necessity, and it can be bought or manufactured at home for a nominal sum. Many a guest has been well nigh martyred at table with a fire in the rear and sunlight to the fore, whose meals might have been made a delight by a careful adjustment of shades and blinds and the judicious intervention of a screen between chair and grate. Doors must needs be left open as servants pass back and forth, and a screen between the mistress's chair and the door may save her from many an annoying influenza.

A thick, white cloth of felt or Canton flannel should be spread over the table before it is "set." This not only protects the polish of the table top, but makes the linen cloth lie much better, and appear to the best advantage.

Table linen should, so far as possible, be spotless. Fine damask is costly, but a clean, coarse cloth looks better than a fine one soiled and tumbled. It is true that table linen is worn more by washing than by use. Still it must be washed—at least that is the American theory. I have sat at a table in Saxony where the table linen bore the date of more than a century before, but there the wash was perhaps a semi-annual affair, and a breakfast cloth was made to serve from Easter till July, breakfast being only a simple meal of a roll and a cup of coffee.

We can lay down no further rule for the changing of table linen. It is perhaps better to keep breakfast and dinner cloths separate, the finer for dinner, that being the more formal meal; tea and luncheon can be served, if one wishes, without table cloth. If care be taken to lay a carving cloth or napkin under the meat platter, or a tea cloth where tea or coffee are to be poured, breakfast and dinner cloths can be kept fresh longer.

Some writers more nice than wise sneer at napkin rings, implying that no table linen should be used more than once without washing. But there are few families of any size that can afford such lavish laundry work. A family of six would require twenty-one dozen napkins in constant use, if given out fresh each meal. When the same napkin must serve for more than one meal, a napkin ring is the simplest and surest way of securing each person his own. Of course rings are only for family use, not for the transient guest, and they would be out of place at a dinner party.

Table cloths should be done up with a suspicion of starch, not enough to stiffen them, but with only so much as will make them iron well. Heavy linen looks and wears better than light. Large napkins are for dinner use. Delicate doilies of fine drawn linen work or silk embroidery are laid under finger bowls to protect the choice china plates on which the bowls rest. This doily should be laid to one side with the bowl, it should not be used as a fruit doily. I have seen an absent minded man roll up in a crumpled heap one of these delicate lace affairs costing five dollars, perhaps, and then carelessly wipe hands and moustache with it, while the mistress of the house looked on with an assumed placidity which spoke volumes for her powers of self-control.

The finger bowl is not an elegant affectation, but is a genuine comfort where fruit or sweets are served. Fruit napkins should be used to save large damask ones from stains.

If the first requirement for a well ordered table is cleanliness in damask and dishes, the second is tidiness and regularity of arrangement. If mats are placed under hot dishes let them lie on the square, and let the plates be put on at regular intervals, and in a straight line. A hotel waiter who flings plates and plated ware at the table by a dextrous twirl of the wrist is no model for the home table setter. Spoons for soup and dessert should lie to the right of the plate, knives above, forks to the left; this is the time honored usage, and it makes the labor of serving dessert easier if all knives, spoons and forks to be used during the meal are laid at the first by each plate. Tumblers stand to the right a little above the plate, butter-plates in a corresponding position to the left. Avoid the use of what are popularly known as "individual" dishes upon the table, such as butter plates, salt cellars, sauce plates, and so forth. This is another hotel fashion that should not find its way into the home. It is better to use a larger plate and take a greater variety of food upon it. The little butter dishes are really needed only with warmed plates; and beans, peas, corn, and other vegetables in separate dishes, about a dining plate, make a table look very untidy, and make extra and unnecessary work for the dishwasher. An English lady who visited me a year ago took home to London with her as an American curiosity a set of butter plates which, so she writes, she has not yet found opportunity to use, not having had any American visitors.

Steel knives are better, and where meats are to be served are more desirable in every way than plated ones, the latter being a device to save the labor of “scouring” with Bristol brick.

Flowers or fruit are never out of place upon the dining room table; a showy *épergne* is not necessary, for a pretty growing plant always makes a good center piece, and a single rose in a slender glass adds flavor to the best cooked meal. My grandmother of blessed memory used to say that the simpler the meal the more pains should be taken to serve it daintily. Broiled salt pork and baked potatoes, according to her theory, could be so bravely set out upon the table as to make a meal fit for gods and men. A parsley bed is of special service in decking out a simple dinner, and celery tops are not to be despised.

The heads of the household should face each other from the ends and not the sides of the table, if the meals are served English fashion, vegetables and meat being placed upon the table. No table can be set with any air of elegance when the meat platter or the tea equipage stand in the middle of one side. It makes comparatively little difference, however, when meals are served *à la Russe*, that is with meats and vegetables placed at side tables and passed by servants, while only fruit, bon-bons and ornamental dishes appear upon the board. The latter fashion seems to be obtaining in America, and an intelligent diner-out remarked in my hearing the other day, that fifty years hence no meats at all would be carved at table. This Russian fashion is pretty and wholly luxurious, as it removes all possible demands for service or helpfulness from those seated at table, and devolves it all upon servants. The fashion requires more and better trained servants than most of us have at command.

The bane of modern entertainments is the enormous number of courses that style makes essential. Women with but one or two servants at the most feel called upon to give luncheon or dinner parties, and course follows course, many of them sent away scarce tasted, and the home silver and china not sufficing for the occasion, must be eked out by borrowing or by expeditious washings between the courses. The giving of such entertainments by persons of moderate means exhausts nerves as well as purse. Let us wisely give up aping rich people’s ways, and aim for simplicity in our table service.

Colored table ware is cheerier upon the table than white. Very pretty English or American sets can be obtained at low prices. The Canton china (willow pattern) comes in good shapes, is of good color and standard design, and single pieces can always be bought to replace what has been broken, but

“Porcelain by being pure is apt to break,”

Or at least to chip at the edges, and for every-day use pretty crockery is good enough unless a painstaking and cautious hand wields the dish-mop. The more covered with decoration (design and color being good) the prettier will be the effect of the ware when in use.

It is not at all necessary to have all the dishes upon the table of the same style and pattern. Harlequin sets can often be brought together so as to combine harmoniously, and pretty single pieces can be bought marvelously cheap. Amateur painted china is generally too costly for daily use, and when good should be treated with respect.

Plain silver is on the whole better for plain livers than that which is more elaborately ornamented, and absolutely plain solid silver forks and spoons can never be out of taste, and can easily be kept tidy with whiting. Electro-silver and patent cleaners of that ilk injure silver and are ruinous to plated ware.

The beauty of silver and pottery depends first upon their form and adaptation to use; secondly, upon their decoration. Delicate chasings and thin *repoussé* work are naturally as appropriate to silver as good shapes and *flat* decoration are to earthen ware.

As to glass, there is a crystal craze at present, and “hob-nail” glass glitters on all tables. Miss Lucy Crane, in her lecture on “Form,” says (and I quote freely because her words are timely):

“As the beauty of glass consists in its transparency and lightness, and its capability of being twisted or blown or moulded into a multitude of delicate forms, it early occurred to the manufacturing mind that if made thick and solid, and cut into facets it would resemble crystal; and thus it has come to

be a fixed idea that hard glitter is its most valuable quality, so it is made inches thick, and pounds heavy, to enhance its brilliancy; and being one of the most fragile of substances, it must be engraved with people's crests and monograms as if it were intended to carry down the name of the family for generations to come! Being of its nature transparent, it must be rendered opaque of set intention by coloring matter, and then painted and gilded! Since at its strongest glass can never be anything but fragile, at least let it keep the beauty belonging to fragility; since it is naturally transparent, let the light be seen streaming through it, sometimes delicately tinted, sometimes iridescent, and, instead of being cut, let it be blown and twisted into the thousand delicate shapes to which it easily lends itself, and of which in the Venetian glass of a bygone day, and in its present revival, there are such delightful examples."

I saw last evening a handful of flasks on their way to the laboratory, whose soap bubble effects were far more beautiful than all the cold glitter of all the "hob-nail" ware that Sandwich has ever produced.

In a boarding house it may economize labor to set the table over night, but it is pleasanter and more homelike to have it set fresh and clean with the morning light; beside, to have the dining table clear of an evening is often a great family convenience.

The dining room affords grand opportunity for the domestic artist. The bread board, bread and carving knife handles, salad fork and spoon, all offer employment to the carver's tool, to say nothing of cabinet, sideboard and over-mantel. Tiles for tea pot rests and all sorts of china call for the decorator's skillful brush, while tea cloths and coseys, doilies, mats, centerpieces and carving cloths all await the embroiderer's needle.

Arise, my young readers, and take your tools in hand, for home work is the fairest adorning of the homelike house.

MEXICO.

Mexico is a country reaching from the Gulf on its eastern coast to the Pacific Ocean, almost 2,000 miles, with a breadth varying from 140 to 750 miles. The whole territory of Montezuma, at the time of the Spanish conquest, was not less than 1,600,000 square miles, more than one half of which has been obtained by the United States by purchase, enforced treaties, or otherwise. The plains on the coast are low, marshy, and in the summer and autumn malarial diseases are very prevalent. Strangers can visit the place with safety only about four months in the year, when severe northern gales cool the heated atmosphere and dissipate the seeds of disease.

There are 6,000 miles of coast line, but, considering its extent, it does not furnish many good harbors.

The main body of the land is an elevated plateau, traversed by chains of mountains, some of which are of extraordinary height. The eastern Cordillera, or chain, that runs nearly north from the initial point has an elevation of 6,000 feet, the western nearly 10,000. Traversing the longitudinal range, there are several cross ranges containing some of the highest volcanoes on the continent. They are all quiescent now, and none of them have been active during the present century. There are not many lakes, and none that are very large. The basins of some, though of sufficient extent, are so arid, and evaporation is carried on so rapidly that the water in them has, at times, quite disappeared. Neither are the rivers of much importance as thoroughfares. The Rio Grande, forming the boundary between Mexico and Texas, is the longest (1,500 miles), but navigable only for a short distance. Those in the mountain region are impetuous torrents, larger near their source than afterward, as they lose more by absorption, in passing through arid portions of the table-lands, than they gain by drainage, except in the rainy season. After plowing deep furrows, and cutting out immense ravines among the foot hills of the mountains, some are partly exhausted, drawn into reservoirs and canals constructed for purposes of

irrigation, and spread out into sluggish bayous, of no great depth, before they reach the sea. The lack of navigable streams has been seriously felt.

Climate, other things being equal, decides the flora of a country, and in this respect Mexico has many advantages. Were the country level from the Gulf to the ocean, it would have mostly a tropical climate, and produce only the vegetation of the tropics. But, rising in successive stages to a height of 19,720 feet, the temperature changes with the elevation, and a large portion enjoys the climate of the temperate zones. The low lying region near the coast, called the "hot country," has a rich soil, a humid atmosphere, and abundant rains, that perpetually nourish a rank tropical vegetation. At an elevation of 3,000 feet we reach a delightful zone where the extremes of heat and cold are unknown, the temperature ranging from fifty to eighty-six degrees. Here the forms of vegetable life, mingling those of the lower and upper regions, have a charming variety. Crossing this wide belt, with its luxuriance in things of surpassing beauty and usefulness, and advancing gradually till the mountains begin to show their rugged forms, at an elevation of 8,000 feet a colder climate is reached, with a corresponding change in the vegetation that now ranges from the corn, barley, and other useful cereals and hardier fruits to the cryptogamia of the mountain top. Take it all through, from coast to mountain, it is quite safe to say Mexico has a flora not excelled by any other country of the same dimensions. And it has increased with the advance of civilization. Many plants, flowering shrubs, and fruit-bearing trees that were not indigenous, but successive contributions from the Old World, have a vigorous growth, and produce abundantly. Apples, pears, peaches, cherries, oranges and grapes, with a variety of choice East India fruits, are widely distributed through the country. In the coast region, and to an elevation of about 1,500 feet, they have cotton, cocoanuts, cocoa, cloves, vanilla, nutmegs, peppers, and other spices of commerce, beside the fruits of nearly all tropical countries of the east and west. Higher up they have sugar, coffee, indigo, rice, tea, bananas, and an abundant supply of edible roots, such as yam, arrow-root, sweet potato, and all the fruits of America, Central Asia, and Barbary.

From a partial catalogue of the productions of the country there is evidence that its agricultural possibilities are very great. Nearly all fruits and grain, indeed, nearly all plants that grow, are either indigenous to the country or may find a congenial home within its limits. Some parts of the

upland require irrigation to make them productive, and, if the dry season is prolonged, water must be stored in basins for the use of stock. The neglect of this, especially where the land has been long cleared, causes barrenness, and gives the country a desolate appearance.

The agriculture of the country has never been of a high order, though the Aztecs, at the time of the Spanish invasion, were an agricultural people, and about as well acquainted with the arts and processes of husbandry as most nations of the East were at that day. Having incorporated in their communities the shattered remains of the old Tolteck tribes they had acquired considerable civilization, and were not, as the invaders supposed, rude nomads, or even herdsmen, but cultivators of the soil, and fixed in the possession of their estates. Theirs was not a skillful husbandry, since necessity, mother of inventions, had not greatly improved either their methods or their instruments. They had no plows, harrows, or cultivators, but used hoes, knives, and sickles made of copper. In planting, the earth was loosened with a hoe or stick, and the seed, when dropped, covered with the foot.

The present state of agriculture, though much improved, is still very inferior, and the production, reported in the last census, \$177,451,985, might, from the same areas, be greatly increased. Before the recent advent of railroads those far in the interior had no adequate means for exporting the excess of their products, and little inducement to raise more than they needed to consume.

Mexican forests furnish in abundance nearly, if not all, the useful timber trees of the north, and those valuable woods that grow only in the tropics. Some sixty varieties used for timber are mentioned, and twenty suitable for the finest style of interior finishing and furnishing.

The mines of Mexico have long been famous, and are not surpassed in richness by those of any other country in the world. Early in the fifteenth century the inhabitants had accumulated wealth from that source, and the glitter of their gold led the avaricious Spaniards to undertake the conquest of the country. Just how long the mines had been worked before the invaders came is not known. After a change of owners, and the improved methods they adopted, the product was greatly increased, and ever since,

though subject to many interruptions on account of political disturbances, it has been larger than in any other country except the United States. The Spanish settlers at once engaged in working the mines of Tasco, Pachuca, Zacatecas, and Guanajuato. Cortes selected for himself and worked the gold mines of Techuantepec, and the silver mines of Zacatecas, that were found productive. The mine at Real del Monte, near the city of Mexico, has yielded largely, and enriched several successive owners. And the principal vein at Guanajuato, noted for its richness, is described as ten yards wide, and has been worked a distance of more than eight miles. In the early part of this century the annual product of these mines exceeded twenty-five million dollars, and they seem inexhaustible. The whole of the gold and silver taken from the mines of Mexico up to 1870 was estimated at \$4,200,000,000. The seven principal mines of San Luis Potosi are said to be very productive, and the whole of Sinaloa abounds in silver mines. In Sonora there are one hundred and forty-four operated, chiefly producing gold, and a much larger number in which, though productive, work is suspended. Many large mining districts are simply located, and their development delayed, awaiting more ready means of access to them. That country alone, probably, could furnish the world a full supply of the precious metals for centuries, or until they become as plenty and cheap as they were in Jerusalem in the time of Solomon's reign. Mexico has not only mines of gold and silver, but the country abounds in other minerals of no less importance. Iron, tin, copper, lead, mercury, cinnabar, and nearly all the known metals are more or less abundant. Coal is found in three or four districts, but to what extent, or of what quality we are not informed. The products of the coal fields, and their rich quarries, and of the oil belts, can be but little known till their facilities for transportation are improved.

The roads constructed as thoroughfares of travel and commerce will modify the industries of the country through which they pass. Mining and stock raising, already extensive, will be increased. Farming and farms, such as we have in the States, will be common, and, as the resources of the country become better known, many enterprising men will be attracted to the Mexican plateaux; society will improve, the reign of superstition will cease, and a free government for an intelligent Christian people, though for a time struggling against chronic tendencies to revolution, will become established, and strong as it is liberal.

Mexico encourages immigration, but, naturally enough, prefers those of the Latin race, as more like the native population. Still, having friendly relations with the United States, and greatly improved opportunities for intercourse, prejudices will be overcome, barriers that have hindered immigration taken down, and perfect liberty of conscience proclaimed through all the land, to all the inhabitants thereof.

TWO SEAS.

BY ADA IDDINGS GALE.

Are not those wild steeds champing on the beach,
Rearing and splashing on the lonesome shore,
The main land seeking frantic'ly to reach,
Their white manes gleaming like the frost wreaths hoar?

Steeds of the sea are they that tireless ever
Beat with their sounding hoofs the hard sea sand,
Lashed onward by the blast, with fierce endeavor
They vainly seek the quiet of the land.

Type of that wild unrest that fills the soul:
The waves of longing, mad desire, and strife,
Whose undertone of sorrowfullest dole
Is the sad voicing of the sea called Life.

A type and yet unlike—there is a shore
Where the wild sea forgets the tempest's breath,
And rests in lullful silence evermore
Upon the wide, white, shining strand of death.

O perfect peace! O blessed mystery!
Where waves of longing cease their gainless quest,
And on the still sands of eternity
Do melt away in an eternal rest.

NEW ORLEANS WORLD'S EXPOSITION.

BY BISHOP W. F. MALLALIEU, D.D.

London and Paris, Vienna, Philadelphia and New Orleans share the honor of having been selected as sites for the grandest displays of which modern civilization is capable. This far-away city of the Southeast was selected in view of the fact that it is the great metropolis of a vast and rapidly developing portion of the Union, and to emphasize the fact that the time has come when the past, with its mistakes and antagonisms should be left behind, and also to encourage the rising industries of the entire South. The general government did well when it extended most generous financial aid to the enterprise. And should further need of such assistance be developed, it is to be hoped that enough will be supplied to make the Exposition a complete success.

The formal opening took place on December 16, 1884, by the President of the United States. True, he was not present, and yet the touch of his fingers set in motion the engine that drives a thousand whirling gears and pulleys. Fifty years ago it would have taken President Jackson a month to travel from Washington to New Orleans, but now, quicker than the revolving planet turns upon its axis, the President, standing in his office in Washington, executes his will in a city a thousand miles away. This world used to be twenty-four thousand miles in circumference, and it took six months to make a voyage around it. Now it has become so small there are no distant lands; we are all neighbors, and crowded at that, and thought, which is a part of man and the best part, travels round the world in the twinkling of an eye. It is a great thing to live on so small a world in such an age as this. Nowhere do such thoughts more forcibly impress themselves upon the observer than in a World's Exposition, for here, side by side in friendly rivalry, are the people and productions of almost all the nations of the earth. The Chinaman is here with his hideous gods and all sorts of queer things, from ivory chopsticks to the most elaborate porcelains. The men of

Japan are found wherever there is an honest dollar to be made. They bring things to show and to sell. With their thin lips and sharp pointed noses, and keen, bright eyes, they remind one of the shrewdest types of Yankee peddlers. Nobody expects to get the better of one of these Yankees from the land of the rising sun. Their ingenuity is surprising, and their powers of imitation are nearly equal to those of the Chinese. With the inspiration which comes with Christianity, it is safe to predict that a future of great promise is the portion of this nation. The ubiquitous Turk is here with the same articles, or their duplicates, that he has had in every exposition, and which he is gradually introducing into state fairs. These institutions of the present age must greatly stimulate the small industries of the Turkish Empire, though some people have thought the Turks at Philadelphia were, for the most part, born in Ireland, and these of New Orleans are supposed to be native Creoles, but still they sell olive wood paper weights, paper cutters, work boxes and trinkets of various sorts, said olive wood having the reputation of coming from Jerusalem, and, to support the reputation, being inscribed with divers Hebrew letters which the sellers are unable to decipher. Of course the European nations are represented, but not to so great an extent as at Philadelphia, and not so fully as will be the case a month later. The foreign countries best represented are our next door neighbors. Here is Jamaica, true to its past and present, with an exhibition of all sorts of rum, from thirty years old and less, in bottles and barrels of all shapes. It is put up with a nicety and even elegance which would be worthy of something better. Then she sends sugar and molasses, dye woods, coffee, cocoa, and skins dressed and undressed, with samples of varied workmanship in several departments. Mexico sends the military band of the Eighth Regiment of cavalry, more than fifty pieces, and it does credit to that Republic. There is an air of Spain about all the productions of Mexico, whether it be the crude ore from her mines of gold and silver, or the richly caparisoned saddles, which in beauty and comfort are unsurpassed. Honduras, both Spanish and British, Guatemala, and Central America, add largely to the extent and attractiveness of the display. No one can carefully study the exhibits of these four last named countries without being profoundly impressed with the idea that they must possess a wealth of undeveloped resources which will, in the near future, attract the attention of the civilized world. It is manifest that they have a soil of exuberant fertility, and a climate that is free from the cold rigors of the north and even from all

dangers of frost, and that all circumstances offer the promise of the maximum of results for the minimum of toil and capital. It seems as if a good many of the physical conditions of the Garden of Eden were still retained by these favored countries.

Nearly, if not quite all the states of our Union are represented, though it is to be regretted that some of them, especially Massachusetts and Pennsylvania, are deserving of severe criticism for the very meager displays which they offer. The people of Massachusetts will have more cause for shame than pride when they visit the spot where their activities and achievements should be fairly and fully set forth. There is no excuse for such a failure. Even little Rhode Island does better than her proud neighbor. It is a Rhode Island Harris-Corliss engine that drives the machinery, and the same State sends one of the grandest locomotives that ever ran on rails. Connecticut, the land of notions and wooden nutmegs, makes a fine show of her thread manufactures. The whole process, from preparing the raw cotton to selling the thread in spools, is displayed before the eyes of the admiring spectators. Not a few of the Southern people are led to ask, as they see the thread making and, close beside it, the weaving of cotton cloth, why should we send the cotton we raise to the North, especially to the most distant eastern corner of the North, and after the people there have made it into thread and cloth bring the same cotton back again? Why pay them for transporting it both ways and also for manufacturing it? It is well for them that they are asking such questions. When people begin to inquire it is a sure sign that they are getting ready to act. Soon we may expect to find them making their own cloth and thread where the cotton is grown.

The great West is here in full force, the states west of the Mississippi being especially prominent. It is not long since Kansas and Nebraska were both included within the limits of "The Great American Desert," on whose sandy soil it was said not even grass could grow. But now from those same arid plains come the best of corn and wheat, and all the other cereals, with fruits and vegetables that are truly surprising. Such potatoes as Oregon and Colorado send need at least such hills as those in which eastern farmers raise similar crops. Think of potatoes ten inches long, six inches wide, and four thick. But time and space would alike fail to specify the abundance and variety of the horticultural and pomological products of the West, this

including all west of the Alleghenies, and especially west of the Father of Waters.

One of the most important sources of national prosperity, growth, and riches is to be found in our mineral deposits. Here we see rich specimens of almost every known mineral, and all found within our own borders. Within the list are tin, zinc, copper, silver, gold, iron and coal, with unnumbered others; but these mentioned are the principal, and these are the factors which enter largely into all problems of modern progress and civilization; they add to the riches, if not the wealth of any people; and wisely used, they will add to the wealth as well as riches. The central and eastern portions of the Union abound in coal and iron; these give strength and stability to the enterprise and industry of a people. The Rocky Mountain range, in all its length, from its outlying spur reaching through Alaska to Behrings Strait on the north, to the Mexican border on the south, is full of gold and silver. These deposits excite the ambition and stimulate the energies of the people; and it is sure that the fact just stated will help the American people to find a solution to the disgusting problem presented by Mormons. The heart of the Rocky Mountains will not always be dominated by the most virulent enemies of all that is truly American and Christian. The forests, with the endless variety of woods they produce, never made a better showing than at this Exposition. From Maine to California, and from Florida to Dakota, various woods gathered from the plains, the mountains and the swamps show the abundant supply with which the country is provided. The specimens are prepared so that the trunk of the tree, with the bark covering it, the wood showing the grain polished, and varnished and unvarnished, can all be seen at a glance.

To most people of middle age or beyond, the collection of machinery is peculiarly interesting. Young people have no personal knowledge of the extraordinary progress of invention within the last twenty-five or thirty years. Thirty years ago men and women were reaping the ripened grain just as the Greeks and Romans did 700 B. C., and just as the servants of Boaz did on the plains of Bethlehem 1100 B. C., and, in fact, just as Noah and his family did when they raised the first crop after they left the ark. But there has been a revolution in the implements of husbandry. A crooked stick is no longer used as a plow, but in the place of the stick are plows of all shapes and sizes, gold mounted and nickel plated, as ornamental as a parlor

piano. The rude hoe is superseded by all sorts of cultivators. Planting is done by machinery, elaborate, exact, scientific and elegant. The great Daniel Webster when asked as to the best way to hang a scythe replied the best way he had ever found (and he was brought up on a farm) was to hang it over the limb of a tree. If he could see these mowers and the many other machines to make hay, he would conclude that he had reached the millennium as far as hay making is concerned. So, too, the sickle has given way to the machinery drawn by a span of horses, that can almost do the work required on a trot. The machine reaps, gathers up and binds the bundles. Not long ago all threshing was done by tying two straight sticks together with a string, the best string was an eelskin dried and tanned, and then the farmer, in dust and solitude, would pound away at the straw laid out upon the barn floor; but here is a machine that will thresh and winnow wheat as fast as six men can toss in the bundles to the man who feeds, and it will take as many more to remove and stack the straw. And so it is with the whole business of farming. What is true of the processes is equally true of almost every other manual industry. It is a revelation of wonders to walk about amid these exhibits of machinery, and remember that to all intents and purposes the results we behold are the achievements of the last fifty, and in most cases of the last thirty years. And it is equally remarkable that most of these inventions are the offspring of American thought.

It is most natural for every thoughtful person to ask, how is this and why? The ready and superficial answer is that "Necessity is the mother of invention," and that the American people, by the conditions of life surrounding them, have been compelled to invent. But surely such an answer can not be considered satisfactory. There are two events in modern times that no philosopher, physicist, ethnologist or theologian has up to this time fully measured, and much less has been able to estimate their relation to the future of humanity. The first of these events is the vast migration of the peoples of the Old World to the New, by which, within the last sixty years 12,000,000 of human beings, most of them young men and women, have left Europe to make their homes in the United States. God only knows the importance and significance of this movement. The second marvelous event of these days in which we live is the sweep and triumph of invention. It is worth considering that the steam power of the United States represents more than the entire muscular force of all the able-bodied men in the world.

And the improvements in machinery represent immeasurable conquests of mind in the realm of matter. It does not take omniscience to apprehend, to some extent, the fact that these things must affect the destiny of the whole family of mankind. With such thoughts as these in mind one walks amid these minute or ponderous contrivances for the application of power, with something of the reverence and wonder felt by Moses when he stood in the presence of the bush that burned but was not consumed. It is evident that God, the Eternal Ruler of all things, is in the midst of these "flying wheels."

One of the most interesting exhibits is that made of the live stock. The spirited, clean-limbed trotting stock of Kentucky is here. The little Shetland ponies are side by side with the vast Normans. Some of the full grown ponies are so small that a strong man could easily toss one of them to his shoulders, but a Norman that weighs more than 2,000 pounds is altogether a different creature. The Clydesdales may be good for draft horses, but their enormous fetlocks so disfigure their feet and legs as to make them appear homely and uncouth. The Normans and Percherons do not have this disfigurement. They are magnificent in size, some of them black and glossy as anthracite coal, others are deep bay, almost a rich mahogany color, others are dapple gray, from very dark to very light, and two of them are as white as milk. To any one who loves horses this show is worth the travel of a thousand miles. It would make the heart of Rosa Bonheur glad to walk through the stables; and if the finest of the horses could be grouped together under her artistic eye she would have all she could wish for one of her famous pictures. These, or such as these, Job had in mind when he wrote: "Thou hast given the horse strength, thou hast clothed his neck with thunder. He mocketh at fear and is not affrighted, he rejoiceth in his strength."

Nothing less, in every praiseworthy point, is the exhibit of horned cattle. Short Horns, Herefords, Devons, Jerseys, Holsteins, Galloways, vie with each other in size and beauty. One ox weighs 2,990 pounds, and many of them exceed 2,000 pounds. They are thoroughbreds, or carefully crossed, and it is doubtful if finer specimens could be obtained, even in the original habitats of the respective breeds. But I need not write of jacks and jennies, of mules, and sheep, and hogs, they are all here, after their kind, and worthy of admiration for the perfection they display as the result of painstaking skill.

The educational interests are variously represented, and many of the cities and educational societies, and even private or denominational schools find space to show the methods and results of each. The Freedman's Aid Society of the Methodist Episcopal Church makes a creditable display. The same is true of the American Missionary Society, and of several Roman Catholic institutions. The facts, however, seem to show that comparatively little progress has been made in the science of education in the last twenty years. Whether we have reached the ultimatum, so far as methods are concerned, is the question. The child is yet to be born who knows his letters without being taught them. The capital of each, at start, is nothing, and only one thing can be learned at a time, and the human brain is only capable of a certain amount of work. These are some of the limitations a good many educators are inclined to overlook, and yet they will continue to confront all practical people as long as the world stands. Would it not be well, at about this time, for visionary people, with all sorts of educational vagaries, to halt for a little while and inquire if a thorough, plain, fundamental education is not the desideratum for the great majority of the youth of every land. A good part of modern education partakes of the frivolous character of the times. Substantial, honest, common-sense education is vastly better than the illusions and flippancies of sentimental theorists.

Speaking of the Freedman's Aid Society as above, reminds one that the colored people are admitted to participate in the Exposition. Well, the world moves. We are not where we were twenty-five years ago. We are coming up out of the wilderness. Shall we come "full as the moon, clear as the sun, and terrible as an army with banners?" Yes, if we come in God's order. No, if we do not. It always pays to do right. It never pays to do wrong. No curse ever comes causeless. It is sometimes worth remembering that the 7,000,000 colored people in the United States own on an average property to the amount of \$14, and it is not long since they started with nothing. They will send some missionaries to Africa, but most of them will live and die with us, and where we are buried there will they be buried. It is time we recognized the fact that our God is their God. Let us all rejoice that they have a place in the World's Exposition in New Orleans. We need them, they need us. Why not recognize our brotherhood with them, and then together consecrate ourselves to the glorious task of making this land the first and foremost of all the world in the possession and exemplification of all

Christian, and manly, and patriotic graces? And why not join all forces, North, South, East and West in one sublime and divinely led effort to carry the untold blessings of education, morality, freedom, and Christianity to all peoples who still sit amid the shadows of tyranny, superstition, poverty and ignorance? This World's Exposition will reach its highest and grandest legitimate possibilities just in proportion as it shall help forward these desires of all good men and these plans and purposes of the World's Redeemer.

GEOGRAPHY OF THE HEAVENS FOR MARCH.

BY PROF. M. B. GOFF,
Western University of Pennsylvania.

ECLIPSES.

In the early ages of the world eclipses were regarded as alarming deviations from the established laws of nature, presaging great calamities, as famines, pestilences and earthquakes; and among heathen and superstitious peoples, as evidence of the displeasure of the Deity, or deities. Herodotus tells of an eclipse of the sun occurring in 585 B. C., which put an end to a battle between the Medes and Lydians, who were so terrified by the day turning suddenly into night, that the contending armies ceased fighting and concluded a peace which was cemented by a twofold marriage. Another total eclipse of the sun occurred on March 1, 557 B. C., which so terrified the defenders of the Median city Larissa, that they withdrew from its walls, thus permitting it to fall into the hands of its besiegers, the Persians.

Among the Hindoos, it is imagined that the moon, as it covers from sight the face of the sun, is a huge dragon which devours our luminary, and can only be compelled to disgorge and then driven away “by the beating of gongs and rending the air with discordant screams of terror and shouts of vengeance.”

An eclipse of the moon, March 1, 1504, was employed by Columbus to obtain provisions for himself and his starving companions. Having been wrecked on the coast of Jamaica, the natives refused him supplies. Knowing that an eclipse of the moon was about to take place, he informed them that the Great Spirit was displeased with them on account of their ill-treatment of the Spaniards, and would manifest his displeasure by shutting out the light of the moon. When the eclipse occurred, the Indians, terrified by the

sight, hastened to him with abundant supplies, beseeching him to intercede with the Great Spirit in their behalf.

At the present day we look upon these wonderful events as the results of natural causes, whose operations have long since been explained. We have learned that an eclipse of the sun is merely the moon coming between the earth and the sun, thus shutting off from the former all or a portion of the light of the latter; that this event may occur as often as five times, and never less than twice in one year; that it can only occur at time of new moon; that it occurs only in limited portions of the earth at any one time, and hence, that although happening so often, for any given place it is a comparatively rare event—especially the last two of the three kinds, *partial*, *annular*, and *total*; and that the portion of the earth affected by a total eclipse does not exceed 170 miles in diameter; or, in other words, the width of the moon's shadow, when it falls perpendicularly on the earth's surface, is not more than 170 miles. We have learned, also, that an eclipse of the moon is occasioned by the earth coming between the moon and the sun; that this event can not occur more than twice in any one year, and may not occur even once; that it happens always at full moon; that it can be seen in all parts of the earth where the moon is above the horizon at the time of the occurrence; and for this reason, although it only happens in the ratio of 29 to 41 as compared with eclipses of the sun, yet there are more lunar than solar eclipses visible in any given place.

During the present month we shall have two eclipses, one of the sun and one of the moon.

THE SUN

Will be eclipsed on the 16th, first contact taking place in longitude $136^{\circ} 49.3'$ west and latitude $13^{\circ} 25.3'$ north, at 8:26 p. m., Washington mean time; and the last contact in longitude $32^{\circ} 58.3'$ west and latitude $49^{\circ} 0.8'$ north, at 1:22 a. m. on the 17th. The central eclipse will begin in longitude $156^{\circ} 39.5'$ west and latitude $35^{\circ} 54.5'$ north at 9:48 p. m., and end in longitude $15^{\circ} 4.6'$ west, latitude $71^{\circ} 24.1'$ north, at 12:00, midnight. The path of the central eclipse in North America will be about 35 miles wide, and will take a northeasterly course from a point near Cape Mendocino on the western coast of California, and will embrace Weaverville, Cal., Idaho

and Boise cities in Idaho; Bannock City and Gallatin, Montana; cross the boundary line between the United States and the British Possessions in longitude about 105° west; pass through the central part of Hudson's Bay; cross Hudson Strait, Davis's Strait, and Greenland, ending as above in longitude $15^{\circ} 4.6'$ west, latitude $71^{\circ} 24.1'$ north. As this is an annular eclipse, the shadow of the moon being too short to reach the earth, parties located in the path named will see the edge of the sun like a bright ring around the dark shadow of the moon. Persons outside of this path will see the sun more or less eclipsed, dependent on their position. The beginning and end of the eclipse at a number of places in the United States is given below, in the local time of the cities mentioned: Bangor, Me., begins at 12:48 p. m., ends at 3:23 p. m.; Concord, N. H., begins at 12:32 and ends at 3:07 p. m.; at Montpelier, Vt., lasts from 12:26 to 3:03 p. m.; Boston, Mass., from 12:36 to 3:09 p. m.; Providence, R. I., from 12:33 to 3:05 p. m.; Hartford, Conn., from 12:25 to 2:58 p. m.; New York, 12:17 to 2:50 p. m.; Trenton, N. J., 12:13 to 2:45; Pittsburgh, Pa., 11:38 a. m. to 2:18 p. m.; Wilmington, Del., from 12:07 to 2:40 p. m.; Charleston, S. C., from 11:37 a. m. to 2:04 p. m.; Columbus, Ohio, 11:20 a. m. to 2:01 p. m.; Detroit, Mich., 11:21 a. m. to 2:04 p. m.; Indianapolis, Ind., 11:02 a. m. to 1:45 p. m.; Chicago, Ill., 10:55 a. m. to 1:40 p. m.; Jefferson City, Mo., 10:24 a. m. to 1:09 p. m.; Lawrence, Kan., 10:07 a. m. to 12:52 p. m.; Omaha, Neb., 10:04 a. m. to 12:51 p. m.; St. Paul, Minn., 10:26 a. m. to 1:13 p. m.; Des Moines, Ia., 10:18 a. m. to 1:04 p. m.; Janesville, Wis., 10:47 a. m. to 1:33 p. m.; Santa Fé, New Mex., 8:59 to 11:49 a. m.; Wheeling, W. Va., 11:32 a. m. to 2:13 p. m.; Washington, D. C., 11:58 a. m. to 2:31 p. m.; Louisville, Ky., 11:03 a. m. to 1:44 p. m.; Denver, Col., 9:10 a. m. to 12:01 p. m.; Bismarck, Dakota, 9:44 a. m. to 12:33 p. m.; New Orleans, La., 10:28 a. m. to 1:08 p. m. Our usual notes for the sun are as follows: Rises on the 1st at 6:33; on the 16th, at 6:09; and on the 30th, at 5:45 a. m.; and sets on the corresponding days at 5:51, 6:10 and 6:22 p. m. respectively. Spring begins on the 20th at 5:21 a. m.; northward movement, $12^{\circ} 6'$.

THE MOON

Will be partially eclipsed on the 30th, entering the earth's shadow at 9:50 a. m. and leaving it at 1:02 p. m. Magnitude of the eclipse, .886. As the moon does not rise with us on this date till between 6:00 and 7:00 p. m. it is

evident that the eclipse will not be visible in the United States. It will be visible, however, in the western Pacific Ocean, Asia, and the eastern portions of Europe and Africa. On the 1st, moon rises at 6:42 p. m.; on the 15th, at 5:29 a. m.; and on the 31st, at 7:39 p. m. It presents the following phases: Last quarter, 8th, 1:46 p. m.; new moon, 16th, 12:28 p. m.; first quarter, 23d, 12:15 p. m.; full moon, 30th, 11:32 a. m. Farthest from earth, 9th, 3:12 p. m.; nearest earth, 23d, 3:54 p. m.; least elevation, 9th, $30^{\circ} 17' 23''$; greatest elevation, 23d, $66^{\circ} 41' 16''$ (in latitude $41^{\circ} 30'$ north).

MERCURY

Has a direct motion of $52^{\circ} 59' 4''$; increase in diameter, one second; on 7th, at 9:00 a. m., $1^{\circ} 3'$ south of Mars; 13th, at 1:00 p. m., in superior conjunction with the sun; 16th, at 8:02 p. m., $1^{\circ} 37'$ south of the moon; 28th, at 4:00 a. m., nearest the sun. On the 1st, 16th and 30th, rises at 6:23, 6:26 and 6:22 a. m. respectively; and sets on same days at 4:51, 6:16 and 7:42 p. m. Can be seen with naked eye on the last few evenings of the month.

VENUS

Continues as morning star throughout the month, but makes little display, both on account of her distance from us and her proximity to the sun. Her diameter diminishes from $10.6''$ to $10''$, and her time of rising is as follows: On the 1st, 6:02 a. m.; on the 16th, 5:51 a. m.; on the 30th, 5:36 a. m.; on the 6th, at 6:00 a. m., she is farthest from the sun; on 15th, at 1:42 p. m., $3^{\circ} 32'$ south of moon; 27th, at 10:00 p. m., $36'$ south of Mars. Her motion is direct and equals $37^{\circ} 23' 30''$.

MARS

Rises on the 1st at 6:32 a. m. and sets at 5:24 p. m.; on the 16th, rises at 6:00 a. m. and sets at 5:28 p. m.; on the 30th, rises at 5:33 a. m., sets at 5:25 p. m. Motion direct and amounts to $22^{\circ} 25' 38''$; diameter, $4.2''$; on 7th, at 9:00 a. m., $1^{\circ} 3'$ north of Mercury; 16th, at 12:50 a. m., $2^{\circ} 34'$ south of moon; 27th, at 10:00 p. m., $36'$ north of Venus.

JUPITER

Lessens his diameter two seconds, and makes a retrograde motion of $2^{\circ} 56'$. On 27th, at 9:57 a. m., is $4^{\circ} 40'$ north of the moon. He rises on the 1st at 4:42 p. m. and sets on the 2d at 6:07 a. m.; rises on the 16th at 3:35 p. m. and sets on the 17th at 5:09 a. m.; rises on the 30th at 2:34 p. m. and sets on the 31st at 4:10 a. m.

SATURN,

As a telescopic object, is still improving, and his time of setting permits him to be viewed with less than usual inconvenience. On the 2d he sets at 1:44 a. m.; on the 17th at 12:49 a. m.; and on the 30th at 11:55 p. m., affording thus all the evening for observations. On the 7th, at 3:00 p. m., he is “in quartile,” or 90° east of the sun; on 22d, at 10:28 a. m., $3^{\circ} 56'$ north of the moon.

URANUS

Retrogrades $1^{\circ} 12' 23''$; his diameter remains at $3.8''$; on 2d, at 11:59 a. m., he is $1^{\circ} 6'$ north of moon; 21st, at 3:00 a. m., in opposition to the sun (on the other side of the sun from the earth); 29th, at 7:05 p. m., $1^{\circ} 13'$ north of moon; sets on the 1st at 7:31 a. m.; on the 16th, at 6:30 a. m.; on the 31st, at 5:30 a. m. Morning star till the 21st; after that evening star.

NEPTUNE,

With his diameter of $2.6''$ moves some $44' 46''$ of arc in his orbit, which is not so slow after all when we consider that his average absolute motion is 3.36 miles per second, and that his aggregate for the 31 days of this month is a little less than nine million miles. His right ascension on the 1st is 3 hours, 15 minutes, 18 seconds, and his declination $16^{\circ} 17' 57''$ north. He sets on the 1st at 11:33 p. m.; on the 16th at 10:29 p. m.; and on the 30th at 9:43 p. m.—an evening star.

HOW TO WIN.

BY FRANCES E. WILLARD,
President National W. C. T. U.

CHAPTER I.

Long ago, and long ago it was, in the days when I used proudly to write "School Teacher" after my name, I bought a certain book for the express purpose of reading it to "the girls I've left behind me." The book is one beloved by train boys, of which they and other venders have sold so many that the latest "dodgers" read, "Twentieth thousand now in press." It is sensible in matter, attractive in style, and goes by the enticing name of "Getting on in the World." Naturally enough it was written in Chicago, and like most "Garden City" notions, is "a success." But the trouble with this volume was that it didn't fill the bill. I wanted to read it to "my girls," to stir up their pure minds by way of remembrance that "life is real, life is earnest," and the rest of it. But as I scanned its bright and pleasant pages I found out—what do you think I found? Why, that with the light of a new dispensation blazing in upon him, and the soprano voices of several million "superfluous women," crying, "Have you no *work* for me to do?" this honored author had written never a word about creation's gentler half! His book contained 365 pages, but if you had read a page each day, all the year round, you wouldn't have found out at last that such a being as a woman was trying to "get on" in this or any other world. Not a bread-winning weapon had he put into the hand of the neediest among us, nor had he, even in a stray chapter or "appendix," taken us off by ourselves and drawn us a diagram of "our sphere."

I was so pained by this that I wrote Prof. Matthews (the gifted author, and my personal friend), asking him why he had thus counted out the women folks in his book upon success in life. I even ventured to hypothecate his reason, saying to him:

“DEAR SIR:—I do not think you did this with malice aforethought, or from lack of interest in our fate, but simply and only because, like so many of our excellent brethren, you ‘done forgot all about us,’ as *Topsey* would say.”

Whereupon came a prompt and gracious reply, with the frank and manly admission:

“You guessed aright. I simply forgot to speak of women.”

Now, you perceive, it set me thinking—this obliquity of mental vision, which had led a writer so talented and wise to squint thus at the human race, seeing but half of it. I recalled the fact that, into most families, are born girls as well as boys; nay, as many an over-burdened *pater familias* can testify, they come not unfrequently in largely superior, if not exclusive numbers. Having, also, at a remote period of my history, belonged to the same helpless fraternity, I was haunted by the wish that I might write a sequel to the Professor’s excellent book, talking therein to girls and women about success in life. Perhaps my time has come; perhaps, in the generous pages of *THE CHAUTAUQUAN*, whose editor is so tolerant of the “strong minded” sisterhood, I have the largest audience that has yet consented to listen to my “views.” Anyhow, I mean, in these newly acquired pages to talk to girls of “How to Win” in something besides the sense treated of in books of etiquette and fashion magazines, or systematically taught in dancing schools.

And now, my dears, if you are patient and my small assistant keeps me in lead pencils, I shall try to show that if every young woman held in her firm little hand her own best gift, duly cultivated and made effective, society would not explode, the moon would not be darkened, the sun would still shed light. Somehow, dear girls, when I see an audience of young men, they remind me of a platoon of soldiers, marching with fixed bayonet, to the capture of their destiny. An assembly of young women, on the other hand, recalls a flock of lambs upon a pleasant hillside. They frisk about and nibble at the herbage and lie down in the sun, while above them soars the devouring eagle of their destiny, sweeping in concentric rings through the blue air, and ready to pounce down upon them, while the meek little innocents turn their white faces upward and mildly wonder “what that

graceful creature is up yonder?" They remind me, too, of the reply given by a bright young friend of mine to the solemn exhortation that she should "make the most of life."

"Humph!" she exclaimed with a rueful grimace, "I have no chance, for life is busy making the most of me!"

The trouble is, we women have all along been set down on the world's program for a part so different from the one we really play upon its stage. For instance, the program reads: "Woman will take the part of Queen in the Drama of Society," but often times, before the curtain falls, the stage reveals her as a dressmaker, a school teacher, perchance that most abused of mortals, a reformer! The program reads: "This august actress will be escorted to the stage by Man, her loyal and devoted subject, to whom has been assigned the part of shielding her from the glare of the footlights, and shooting anybody in the audience who dares to hiss." But, alas! oftentimes the stage reveals her coming in alone, dragging her own sewing machine, while her humble and devoted subject, with tailor's goose in one hand and scissors in the other, indicates by energetic pantomime his fixed intention to drive her speedily behind the scenes. The program, my beloved innocents, attires you all in purple and fine linen and bids you fare sumptuously every day, but not infrequently the stage reveals you attired in calico gowns, and munching your hard-earned crackers and cheese. The world's theory furnishes every young lady that draws breath, with a lover, loyal and true, but the world's practice shoots him on the battlefield, or poisons him with alcohol and nicotine until he can only "rattle around" through life in the place God meant him to fill within home's sacred sanctuary. It is just this discrepancy that I complain of, and the generous age we live in is complaining of it with a thousand tongues, so that "the logic of events" that happen, instead of events that ought to happen, is impelling toward nobler fortunes that phenomenal creature whom a French author has called "the poor woman of the nineteenth century."

Naturally enough, in thinking over the "case," I contrast your aims in life with what were once my aims, your outlook upon life with mine. The other day—a rainy one, you may be sure—I brought from the vasty deep of the family garret some of my girlish journals, which I was curious to compare with the diary of a friend and former pupil at Evanston. Let me give you a

few parallel passages because of the lesson they teach. My pupil (aged sixteen) writes thus:

“Was registered this day a member of the Freshman class in the Northwestern University. The president advises me to take the classical course, and I’ve made up my mind to try it.”

From mine at fifteen years I read:

“Caught a blue jay in my trap out in the hazel thicket. I knew he wasn’t “game” and let him go. The school house in our district is finished at last. A graduate of Yale College, and former tutor at Oberlin, is to be our teacher. I shall attend regularly, visiting my traps on the way.”

Later:

“Sister and I got up long before light to prepare for the first day at school. We put all our books in mother’s satchel; had a nice tin pail full of dinner. I study arithmetic, geography, grammar, reading and spelling, which takes up every minute of my time. Stood next to Pat O’Donahue in spelling, and Pat stood at the head.”

From my pupil’s diary, a few months later, take this extract:

“I am thinking seriously about my future. Perhaps this is premature, for I am only in my freshman year, but I have just about decided that I’ll study medicine.”

From mine, at a similar age (you see precocity was not among my failings):

“Sister was sick, and I brought out all my little bottles of sugar, salt and flour. Besides these medicines, I dosed her with pimientoes and poulticed her with cabbage leaves, but she grew no better, quite fast, so mother called another doctor. Dear me, if I were my brother, instead of being only a girl, we’d soon see whether I’ve a talent for medicine or not.”

From my young friend I quote again:

“I am greatly interested in the question for debate in our literary society this week, especially as I am chief disputant on the affirmative. It reads as follows: *Resolved*, That the votes of women are needed to help put down the liquor traffic.”

From mine:

“It is election day and my brother is twenty-one years old. How proud he seemed as he dressed up in his best clothes and drove off with father to vote for John C. Fremont, like the sensible ‘Free Soiler’ that he is! My sister and I stood at the front window and looked out after them. Somehow I felt a lump in my throat, and then I couldn’t see their wagon any more, things looked so blurred. I turned to Mary, and she, dear little innocent, seemed wonderfully sober, too. I said: ‘Wouldn’t you like to vote as well as Oliver? Don’t you and I love the country just as well as he, and doesn’t the country need our ballots?’ Whereupon she looked scared, but answered, ‘Of course we do, but don’t you go ahead and say so, for then we should be called strong minded.’”

From my pupil at seventeen I quote once more:

“The recent articles by members of the ‘Women’s Congress,’ some people would call radical, but they express precisely my opinions on the dress question. It is time for me to assume the garb of a young lady, but upon two things I am determined: First, I will never trail my garments on a filthy pavement while I live. If I am the only young lady in this university, who, when she walks, wears walking costume, I will still be true to my individual sense of cleanliness and taste. I will also carry the jewel of an *unpunctured ear* through life, though, by so doing, I oblige Mr. Darwin to confess ‘a missing link’ between me and my evolutionary ancestors.”

Finally, from mine:

“This is my seventeenth birthday, and the date of my martyrdom. Mother insists that at last I *must* have my hair ‘done

up woman fashion.' She says she can hardly forgive herself for letting me 'run wild' so long. We had a great time over it all, and here I sit, like another Samson, 'shorn of my strength.' That figure won't do, though, for the greatest trouble with me is that I never shall be shorn again! My 'back hair' is twisted up like a corkscrew; I carry eighteen hair-pins; my head aches, my feet are entangled in the skirt of my new gown. I can never jump over a fence again so long as I live. As for chasing the sheep down in the shady pasture, it's out of the question, and to climb to my 'Eagle's Nest' seat in the big burr oak would ruin this new frock beyond repair. Altogether, I recognize the fact that 'my occupation's gone.'"

My readers smile at this, but they may be assured there are such blots upon the page where it was written, as briny drops alone can make.

You see, dear friends, from this contrast I have drawn, showing a glimpse of past and future in two eager, young lives, how fast this world is getting on. What is the difference in the outlook of your life that is, and mine that used to be? Let us consider: I was a daring sort of girl; you are the sort of girls who dare. I had aspiration; you have opportunity. I breathed an atmosphere laden with old time conservatisms, from which my glorious mother's liberality of soul was my one safety valve of deliverance. But you are exhilarated by the vital air of a new liberty. "The world is all before you, where to choose." If I required but little of myself, it was because the world required so little of me. No college of first rank in east or west—save noble old Oberlin and generous Antioch—could have been coaxed to count me in when she made up her jewels. Briefly, public opinion proposes to give you a chance. It proposed to let me shirk for myself. It means to put a shield in your left hand and a sword in your right. It let me go forth, as best I could, to beat the air with unarmed hands, or to sharpen my weapons on the field and in plain sight of the enemy.

Society set before me very few incentives, and commended to me only the passive virtues. Indeed, she never really bestirred herself on my behalf at all, save that she ceased not in story and poem, by sermon and song by precept and example, and (most cogently of all) by setting no other hope before me to ground me, so far as she was able, in the philosophy that

sustained the illustrious *Micawber*. “Now my daughter,” thus was she wont to speak, “do you but be docile and obedient, as a young woman should, and something, something very particular indeed will most assuredly turn up.”

But I learned early to distrust a Mentor who took so little cognizance of the imperious ardor of my youth; who was so stupidly oblivious of the varied possibilities in brain and hand and heart, and so I began early to follow out my own devices as to a plan of character and work. Would that the generous impulse of your enthusiasm, guided by your broader opportunity, might

“Give me back the wild pulsation
That I felt before the strife,
When I heard my days before me,
And the tumult of my life.”

More anon.

EVANSTON, January 31, 1885.

NOTES ON POPULAR ENGLISH.

BY ISAAC TODHUNTER.

I have from time to time recorded such examples of language as struck me for inaccuracy or any other peculiarity; but lately the pressure of other engagements has prevented me from continuing my collection, and has compelled me to renounce the design once entertained of using them for the foundation of a systematic essay. The present article contains a small selection from my store, and may be of interest to all who value accuracy and clearness. It is only necessary to say that the examples are not fabricated; all are taken from writers of good repute, and notes of the original places have been preserved, though it has not been thought necessary to encumber these pages with references. The italics have been supplied in those cases where they are used.

One of the most obvious peculiarities at present to be noticed is the use of the word *if* when there is nothing really conditional in the sentence. Thus we read: "If the Prussian plan of operations was faulty, the movements of the crown prince's army were in a high degree excellent." The writer does not really mean what his words seem to imply, that the excellence was contingent on the fault; he simply means to make two independent statements. As another example we have: "Yet he never founded a family; if his two daughters carried his name and blood into the families of the *Herreras* and the *Zuñigos*, his two sons died before him." Here again the two events which are connected by the conditional *if* are really quite independent. Other examples follow: "If it be true that Paris is an American's paradise, symptoms are not wanting that there are Parisians who cast a longing look toward the institutions of the United States."

Other examples, differing in some respects from those already given, concur in exhibiting a strange use of the word *if*. Thus we read: "If a big book is a big evil, the 'Bijou Gazetteer of the World' ought to stand at the summit of excellence. It is the tiniest geographical directory we have ever

seen.” This is quite illogical; if a big book is a big evil, it does not follow that a little book is a great good. “If in the main I have adhered to the English version, it has been from the conviction that our translators were in the right.” It is rather difficult to see what is the precise opinion here expressed as to our translators; whether an absolute or contingent approval is intended. For the last example we take this: “... but if it does not retard his return to office it can hardly accelerate it.” The meaning is, “This speech can not accelerate and may retard Mr. Disraeli’s return to office.” The triple occurrence of *it* is very awkward.

An error not uncommon in the present day is the blending of two different constructions in one sentence. The grammars of our childhood used to condemn such a sentence as this: “He was more beloved but not so much admired as Cynthio.” The former part of the sentence requires to be followed by *than*, and not by *as*. The following are recent examples: “The little farmer (in France) has no greater enjoyments, if so many, as the English laborer.” “I find public school boys generally more fluent, and as superficial as boys educated elsewhere.” “Mallet, for instance, records his delight and wonder at the Alps and the descent into Italy in terms quite as warm, if much less profuse, as those of the most impressible modern tourist.” An awkward construction, almost as bad as a fault, is seen in the following sentence: “Messrs having secured the coöperation of some of the most eminent professors of, and writers on, the various branches of science....”

A very favorite practice is that of changing a word where there is no corresponding change of meaning. Take the following example from a voluminous historian: “Huge pinnacles of bare rock shoot up into the azure firmament, and forests overspread their sides, in which the scarlet rhododendrons sixty feet in *height* are surmounted by trees two hundred feet in *elevation*.” In a passage of this kind it may be of little consequence whether a word is retained or changed; but for any purpose where precision is valuable it is nearly as bad to use two words in one sense as one word in two senses. Let us take some other examples. We read in the usual channels of information that “Mr. Gladstone has issued invitations for a full-dress Parliamentary *dinner*, and Lord Granville has issued invitations for a full-dress Parliamentary *banquet*.” Again we read: “The government proposes to divide the occupiers of land into four categories;” and almost immediately

after we have “the second class comprehends...”: so that we see the grand word *category* merely stands for *class*. A writer recently in a sketch of travels spoke of a “Turkish gentleman with his *innumerable* wives,” and soon after said that she “never saw him address any of his *multifarious* wives.” One of the illustrated periodicals gave a picture of an event in recent French history, entitled “The National Guards Firing on the People.” Here the change from *national* to *people* slightly conceals the strange contradiction of guardians firing on those whom they ought to guard.

Let us now take one example in which a word is repeated, but in a rather different sense: “The grand duke of Baden sat *next* to the emperor William, the imperial crown prince of Germany sitting *next* to the grand duke. *Next* came the other princely personages.” The word *next* is used in the last instance in not quite the same sense as in the former two instances; for all the princely personages could not sit in contact with the crown prince.

A class of examples may be found in which there is an obvious incongruity between two of the words which occur. Thus, “We are more than doubtful;” that is, we are *more than full* of doubts: this is obviously impossible. Then we read of “a man of more than doubtful sanity.” Again we read of “a more than questionable statement;” this is I suppose a very harsh elliptical construction for such a sentence as “a statement to which we might apply an epithet more condemnatory than *questionable*.” So also we read “a more unobjectionable character.” Again: “Let the Second Chamber be composed of elected members, and their utility will be *more than halved*.” To take the *half* of anything is to perform a definite operation, which is not susceptible of more or less. Again: “The singular and almost excessive impartiality and power of appreciation.” It is impossible to conceive of *excessive impartiality*. Other recent examples of these impossible combinations are, “more faultless,” “less indisputable.” “The high antiquity of the narrative can not reasonably be doubted, and almost as little its *ultimate Apostolic origin*.” The ultimate origin, that is the *last beginning*, of anything seems a contradiction. The common phrase *bad health* seems of the same character; it is almost equivalent to *unsound soundness* or to *unprosperous prosperity*. In a passage already quoted, we read that the czar “gave *audience* to numerous *visitors*,” and in a similar manner a very distinguished lecturer speaks of making experiments “*visible* to a large *audience*.” It would seem from the last instance that our language

wants a word to denote a mass of people collected not so much to hear an address as to see what are called experiments. Perhaps if our savage forefathers had enjoyed the advantages of courses of scientific lectures, the vocabulary would be supplied with the missing word.

Talented is a vile barbarism which Coleridge indignantly denounced; there is no verb *to talent* from which such a participle could be deduced. Perhaps this imaginary word is not common at the present; though I am sorry to see from my notes that it still finds favor with classical scholars. [Webster says: “This word—which is said to be of American origin—has been strongly objected to by Coleridge and some other critics, but as it would seem, upon not very good grounds, as the use of *talent* or *talents* to signify mental ability, although at first merely metaphorical, is now fully established, and *talented*, as a formative, is just as analogical and legitimate, as *gifted*, *bigoted*, *turreted*, *targeted*, and numerous other adjectives having a participial form, but derived directly from nouns, and not from verbs.”—ED. THE CHAUTAUQUAN.]

Ignore is a very popular and a very bad word. As there is no good authority for it, the meaning is naturally uncertain. It seems to fluctuate between *wilfully concealing* something and *unintentionally omitting* something, and this vagueness renders it a convenient tool for an unscrupulous orator or writer.

The word *lengthened* is often used instead of *long*. Thus we read that such and such an orator made a *lengthened* speech, when the intended meaning is that he made a *long* speech. The word *lengthened* has its appropriate meaning. Thus, after a ship has been built by the Admiralty, it is sometimes cut into two and a piece inserted; this operation, very reprehensible doubtless on financial grounds, is correctly described as *lengthening* the ship. It will be obvious on consideration that *lengthened* is not synonymous with *long*. *Protracted* and *prolonged* are also often used instead of *long*; though perhaps with less decided impropriety than *lengthened*.

A very common phrase with controversial writers is, “we *shrewdly* suspect.” This is equivalent to, “we *acutely* suspect.” The cleverness of the

suspicion should, however, be attributed to the writers by other people, and not by themselves.

The simple word *but* is often used when it is difficult to see any shade of opposition or contrast such as we naturally expect. Thus we read: "There were several candidates, *but* the choice fell upon of Trinity College." Another account of the same transaction was expressed thus: "It was understood that there were several candidates; the election fell, *however*, upon of Trinity College."

The word *mistaken* is curious as being constantly used in a sense directly contrary to that which, according to its formation, it ought to have. Thus: "He is often mistaken, but never trivial and insipid." "He is often mistaken" ought to mean that other people often mistake him; just as "he is often misunderstood" means that people often misunderstand him. But the writer of the above sentence intends to say that "He often makes mistakes." It would be well if we could get rid of this anomalous use of the word *mistaken*. I suppose that *wrong* or *erroneous* would always suffice. But I must admit that good writers do employ *mistaken* in the sense which seems contrary to analogy; for example, Dugald Stewart does so, and also a distinguished leading philosopher whose style shows decided traces of Dugald Stewart's influence.

I should like to ask why a first charge is called a *primary* charge, for it does not appear that this mode of expression is continued. We have, I think, second, third, and so on, instead of *secondary*, *tertiary*, and so on, to distinguish the subsequent charges.

Cobbett justly blamed the practice of putting "&c." to save the trouble of completing a sentence properly. In mathematical writings this symbol may be tolerated because it generally involves no ambiguity, but is used merely as an abbreviation, the meaning of which is obvious from the context. But in other works there is frequently no clue to guide us in affixing a meaning to the symbol, and we can only interpret its presence as a sign that something has been omitted. The following is an example: "It describes a portion of Hellenic philosophy; it dwells upon eminent individuals, inquiring, theorizing, reasoning, confuting, &c., as contrasted with those

collective political and social manifestations which form the matter of history....”

A recent cabinet minister described the error of an Indian official in these words: “He remained too long under the influence of the views which he had imbibed from the board.” To imbibe a view seems strange, but to imbibe anything from a board must be very difficult. I may observe that the phrase of Castlereagh’s which is now best known, seems to suffer from misquotation; we usually have “an ignorant impatience of taxation;” but the original form appears to have been, “an ignorant impatience of the relaxation of taxation.”

The following sentence is from a voluminous historian: “The *decline* of the material comforts of the working classes, from the effects of the Revolution, had been incessant, and had now reached an alarming *height*.” It is possible to ascend to an alarming height, but it is surely difficult to decline to an alarming height.

“Nothing could be more one-sided than the point of view adopted by the speakers.” It is very strange to speak of a point as having a side; and then how can *one-sided* admit of comparison? A thing either has one side or it has not; there can not be degrees in one-sidedness. However, even mathematicians do not always manage the word *point* correctly. In a modern valuable work we read of “a more extended point of view,” though we know that a point does not admit of extension. I suppose that what is meant is, a point which commands a more extended view. “Froschammer wishes to approach the subject from a philosophical standpoint.” It is impossible to *stand* and yet to *approach*. Either he should *survey* the subject from a *stand-point*, or *approach* it from a *starting-point*.

A large school had lately fallen into difficulties owing to internal dissensions; in the report of a council on the subject it was stated that measures had been taken to *introduce more harmony and good feeling*. The word *introduce* suggests the idea that harmony and good feeling could be laid on like water or gas by proper mechanical adjustment, or could be supplied like first-class furniture by a London upholsterer.

A passage has been quoted with approbation by more than one critic from the late Professor Conington’s translation of Horace, in which the

following line occurs:—

After life's endless babble they sleep well.

Now the word *endless* here is extremely awkward; for if the babble never ends, how can anything come after it?

To digress for a moment, I may observe that this line gives a good illustration of the process by which what is called Latin verse is often constructed. Every person sees that the line is formed out of Shakspeare's "After life's fitful fever he sleeps well." The ingenuity of the transference may be admired, but it seems to me that it is easy to give more than a due amount of admiration; and, as the instance shows, the adaptation may issue in something bordering on the absurd.

The language of the shop and the market must not be expected to be very exact: we may be content to be amused by some of its peculiarities. I can not say that I have seen the statement which is said to have appeared in the following form: "Dead pigs are looking up." We find very frequently advertised, "*Digestive* biscuits"—perhaps *digestible* biscuits are meant. In a catalogue of books an "Encyclopædia of Mental Science" is advertised; and after the names of the authors we read, "invaluable, 5s. 6d.;" this is a curious explanation of *invaluable*.

The title of a book recently advertised is, "Thoughts for those who are Thoughtful." It might seem superfluous, not to say impossible, to supply thoughts to those who are already full of thought.

The word *limited* is at present very popular in the domain of commerce. Thus we read, "Although the space given to us was limited." This we can readily suppose; for in a finite building there can not be unlimited space. Booksellers can perhaps say, without impropriety, that a "limited number will be printed," as this may only imply that the type will be broken up; but they sometimes tell us that "a limited number *was* printed," and this is an obvious truism.

Some pills used to be advertised for the use of the "possessor of pains in the back," the advertisement being accompanied with a large picture representing the unhappy capitalist tormented by his property.

Pronouns, which are troublesome to all writers of English, are especially embarrassing to the authors of prospectuses and advertisements. A wine company return thanks to their friends, “and, at the same time, *they* would assure *them* that it is *their* constant study not only to find improvements for *their* convenience....” Observe how the pronouns oscillate in their application between the company and their friends.

In selecting titles of books there is room for improvement. Thus, a *Quarterly Journal* is not uncommon; the words strictly are suggestive of a *Quarterly Daily* publication. I remember, some years since, observing a notice that a certain obscure society proposed to celebrate its *triennial anniversary*.

A few words may be given to some popular misquotations.

“He that runs may read” is often supposed to be a quotation from the Bible; the words really are, “He may run that readeth,” and it is not certain that the sense conveyed by the popular misquotation is correct.

A proverb which correctly runs thus: “The road to hell is paved with good intentions,” is often quoted in the far less expressive form, “Hell is paved with good intentions.”

“Knowledge is power” is frequently attributed to Bacon, in spite of Lord Lytton’s challenge that the words can not be found in Bacon’s writings.

It seems impossible to prevent writers from using *cui bono?* in the unclassical sense. The correct meaning is known to be of this nature: suppose that a crime has been committed; then inquire who has gained by the crime—*cui bono?* for obviously there is a probability that the person benefited was the criminal. The usual sense implied by the quotation is this: What is the good? the question being applied to whatever is for the moment the object of depreciation. Those who use the words incorrectly may, however, shelter themselves under the great name of Leibnitz, for he takes them in the popular sense; see his works, vol. v., p. 206.

The *Times*, commenting on the slovenly composition of the queen’s speeches to Parliament, proposed the cause of the fact as a fit subject for the investigation of our *professional thinkers*. The phrase suggests a delicate reproof to those who assume for themselves the title of *thinker*, implying

that any person may engage in this occupation just as he might, if he pleased, become a dentist, or a stockbroker, or a civil engineer. The word *thinker* is very common as a name of respect in the works of a modern distinguished philosopher. I am afraid, however, that it is employed by him principally as synonymous with a *Comtist*.

The *Times*, in advocating the claims of a literary man for a pension, said, "He has *constructed* several useful schoolbooks." The word *construct* suggests with great neatness the nature of the process by which schoolbooks are sometimes evolved, implying the presence of the bricklayer and mason rather than of the architect.

[Dr. Todhunter might have added *feature* to the list of words abusively used by newspaper writers. In one number of a magazine two examples occur: "A *feature* which had been well *taken up* by local and other manufacturers was the exhibition of honey in various applied forms." "A new *feature* in the social arrangements of the Central Radical Club *took place* the other evening."]—*Macmillan's Magazine*.

THE CHAUTAUQUA SCHOOL OF LIBERAL ARTS.

BY CHANCELLOR J. H. VINCENT, D.D.

Beyond the “Inner Circle,” which leads to the “Upper Chautauqua,” we come to the Uppermost Chautauqua—the University proper, with its “School of Liberal Arts,” and its “School of Theology.” Here we find provision made for college training of a thorough sort. Students all over the world may turn their homes into dormitories, refectories, and study rooms, in connection with the great University which has its local habitation at Chautauqua. Thus “hearers” and “recipients” in the Assembly, “readers” in the C. L. S. C., “student readers” in the “inner circle”—the “League of the Round-Table,” may go beyond, even to the School of Liberal Arts, the *bona fide* College of Chautauqua.

Chautauqua exalts the college. She believes that the benefits of a college training are manifold.

1. The action by which a youth becomes a college student—the simple going forth—leaving one set of circumstances and voluntarily entering another, with a specific purpose—is an action which has educating influence in it. It is a distinct recognition of an object and a deliberate effort to secure it. The judgment is convinced, the will makes a decision, and corresponding action follows. We have the thought, the aim, the standards, the resolve, the surrender, and the embodiment of all in an actual physical movement. There must follow these activities a reflex influence on the youth himself. It becomes a “new birth” in his life. He has gone to another plane. His everyday conduct is modified by it. He looks up and on. According to the standard he has set, the idea he entertains of education, and the motives which impel him will be the subjective effects of his action—the real power of his new life.

2. There is educating power in the complete plan of study provided in the college curriculum, covering as it does the wide world of thought, distributed over the years, with subdivisions into terms, with specific assignments of subjects, with a beginning and an ending of each division, and many beginnings and endings, with promotions according to merit, and final reviews, recognitions, and honors. There is great value in the enforced system of the college. It tends to sustain and confirm new life, begun when the student made his first movement toward an institution.

3. The association of students in college life is another educating factor. Mind meets mind in a fellowship of aim, purpose, and experience. They have left the same world; they now together enter another world. They look up to the heights and to the shining of crowns which await the gifted and faithful. They are brothers now—one “alma mater” to nourish them. They sing their songs—songs which, although without much sense, have power to awake and foster sympathy. Even a man of sense loves to listen to them. He laughs at the folly, and, though himself a sage, wishes he were one of the company of singers. The laws of affinity work out. Soul inspires soul. Memories grow apace. Attachments that endure, adventures seasoned with fun or touched with sadness, absurdities, failures, heroisms, triumphs, are crowded into the four years, and like fruitage of bloom and fragrance from a conservatory may go forth to bless many an hour of wandering, of sorrow, of reunion, of remembrance, in the later years. There was something pathetic in the return of the famous Yale College class of 1853 to their alma mater two summers ago. As they wandered about the scenes of their youth, under the old elms, through recitation rooms and chapel, singing the old songs, reviving the old friendships, recalling faces to be seen no more, no wonder that tears fell down furrowed cheeks from eyes unused to weep. Is there any stronger or sweeter friendship than that born under the ivied towers and spreading elms of college hall and campus?

In college mind meets mind in the severe competition of recitation and annual examination. The bright boy—one of a small class at home, who had it all his own way there—now finds a score or more of leaders whose unvoiced challenge he is compelled to accept, and how he does knit his brow, close his eyes, summon his strength, school his will, force his flagging energies, and grapple problems that he may hold his own, outstrip

his rivals, and win prize and place for the sake of his family's fame and for his personal satisfaction!

There is nothing that so discovers to a youth the weak points of his character as the association of college life. There are no wasted courtesies among students. Folly is soon detected, and by blunt speech, bold caricature, and merciless satire exposed. Sensitiveness is cured by ridicule, cowardice never condoned, and meanness branded beyond the possibility of concealment or pardon. College associations stimulate the best elements in a man, expose weak and wicked ones, and tend to the pruning and strengthening of character.

4. Then there is in college life association with professors and tutors, and this is, I confess, sometimes of little value, as when teachers are mere machines, but in it, at its best, are distinguishing benefits. When teachers are full men, apt men, and enthusiastic men—as college professors, and for that matter all teachers ought to be—the place of recitation soon becomes a center of power. Tact tests attainment, exposes ignorance, foils deceit, develops strength, indicates lines of discovery, and inspires courage. A living teacher supplies at once model and motive. He has gone on among the labyrinths, and up the steep of knowledge; has tried and toiled and triumphed. He sought and he is. And now by wise questioning, by judicious revelation, by skillful concealment, by ingenious supposition, by generous raillery, by banter, by jest, by argument and by magnetic energies, the teacher stirs the student into supreme conditions of receptivity and activity. Such teachers make the college. As President Garfield said: “Give me an old school house, and a log for a bench. Put Mark Hopkins on one end, and let me, as student, sit on the other, and I have all the college I need.” When an institution is able to employ men of superior knowledge, power, and tact, students must be trained, and all their after lives affected by the influence. For memory magnifies the worth of a true teacher, and the hero of the college quadrennium becomes a demigod through the post-graduate years. A dozen men of this mold, if once they could be gotten together, would make a college the like of which has not yet been seen on the planet. Shall Chautauqua one of these days find them?

5. The college life promotes mental discipline. It drills, and drills, and draws out. It compels effort, and effort strengthens. It provides a system of

mental gymnastics. What was difficult at first, soon becomes easy, until severer tests are sought from the very delight the student finds in concentration and persistency. Thus development takes place in the varied faculties of the soul. The student acquires power to observe with scientific exactness, to generalize wisely from accumulated data, to project hypotheses, to watch psychical processes, to reason with accuracy, to distinguish between the false and the true, both in the inner and the outer world; to grasp protracted and complicated processes of mathematical thought; to trace linguistic evolutions—remembering, analyzing, philosophizing; to study the students of the ages, and the products of their genius in art, poetry, jurisprudence, and discovery, in the facts of history and the great principles of sociology. All the powers employed in this manifold work during the college term are trained and thus prepared for work after the college term is ended. It is not so much the amount of knowledge acquired during the four years, as it is the power at will ever after to acquire knowledge, that marks the benefits of the college course.

6. With discipline comes the comprehensive survey of the universe. The college outlook takes the student backward along the line of historical development. It shows him the heights and the depths, the manifold varieties and inter-relations of knowledge. It gives him tools and the training to use them, and a glance at the material on which he is to use them. The student through college is a traveler, sometimes examining in detail, sometimes superficially. He gives a glance and remembers; he takes notes and thinks closely. He sees the all-surrounding regions of knowledge, and although he may make but slight researches in particular lines, he knows where to return in the after years for deeper research and ampler knowledge.

7. College life leads to self-discovery. It tests a man's powers, and reveals to him his weakness. It shows him what he is best fitted to do, and the showing may not be in harmony either with his ambitions or his preconceived notions. A boy born for mercantile pursuits, who comes out of college a lawyer or preacher, proves that the college failed to do its legitimate and most important work for him. Professors who merely glorify intellectual attainment, and who neglect to show students their true place in the world, are little better than cranks or hobbyists. College life is the whole of life packed into a brief period, with the elements that make life

magnified and intensified, so that tests of character may easily be made. It is a laboratory of experiment, where natural laws and conditions are pressed into rapid though normal operation, and processes otherwise extending over long periods of time are crowded to speedy consummation. Twenty years of ordinary life, so far as they constitute a testing period of character are, by college life, crowded into four years. A boy who is a failure then, would, for the same reasons, be a failure through the longer probation, unless the early discovery of peculiar weakness may be a protection against the perils which this weakness involves. Therefore it is a good thing for a youth to subject himself thus early to a testing, for from it may come self-discovery, when latent powers may be developed, and impending evils avoided.

Of other advantages of educational institutions I shall not now speak. They are manifold. Our youth of both sexes, whatever their callings in life, would do well to seek these advantages. Therefore parents, primary teachers, and older persons who influence youth, should constantly place before them the benefits of college education, and inspire them to reach after and attain it. Arguments should be used, appeals made, assistance proffered, that a larger percentage of American youth may aspire after college privileges, or at least remain for a longer term in the best schools of a higher grade. Haste to be rich, restiveness under restraint during the age of unwisdom, inability to regulate by authority at home the eager and ambitious life of our youth, together with false, mercenary notions of parents, who “can not afford to have so much time spent by the young folks in studying, because they must be doing something for themselves”—these are some of the causes of the depreciation and neglect of the American college—a neglect lamentable enough, and fraught with harm to the nation.

Chautauqua lifts up her voice in favor of liberal education for a larger number of people. She would pack existing institutions until wings must be added to old buildings, and new buildings be put up to accommodate young men and maidens who are determined to be educated.

Chautauqua would exalt the profession of the teacher until the highest genius, the richest scholarship, and the broadest manhood and womanhood of the nation would be consecrated to this service.

Chautauqua would give munificent salaries and put a premium on merit, sense, tact, and culture in the teacher's office. She would turn the eyes of all the people—poor and rich, mechanics and men of other, if not higher degree, toward the high school and the college, urging house builders, house owners, house keepers, farmers, blacksmiths, bankers, millionaires, to prepare themselves by a true culture, whatever niche they fill in life, to be men and women, citizens, parents, members of society, members of the church, candidates for immortal progress.

To promote these ends the Chautauqua Literary and Scientific Circle was organized. By its courses of popular reading it gives a college outlook to the uncultivated, and exalts the higher learning. It is, as I have elsewhere said, a John the Baptist preparing the way for seminary and university.

The managers of the Chautauqua movement, however, recognize the fact that there are thousands of full-grown men and women who are at their best intellectually, and who, with some leisure and much longing, believe they could do more than read. They want to study; to study in downright earnest; to develop mental power; to cultivate taste; to increase knowledge, to make use of it by tongue and pen and life. There are tens of thousands of young people out of school by necessities commercial and filial, who are awakened to the power within and the possibilities beyond. They believe they could learn a language, and enjoy the literature of it. They believe they could think and grow, speak and write. They are willing, and eager to try. Out of minutes they could construct college terms. They have will enough, heart enough, brain enough to begin, to go on, to go through, and all this, while the everyday life continues with its duty for this hour and for that. They believe that into the closely woven texture of everyday, home and business life, there may be drawn threads of scarlet, crimson, blue and gold, until their homespun walls become radiant with form and color worthy to decorate the royal chamber—the chamber of their king, God the Father of earnest souls.

Chautauqua denounces the talk of certain rich men about the “poor having their place,” and that it would be “better for working people to confine themselves to work, or at best to understand subjects bearing entirely on their everyday duties in field or shop, and let science and literature alone.” Chautauqua would make working men cultivated, and

give them recreation from manual toil in realms of wonder, taste, science, literature and art. Chautauqua would spread out over the lot of the toiler a dome, vast, radiant, rich and inspiring.

Therefore the Chautauqua School of Liberal Arts has been organized, and chartered with full university powers, for non-resident pupils, who, by correspondence with competent instructors, may study what they please, when they please, and as they please, eliciting suggestion, and giving answer and thesis, taking all the time they need, passing final examination in writing in the presence of witnesses, and having their examination papers subjected to the scrutiny of competent and impartial critics. When, after the required standard in the several departments which constitute the college course has been attained, whether in four, or ten, or fourteen years, the successful candidate shall have his diploma and his degree; and through this window he has constructed out of all these fragments of time—fragments picked up from dusty floor and pavement, from mine, and field, and shop—through this window the light shall shine in its beauty, and people shall see what genius, industry and persistent will can do with the cast away fragments of spare moments and random opportunities.

I have thus described the “Upper Chautauqua.” By reason of the action of the Board of Managers, elsewhere reported, the plan of gradation is slightly changed from that laid down in the previous article on the “Upper Chautauqua,” and the following successive steps are found in the scheme of the Chautauqua University:

1. The ASSEMBLY, including the summer meetings, the “Platform,” “the American Church Sunday-school Normal Course,” the “School of Languages,” and the “Teachers’ Retreat.”

2. The CIRCLE, embracing the “C. L. S. C.”

3. The “INNER CIRCLE,” to which they belong who, having seven seals on their diploma, are members of the “League of the Round-Table.”

4. The “UNIVERSITY CIRCLE,” with its “School of Liberal Arts,” and the “School of Theology.”

NEW HAVEN, CONN., February 6, 1885.

OUTLINE OF REQUIRED READINGS.

MARCH, 1885.

First Week (ending March 8).—1. “College Greek Course,” from page 187 to 216.

2. “Chemistry,” chapters IX and X.
 3. “The Circle of the Sciences,” in THE CHAUTAUQUAN.
 4. Sunday Readings for March 1 and 8, in THE CHAUTAUQUAN.
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Second Week (ending March 16).—1. “College Greek Course,” from page 216 to 239.

2. “Chemistry,” chapters XI, XII and XIII.
 3. “Temperance Teachings of Science,” in THE CHAUTAUQUAN.
 4. Sunday Readings for March 15, in THE CHAUTAUQUAN.
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Third Week (ending March 24).—1. “College Greek Course,” from page 239 to 260.

2. “Chemistry,” chapters XIV and XV.
 3. “Home Studies in Chemistry and Physics,” in THE CHAUTAUQUAN.
 4. Sunday Readings for March 22, in THE CHAUTAUQUAN.
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Fourth Week (ending March 31).—1. “College Greek Course,” from page 260 to 284.

2. “Chemistry,” chapters XVI and XVII.

3. “Studies in Kitchen Science and Art,” in THE CHAUTAUQUAN.

4. Sunday Readings for March 29, in THE CHAUTAUQUAN.

PROGRAMS FOR LOCAL CIRCLE WORK.

FIRST WEEK IN MARCH.

1. Blackboard illustration and full explanation of the Greek theater, special attention being given to the arrangement of the stage. If preferred, charts or pictures can be substituted for the blackboard. As aids to this work Donaldson's "Greek Theater," containing charts and illustrations, and Mahaffy's "Classical Greek Literature" will be found very helpful.

2. Essay—George W. Cable and his Works.

Music.

3. Selection—"The Gorgon's Head," found in Hawthorne's "Wonder Book." This story can be read "turn about" by the members. Reference is made to the headless Gorgon, on page 210 of "College Greek Course."

4. Essay—Shrove Tuesday, or Mardi Gras, as observed in New Orleans.

5. A Paper on Great Salt Mines and Springs.

6. Critic's Report.

SECOND WEEK IN MARCH.

1. Essay—Sir Humphrey Davy.

2. Selection—"An Account of Sappho." By Addison.

3. A Paper on Canadian Winter Sports.

Music.

4. A Half-hour's Quiz on the Readings of the Month.

5. Essay—The Life of Euripides.

6. Question Box.

THIRD WEEK IN MARCH.

1. Fifteen Minutes' Talk on Balloons and their Uses.

2. Selection—"On Great Natural Geniuses." By Addison.

3. Character Sketch—Ignatius Loyola.

4. A Paper on the Athenian Orators.

Music.

5. General Conversation on the News of the Day.

6. The Questions and Answers for the Month in THE CHAUTAUQUAN.

MONTHLY PARLOR MEETING.

Music.

1. Roll call—Quotations from Greek Authors.

2. A Map Exercise. Trace Philip's conquering march, as indicated by Demosthenes in his third Olynthiac oration.

3. Essay—Demosthenes.

Music.

4. An Analysis of Tennyson's "Princess."

5. A Paper on the Famous Women of Greece.

Music.

6. Debate—Resolved, that the effects of the modern theater compare unfavorably with those of the ancient.

Music.

It may not be amiss to follow our programs—which are intended to be merely suggestive—with a very short exposition of our program-philosophy. It is not a heavy philosophy; indeed, it is so simple that we half suspect we may be laughed at for calling it a philosophy at all, but its principles, we believe, are true and useful; as such we offer them. According to our ideas there are four subjects which should be represented on each C. L. S. C. program; first in the list and in importance is the week's or month's reading, its prominent features, its suggestions, its facts, its practical lessons; second, the world's work of to-day, not merely its events of public interest, its schemes and disasters, but its science, invention, art, literature, morals, social life, civilization, its men and its manners; to follow both exercises and clinch what has been suggested, "good talk" ought to be an invariable part of each evening's work. Take care that talk, free, genial, interested talk, follows every performance, or every program, and be sure that always

"Music dwells
Lingering and wandering on as loth to die."

These are the four elements necessary to a good program. As to how they shall be treated we have also a theory. Its first principle is let everything be well done; while thorough, do not go astray in dates and statistics, but go to the point which you desire to make. Then be bright and interesting, the third essential in each performance. Withal, suit your theme and your treatment of it to your audience. Let the subject be of common interest, the matter neither so commonplace as to seem puerile nor so technical as to be "over the heads" of your auditors. Such is our program-philosophy. A better you will undoubtedly formulate by practicing this.

LOCAL CIRCLES.

C. L. S. C. MOTTOES.

“We Study the Word and the Works of God.”—“Let us keep our Heavenly Father in the Midst.”—“Never be Discouraged.”

C. L. S. C. MEMORIAL DAYS.

1. OPENING DAY—October 1.
2. BRYANT DAY—November 3.
3. SPECIAL SUNDAY—November, second Sunday.
4. MILTON DAY—December 9.
5. COLLEGE DAY—January, last Thursday.
6. SPECIAL SUNDAY—February, second Sunday.
7. FOUNDER’S DAY—February 23.
8. LONGFELLOW DAY—February 27.
9. SHAKSPERE DAY—April 23.
10. ADDISON DAY—May 1.
11. SPECIAL SUNDAY—May, second Sunday.
12. SPECIAL SUNDAY—July, second Sunday.
13. INAUGURATION DAY—August, first Saturday after first Tuesday; anniversary of C. L. S. C. at Chautauqua.

14. ST. PAUL'S DAY—August, second Saturday after first Tuesday; anniversary of the dedication of St. Paul's Grove at Chautauqua.

15. COMMENCEMENT DAY—August, third Tuesday.

16. GARFIELD DAY—September 19.

Regularity is necessary to permanency. Whatever undertaking we desire to make a permanent success, we must make regular; whatever we wish to do successfully, we must do regularly. A tiresome, prosaic quality we are apt to consider it, and one which restricts our freedom. The regular return of small duties often makes them annoying, yet in large affairs regularity adds dignity and strength. It is essential for the establishment of any institution. A trite truth this may be, but trite truths are not always applied, and it is for the application of this homily to local circles that we sue.

It is most desirable that your local circle should become durable. Not a club, to which you can run in as you have leisure, or which can be adjourned for other engagements; which shall run this winter, and "perhaps," "if nothing happens," go on next winter. Not at all. There is a higher idea embodied in the plan. The true ambition of each member of a circle should be to make it *the* literary association of the community, the leader in practical ideas, clear thinking, intelligent talk and refined manners; but to reach this goal the circle meeting must be considered too valuable to be omitted for any occasion whatever. Its object is equal to that of any institution in the town. If you wish to develop this idea, to establish your circle, to secure for it recognition as a well founded organization, regularity in meeting and attendance must be secured. It is true that a social or religious event sometimes happens for which courtesy seems to demand an adjournment. In such a case it is quite possible to select another night. The one idea upon which we would insist is that the circle be considered and conducted as a permanent institution, that it be made the intellectual center of your life. How wonderful an impetus to thought and culture is such an organization, only those who lack its influence can tell. Some of the earnest letters which come to us from time to time give a suggestion of what a circle might be to lone readers. Is there not, indeed, in this delightful letter from BULGARIA, a hint of the real value of a circle, a value which we so

often fail to appreciate? It comes from an old Chautauqua friend—Miss Lenna A. Schenck, now a missionary at LOFTCHA, BULGARIA: “How gladly would we report to you from this out-of-the-way corner of the earth the organization of a flourishing local circle. But, alas! alas! we can not boast of even a triangle or a straight line, only a point, a mere dot, but a thoroughly loyal one, keenly enjoying the good things of THE CHAUTAUQUAN, that most welcome and highly prized of all the white-winged friends that come to us by mail. Though so few in number, we keep the vesper hours and the memorial days, and begin each day happily by devoting the time from six to seven in the morning to Chautauqua reading, and so we are inspired by glimpses of charming circles away in the homeland, and by memories of delightful summers with our blessed alma mater, Chautauqua herself. Before another year rolls round, we hope to have at least a local triangle here at Loftcha, and perhaps a Bulgarian translation of some of Chautauqua’s best ‘ideas.’ Many things might be said of our new home and new work, but we remember the delicate suggestion given in the November ‘Local Circle,’ that ‘no one could stay very long,’ so with heartfelt greetings to the class of ’83 and to all good Chautauquans the world over, we bid you adieu.”

Are not such friends of Chautauqua the prophecy of a time when the work shall encircle the earth? Each month brings signs of its growth. Particularly do we notice this month the spread of the work in CANADA. The press is particularly friendly to the movement in the Provinces; for example, the *Educational Weekly*, of TORONTO, quotes the *Globe* of that city as saying: “The Chautauqua Literary and Scientific Circle is now pretty well known. It has been in operation since 1878, and has done a great deal of good. The yearly reunions at Chautauqua have come to be very pleasant and very profitable. We understand that a similar summer resort is to be instituted in Canada, in connection with the reading circles already established in the Dominion. We wish the enterprise all success.” Much of the interest in Toronto is undoubtedly due to the hearty work of Mr. E. Gurney, and Mr. Lewis Peake, president and secretary of the “Central” circle. This circle has recently had the pleasure of hearing a lecture on “Athenian Literature” from Professor Hutton, of the University College. LONDON has also a very flourishing circle, dating from the fall of 1883, when it was organized with a membership of about forty. It is a most

healthy sign of growth, when reorganization finds a circle larger than when it disbanded. The "Central" circle had this fortune. They began the present year with a membership of forty eight. Their plans have been most happy; the vesper services in the Chautauqua song books are used at every meeting, and quotations as responses to roll call; chemical experiments are performed for them by a professor of practical chemistry, who is a member of the circle, and their programs are full of variety. So important to them is their circle that they made Christmas the occasion of a special meeting, at which they used the Christmas vesper and praise service which appeared in THE CHAUTAUQUAN for December. The service was followed by an address and several entertaining exercises. This is exactly the work which enhances the value of the circle, both for the members and for the community. It raises a circle to the point where it becomes the medium through which all extra social occasions may be observed. It makes it not only a reading club, but a factor in the social, religious and intellectual life of a community.

At DARTMOUTH, NOVA SCOTIA, we learn from a local paper, there is also an energetic circle. They have done good work in introducing the C. L. S. C. to the public, securing a notice of a public vesper service, an explanation of the work they are doing, mention of the circles in the vicinity, and following their information by announcing their next meeting with a cordial invitation to the public to be present.

In November last two new circles were formed in MAINE. A "Pine Tree" circle, of twenty-seven members, coming from DOVER and FOXCROFT. These beautiful villages are closely connected by covered bridges—the Piscataquis river flowing between, though it is a hard matter for a stranger to see where one begins or the other ends, so much like one village are they. A friendly way to live, is it not? These classmates have evidently learned what Thackeray found out in London long ago—that "A man ought to like his neighbors, to be popular with his neighbors. It is a friendly heart that has plenty of friends." But we all learn that in the C. L. S. C. The second is the "Simpson" circle at AUBURN, where the Rev. G. D. Lindsay is president. Sixteen enthusiasts make up the circle which, so far, finds the work suggested in THE CHAUTAUQUAN sufficient for its needs.

One of the most interesting and prosperous, though not largest of Chautauqua circles, is the "Baketel" circle, at GREENLAND, N. H. It is named

in honor of its founder and leader, Rev. O. S. Baketel, an old Chautauquan of the class of '82. The organization is very simple. The leader prepares the program for each evening, and the members come promptly. No inflexible rule is adhered to, but as much variety given as possible. That the plan is most successful we know from a recent letter from a friend, in which he says of the work: "Our members vary in age from eighteen years to fifty-three, and none are more enthusiastic than the oldest ones. It makes one of the most interesting gatherings ever brought together in the community, and is furnishing help to some whose advantages in early life were very limited. Every member feels like exclaiming 'All hail C. L. S. C.'"—The "Webster" C. L. S. C., of FRANKLIN, N. H., is enjoying its second year of existence. A good interest was maintained throughout last year, and they began this year's work promptly in October, with twenty-two active members. To them the dining room table has revealed its wonderful power to stimulate sociability and "good talk." They have discovered its genial ways, how it will always stretch to make room for more and still more, and how it seems to be always saying: "Stretch out your arms; don't mind just how you sit. I shield your position, I am here to help you all, to bring you close together, to hold your books, to forbid your parting, to compel you to be a circle." Indeed, we are glad the "Webster" circle has learned the virtues of a dining room for study and for friendliness. Maybe if they but analyzed their devotion to their circle that stout, wooden friend would deserve not a little of the honor, and perhaps, too, it has helped not a little in bringing in the children, which, they write, are crowding into the Chautauqua work until the circle boasts even grandchildren.

The "Clio" club of twenty members at NEWPORT, VT., kindly remembers THE CHAUTAUQUAN with one of the programs used at a recent public meeting. The dainty, tasseled souvenir they send us bears a list of exercises of unusual richness and variety.

MASSACHUSETTS is getting her circles into the press. Scarcely a paper from within her borders comes to our sanctum which does not contain at least one item of Chautauqua import. The *Melrose Journal* of MELROSE reports the organization of a circle of fifteen members in that city.—The *Woburn Journal* notices the work of the circle there in a very appreciative notice: "The fortnightly meetings of the First Woburn Chautauqua Literary and Scientific Circle are being well attended and the exercises are very

profitable intellectually and the students are doing good work. Two weeks ago the Rev. Charles Anderson gave a very interesting talk on Prof. Schliemann's recent explorations in Mycenæ, and Hissarlik, the supposed site of ancient Troy. At some meeting in the near future the Rev. A. E. Winship, a true Chautauquan, connected with the 'New West Education Commission,' a thorough scholar and a very interesting speaker, will lecture on 'Literary Clubs' before the members of the circle."—The *Saturday Union*, of LYNN, speaks of the thorough work their circle is doing in chemistry.—The *Ipswich Chronicle* highly commends the Milton memorial held by the "Masconomo" circle of that city. By the way, the name of this circle brings back an interesting bit of early Massachusetts history. It was the Indian Masconomo, or Masconnomet—from whom the circle is named—who, in 1638, "sold his fee in the soil of Ipswich" for £20, to John Winthrop, Jr. And here was established the town which the Indians called Agawan ("fishing station"), and to which the white men gave the name of Ipswich.—The *Salem Gazette*, too, gives notices of two branches of the C. L. S. C. in that city. About forty members are in each of these societies.—Several new circles we have the pleasure of adding to our visiting book. At MERRIMAC a circle of seventeen members has been formed, with the happy title of the "Hale" circle. The first circle, so far as we know, which has honored itself by assuming the name of our esteemed counselor. They should be glad they waited; so good a name does honor to anybody, and ought to be an omen of future prosperity.—The "Eaton" circle, named in honor of the Rev. G. F. Eaton, begins life with seventy members. Its home is WALTHAM—city of watches. If the spirit of the town is to be the spirit of the circle, wonderful results will certainly be forthcoming.—Last October a few of the many students in the C. L. S. C. in WORCESTER organized a local circle. By the perseverance of these few, others have been persuaded to take the course, until the circle numbers about sixteen. They have taken the name of the "Warren" local circle, in honor of Bishop Warren.—At PROVINCETOWN a company of ten, five ladies and five gentlemen, met on the evening of the sixteenth of December last, to form a local circle. The meetings have occurred every week since; the circle has adopted the name of "Mayflower." The meetings are full of interest, and the members are busy trying to make up the reading of the past months. All are members of the class of '88 except one, who belongs to the class of '85.—SOUTH GARDEN reports a circle organized a year ago, but

which has never been noticed in THE CHAUTAUQUAN before. It is a “Pansy” class—all the fifteen members belonging to the class of ’87.—“Not Chautauquans for four years only, but Chautauquans for life,” the friends at HOLBROOK subscribe themselves. Their motto grew out of the ardor of a lady member of the circle who, when at a recent meeting something was said about a four years’ course, said: “I shall not consider that I have finished the course at the end of four years. I for one am going to be a Chautauquan as long as I live.” A right royal motto, is it not?—The WAKEFIELD circle sends a program of a meeting in which we are glad to notice that present affairs go side by side with discussions of Grecian history and art and literature. The subjects for essays include a “Review of Current Affairs in Massachusetts,” “The Pension Problem,” etc. The history that is making certainly deserves our attention, as well as the history of the past.—NORTH CAMBRIDGE also sends the program which they prepared for the January meetings of the “Longfellow” circle. In addition to their regular work, they added the novel feature of a talk on newspaper work, from a practical newspaper man.—The last of this month’s Massachusetts reports contains a most capital hint. AUBURNDALE is the home of a flourishing circle, which among its other good features has a constitution. One of the articles of this constitution is the suggestion which it will please us to have you all ponder. It reads: “A short report of the condition of our society shall be forwarded twice a year to THE CHAUTAUQUAN.” Do you all take the hint? Perhaps one secret of this energetic article is the nearness of Auburndale to Framingham—so near is it that all the members of the circle went to the Assembly last year. To Massachusetts, too, belongs the honor of the following merry Chautauqua feast, of which a friend from Providence, R. I., has written us: “Spending a few days in ROCKLAND, MASS., I was invited to visit the ‘Sherwin’ Chautauqua Circle, and being a true-blue member of the ‘Clio’ C. L. S. C. of Providence, I was joyful in accepting. The exercises were of a most novel and interesting kind, and unusually pleasing to me, as I was an old acquaintance of Prof. Sherwin. Since this society was instituted, some two years ago, but one representative of the posterity of the circle has been born, and the members of this enterprising circle showed their appreciation of Prof. Sherwin’s noble work in the good cause by naming this gift after him. An elegant gold lace pin had been made to order, with the initials C. L. S. C. neatly engraved upon it, and that evening the presentation was made. After Chautauqua greetings had been

exchanged, the baby Sherwin was called for, and made his appearance, riding on his mother's arm, as wise and dignified in behavior as a youthful Solon. One of the frolicsome Chautauqua dames then read the following formal rhyme:

“There were some fair dames of Chautauqua,
Their possessions were lovely to see,
Between you and me;
They had jewels of gold,
Of value untold,
These elegant dames of Chautauqua;
But children were few,
You scarce find one or two
In the homes of these dames of Chautauqua.
And sad were the dames of Chautauqua
When they read of the Gracchus,
Of Cupid and Bacchus,
The lesson seemed filled up with mocking.
They longed for a son,
So the gods sent them one,
Full of frolic and fun,
Sent a son to these dames of Chautauqua.
Then what joy in the circle Chautauqua!
What pæans were sung,
And Chautauqua bells rung,
To welcome the lad of Chautauqua!
Straight they gave him a name,
Sherwin Burrill the same—
These frolicsome dames of Chautauqua!
Now, they badge him with gold,
So that when he is old,
They can still claim their son of Chautauqua.”

At SOUTH MANCHESTER, CONN., a most encouraging increase of members has taken place. Last year the circle numbered twenty, this year forty-eight. Such growth is full of promise for the future, and yet it is the inevitable

result of enthusiastic members and carefully prepared programs.—The new circle at MANSFIELD CENTER, CONN., numbers ten members. They are expecting a lecture on chemistry soon, from Prof. Washburn, of the North Mansfield Agricultural College.—The “Newfield” C. L. S. C. of WEST STRATFORD, CONN., has recently received the following pleasant letter from “Pansy:”

CARBONDALE, PA., January 6, 1885.

Dear Friends of '87:

My word of greeting to you must commence with an apology. The letter from your secretary found me immersed in work. The holiday season brings upon me a heavy pressure of care, in addition to the usual routine. From the almost hopeless mass of unanswered letters which I have just overturned on my study table, that of your secretary emerges, so I seize it and make a beginning. What shall I say? I might congratulate you on being members of that great literary circle, which verily seems destined to reach out its long arms and encircle the world—but to what purpose would this be?

You already know by experience all, and more than I could tell you of its advantages, and its far reaching influences.

What then, shall I, in this moment of time, say to you who are classmates of mine? Shall I hope that you may be able to pass the Golden Gate and join in the class song of the '87s, and receive your diploma from the hands of the Chautauqua chief, and enjoy all the delights of Commencement day? That indeed I heartily wish. I hope to be there and to clasp hands with you, and give and receive greeting.

But I am conscious while I write, of a higher, stronger, holier hope than that, even that every member of your circle and of all the great Chautauqua Circle may finally pass the Golden Gate that leads to the palace of the King, and receive from him the greeting “well done, good and faithful servants,” and receive

from his hands the crowns laid up for those who are “called, and chosen, and faithful.”

Oh, to be sure of passing safely through the ordeal of examination by the Judge!

When I think of the immense enthusiasm of the C. L. S. C., I am glad. I believe in enthusiasm. I believe in the Chautauqua Literary and Scientific Circle. Yet I wonder, often, whether we, as Christians, can not learn something from the eagerness of many scholars who are not of the royal family, and yet are eager to learn all they can, about our Father’s handiwork in earth and air and sky.

I am writing longer than I meant. I only wanted to say this: Let us make sure of clasping hands at last in our Father’s house.

Yours in His name,

MRS. G. R. ALDEN.—“PANSY.”

Desiring to promote the interests of the C. L. S. C., the Chautauqua circles of Rhode Island, numbering about twenty-five, have united and formed the “Rhode Island Chautauqua Union,” with the following officers: President, Prof. John H. Appleton, A.M., of Brown University; first vice president, the Rev. J. Hall McIlvaine, pastor of the Union Congregational Church; second vice president, Hon. Thos. B. Stockwell, A.M., Rhode Island State Commissioner of Public Schools; third vice president, Levi W. Russell, A.M., Principal of Bridgham School; secretary and treasurer, Wm. D. Porter, D.D.S., all of Providence. There have been three new circles formed this year in PROVIDENCE, R. I., one of them bearing the popular name of “Vincent.” “Hope” circle, formed in 1882, is still in a flourishing condition. They were favored last month with a very interesting and instructive address by Prof. Appleton, on “The Value of the Study of the Natural Sciences.” The executive committee arrange the order of exercises and find the monthly programs in THE CHAUTAUQUAN of great benefit. The circle has now about seventy members, and most of them belong to the C. L. S. C.—Another newly organized circle of Providence is the “Esmeralda Bachelor” circle. It numbers twenty members, all gentlemen,

and they give as the only excuse which it seems to us could be at all valid for forming a circle of bachelors, that they can get more young men into their club by restricting its membership. "Whittier" circle, of the same city, has been reorganized, and a most pleasant item comes to us from them. On the birthday of the beloved poet, the circle sent to him as a souvenir a paper weight of serpentine, from a quarry in Newburg, Mass. It had been cut into a design of oak leaves and acorns. Their remembrance brought back a kindly response from Whittier.—A newly organized circle also exists at RIVER POINT, R. I. It was formed in October last, and numbers thirty-five. Their plan is that laid out in THE CHAUTAUQUAN.

The NEW YORK circles are doing wonderfully energetic work. At JORDAN there is a new and growing organization of twenty-four members.—At MEDINA, one of thirteen, which has already begun to scatter seed, some of it so far away as the Pacific coast, where our Medina members believe they will soon have an offshoot.—The "Wolcott" C. L. S. C. has been organized at WOLCOTT, with over thirty members, who write most enthusiastic words of the benefits they have already received.—At BROCTON the veteran circle, composed of members of the S. H. G. and C. L. S. C., held a delightful Milton memorial.—At ROCHESTER, the circle which is connected with the Academy of Science in that city, wins this appreciative notice from a local paper: "Public sentiment in favor of the Chautauqua movement is spreading with marvelous rapidity. Such certainly is the fact in regard to the circle in this city. There are now upward of forty members enrolled, and beside these a large number of persons attend the semi-monthly meetings who have not yet identified themselves with the regular work. The practical benefit derived from this course of home study becomes more apparent as it is investigated, constantly confirming the wisdom of its founders in setting in motion a plan for the intellectual and moral elevation and culture of thousands who have only spare minutes for such an object."—At GLENS FALLS, not long ago, Chancellor Vincent greeted his C. L. S. C. pupils, delivering his lecture on "That Boy and His Friends," before them, and meeting them afterward at a reception.—At OCEAN GROVE, the circle under Dr. Stokes's genial management is doing admirable work. A delightful social was recently held by the circle at the Sheldon House.—The MARION circle has reorganized this year, strong and hopeful as ever. Says a member: "The 'Inner Chautauqua' is taking a

deeper hold upon us year by year, and we propose that our connection with the C. L. S. C. shall continue indefinitely. We are trying to extend the knowledge of it by distributing the 'Popular Educational Circular,' by inserting an occasional item in our village paper relating to the doings of our circle, as well as by personal conversation with our friends and acquaintances."——The circle at CARMEL has also been reorganized, with seventeen members. Their programs show excellent work.——At SANDY HILL, during the holidays, a special meeting commemorated the season. Among the exercises was a poem on "The Triumph of our Language," which deserves special mention.——The BROOKLYN circle, of Hansom Place M. E. Church, has increased its membership to over one hundred. It owes to the Rev. George E. Reed, its president, the large increase. Having outgrown the capacities of private parlors, they have met lately in those of the church, where, while losing some of its more social elements, there is a far better opportunity for map display and the general working of the monthly class. Following out the assignment of an instruction committee, they find no lack of willing participants. One of the most popular exercises is the five minutes' essay on some person or incident connected with the current reading. In good hands, the information condensed is of the most direct kind, and at its conclusion an opportunity is given the class to ask any questions relevant to the topic. All this is clear knowledge, and has proven one of the most agreeable of their methods.——At FORT PLAIN, the circle carried out on Bryant day a highly enjoyable program. That this circle is enthusiastic, the fact that some of its members come from four miles away, is a proof.——A second New York circle which has enjoyed a visit from Chancellor Vincent, and had the pleasure of tendering him a reception, is that at CHATHAM. Several new members have joined the circle there, the result of the inspiring talk which the Chancellor gave them.——The "Ionian" circle of BURLINGTON, N. Y., is winning friendly attention from the local press, its meetings being noticed, and its exercises commended.——The "Vincent" circle, of TROY, invariably sends out to its members, on its announcements of monthly meetings, some bit of inspiring thought. On the January program we find this sentence, useful, we suspect, for other than Troy readers: "Remember this: In proportion as you put thought and work into these monthly meetings, in that proportion, with high interest, will you draw out in enjoyment and profit."

The local circle of BRIDGEVILLE, PA., was organized November, 1881, with a membership of thirteen. During the intervening three years there have been many changes, but the good work has been steadily going forward. The circle reorganized October, 1884, with eight members, and has taken up the work of the year with increased vigor, the meetings being well attended and very interesting. The monthly meetings are held in the village church, though none of the members live in the village, some having to travel the distance of two miles to attend the meetings.—At READING, the “Cleaver” circle has been reorganized, with double its old membership. Their program they make very interesting, by introducing variety into the exercises.—Nine ladies and gentlemen formed last fall the “Castelian” circle, in PHILADELPHIA. Happy are they to have a large map of Greece. What a treasure it is to a circle these days!

The “Meridian” circle, of WASHINGTON, D. C., has been having a feast of good things. How can it help it? It lives in Washington, and Washington offers peculiar advantages to literary and scientific clubs, not only on account of its immense professional library and large scientific collections in the Smithsonian Institution and National Museum, but also through the *personnel* of these institutions and of the many other scientific bureaus of the government, who, making literature or science their daily vocation, afford a large field from which to draw essayists and experimentalists of a high order. “Meridian” circle has been fortunate in availing itself of these advantages. Last year, during the course in vegetable biology, they had an evening’s instruction in the microscopic examination of bioplasm, by Dr. D. S. Lamb, the eminent anatomist of the United States Medical Museum, who had charge of the autopsy of President Garfield. This year, at their last meeting in November, they had an essay from Mr. Lee Shidy, of the United States Coast Survey, on “The Tides,” a most interesting subject, and most interestingly and ably illustrated and explained.

A seven-years-old circle certainly deserves a warm corner by THE CHAUTAUQUAN’S fireside. Most cordially do we grant it, for we mistrust that a circle so experienced will be unusually good company, and will be able, too, to give us some suggestions of value. It is the “Trojan,” of TROY, OHIO, which claims this rare distinction, and we believe we are not wrong in saying that their history will be of great interest to all. The ‘Trojan’ circle was organized with a large membership in 1878. Eight members graduated

in 1882, five of them being at Chautauqua that season. In the fall of that year the circle increased greatly, and has been growing in interest ever since. Now it numbers thirty-two members. Their plan of work is as follows: They open with singing, and responsive reading from 'Chautauqua Songs.' At roll call each member is expected to respond with a Bible verse. The questions in THE CHAUTAUQUAN on the week's lesson are asked, and also original test questions from some or all of the members, on the readings. Sometimes the circle reads alternately from one of the text-books, or from THE CHAUTAUQUAN. They always have a critic, and a committee of two that gives a digest of the topics of the times, often in the form of questions, which thus makes a pleasant *conversazione*. Memorial days are faithfully remembered and made interesting and attractive by essays, readings, recitations, and music.—At PERRYSBURG, OHIO, ten persons are in the circle, which has been in existence for about four years. One entertaining feature is novel. Occasionally a paper of interesting general news is added to the program. The question box, too, is made a feature of each evening, a practice which is always worth all the work it takes. Memorial days find pleasant observance, the Milton memorial being celebrated with peculiarly pleasing exercises. The circle is rejoicing in their readings, considering them of great benefit.

MICHIGAN advances with a goodly list of new circles this month. At PETOSKEY there are fourteen members formed into a circle. These friends have the invigorating influence of the Bay View summer Assembly to help their work.—At HUDSON, a delightful company of thirty-five has formed the "Carleton" circle, the name being given, of course, in honor of the popular poet, Will M. Carleton, whose birthplace and early home were in Hudson. Round-Tables with genuine "at home" feeling, recitations, select readings, question box, queries, criticisms and quotations make the meetings full of life and variety. The program for an evening is always published in the local papers at least two weeks beforehand, and a report of each meeting is slipped in after each session, so that the people can not forget the existence of the C. L. S. C. At an early meeting our friends are going to take a trip to Naples and return.—Strong organizations have been formed at both KALAMAZOO and SAUGATUCK. At the former place the "Burr Oak" circle has twenty members, and at the latter, a lovely town about two miles up the Kalamazoo River, the circle, though small, is

growing. The use which they make of our columns seems to us very good. "THE CHAUTAUQUAN is our 'guide and counselor,' and though we do not follow closely its outline for local circles, yet we never prepare a program without its aid."——In the land of the arbutus, at TRAVERSE CITY, the "Arbutus" circle, of twenty members, has been organized. A pretty monogram has been designed for them, and it is to be printed upon the sermon paper which the members use for essays and reviews. These contributions are then to be bound in paper covers and filed. An interesting collection it will certainly make. The growth of our language has been furnishing this circle with some interesting topics.

The "Vincent" local circle of LAFAYETTE, IND., has entered upon its fourth year, with forty members, three of whom are C. L. S. C. graduates, but remain active in the work. The president, Prof. Craig, and vice president, Prof. Thompson, both of Purdue University, are thoroughly interested in the work. The program is prepared a month in advance. They are following the suggestions in THE CHAUTAUQUAN, largely. The success of their lecture course last winter left the society with funds sufficient to rent a room, centrally located, for the regular meetings. The vice president, a Professor of Art, recently presented the circle with a terra cotta medallion of Dr. Vincent, his own work. It has been handsomely framed and hung in their room.

From SHELDON, ILL., a friend writes: "We have a local circle of about twenty-five members and great interest is taken in the exercises. We usually follow your program. Not having started until after October 1st, and having been delayed in obtaining our books, has thrown us behind some, still we are making up lost ground better than expected."——At CRETE a circle has started off with twenty-six members—many of them young people, to whom the course has been just what they needed.——ABINGDON also has a society of twenty-three members. Several readers have been there in past years, but not until now has there been a circle. The chemistry readings are furnishing an excellent opportunity for experiments, and the Abingdon circle are fortunate in having a college laboratory to resort to for experiments.

The circle of the Franklin Avenue M. E. Church, MINNEAPOLIS, MINNESOTA, has been formed two years and has not reported until now to

THE CHAUTAUQUAN. They have an interesting class of nineteen members, who are all very zealous in the work. The circle meets every Monday evening to review the week's work, which they are studying after the plan laid out in THE CHAUTAUQUAN.

The circles of MINNESOTA, and, indeed, of the entire north-west, are requested to send a note to Mr. E. P. Penniman, ST. PAUL, MINN., stating whether they will coöperate in a plan for securing a C. L. S. C. day at either the Red Rock camp ground, Lake Minnetonka, or at White Bear Lake. The six circles of St. Paul, those of Minneapolis, Hudson, and Stillwater, have signified their willingness to help carry out this excellent idea. Such a day would be an event of greatest interest and value to the circles in that locality; it would arouse flagging enthusiasm, would give every one present a fund of new ideas, and would spread the plan of home reading in many homes where it is unknown.

We are very much pleased to hear from BLOOMFIELD, IOWA, of a circle, organized in 1882, but which has not before been introduced to our circles. Since its organization its membership has increased from six to fifteen members. The memorial days are observed and much social life enjoyed by the circle which promises that at no late day there will be more than one organization of the C. L. S. C. in their city.

A late number of *The Daily Register*, of MOBILE, ALABAMA, contains an essay on "The Character of Milton," which was read before one of the circles of that city at a recent meeting. Had we space we should gladly reprint this excellent paper. Mobile has two societies reading the Chautauqua course, and we hope that we shall soon receive full reports from them.

A great deal of energy is displayed by the DESOTA, MISSOURI, circle. Few issues of the *Jefferson Watchman* come out without a notice of its meetings. A late number says: "The members of the C. L. S. C. are again busily engaged in their work after their holiday vacation. Two meetings have already been held in the new term, both of which were enjoyable and instructive, and the reading of '85 is well under way. The number of members is about the same as last term, as none after becoming interested in the work seem to have the least inclination to drop out of the circle, but

on the contrary become more and more interested and enthusiastic. The program for the next meeting will be found in another part of this paper.”

KANSAS quite equals Missouri, however, in its enterprising readers. A letter from a reader at WAKARUSA remarks of their circle: “We number but eight members, and are so scattered that our circuit embraces several miles, but having adopted the name ‘Olympian,’ we hope in time to carry off a double prize, one for intellectual attainments, the other for physical prowess exhibited in combat with Kansas mud. Though we have difficulties and discouragements even in our own little circle, we are yet resolute and enthusiastic. At present the Round-Table is the principal feature of our meetings.”——Quite as interesting is a live report from WYANDOTTE: “Although we have not been reported for nearly a year, our circle is not dead, but the interest is increasing, and we are doing better work than ever. Our membership numbers twenty-five, with twenty subscribers to THE CHAUTAUQUAN. In 1884 we held forty-seven meetings, and had an average attendance of twelve. With us, as with nearly all other circles, the great difficulty is to keep from having too much of a sameness in our programs. Thus far we have had good success by giving a committee charge of the literary work, which reports performers and programs a week in advance for regular meetings and three weeks for memorial meetings. We sometimes vary the exercises by devoting an entire evening to one subject. We endeavor to have all roll calls answered with quotations, and stimulate inquiry by having a question box, the contents of which are discussed at each meeting. We observe all memorial days, and they are a never failing source of interest. On Milton memorial the biography of Milton was given by the circle, each member taking up the history where the former one stopped. Each member read a favorite selection from the author, and the variety of selections indicated a variety of taste. We make good use of the Chautauqua songs, and find that the singing of them renders a meeting so much the more interesting, and there is, too, a bond of union in a stirring song. Our members have taken the liberty of naming this circle the ‘Pansy’ circle, as nearly all of us are members of the ‘Pansy’ class.”

We are sorry to “skip” the wide space between Kansas and California, and gladly stop at NORDHOFF, CAL., where we find the “Ojai” circle, which was organized last October. Although they are only seven, they are all in earnest and full of the Chautauqua spirit. They meet once each month, at the

homes of the members. They are all busy people, but are glad to *make* time for the C. L. S. C. reading, which they find adds a charm to busy lives. They hope to be able to persuade many of their friends to join them.

THE C. L. S. C. CLASSES.

CLASS OF 1885.—“THE INVINCIBLES.”

“Press on, reaching after those things which are before.”

OFFICERS.

President—J. B. Underwood, Meriden, Conn.

Vice President—C. M. Nichols, Springfield, Ohio.

Treasurer—Miss Carrie Hart, Aurora, Ind.

Secretary—Miss M. M. Canfield, Washington, D. C.

Executive Committee—Officers of the class.

Class badges may be procured of either President or Treasurer.

A very pretty program comes to us from a loyal “Invincible” of Toledo, Ohio, the Rev. H. M. Bacon, the president of the “Bryant” circle of that city. The program contains a six months’ outline of work. It bears the mottoes, the dates of regular meetings, and the memorial days. A kind of C. L. S. C. calendar which we imagine any one having had would find it hard to do without.

Miss Kimball writes our secretary, Miss Canfield, of Washington, that the Invincibles—true to their name—are making a splendid record, and that the class standing is excellent. She says: “I think the Invincibles may well be proud of their record. The prospect is that the class will stand fully as high, in proportion to its size, as any of the other classes. Of course we can not expect the actual number of graduates to reach that of other classes, as

the whole recorded membership is much smaller.” Let this encouraging news help us to “press on,” and, classmates, see to it that all members of your local circles, who rightfully belong with the ’85s, have their memoranda completed and sent in by the first of July.

Of those who expect to receive diplomas at Chautauqua, forty-one, representing fifteen states, Canada and the District of Columbia, have responded to the request to send their names to the secretary. Let us hear from you all, that the list for “roll call” may be complete.

One ’85, who writes he lives alone with his brother “away out in the backwoods of California,” regrets he can not be present at Chautauqua, but hopes to receive his diploma at Monterey. From the Atlantic to the Pacific the pulse of the C. L. S. C. is beating strong and steady.

CLASS OF 1886.—“THE PROGRESSIVES.”

“We study for light, to bless with light.”

CLASS ORGANIZATION.

President—The Rev. B. P. Snow, Biddeford, Maine.

Vice Presidents—The Rev. J. C. Whitley, Salisbury, Maryland; Mr. L. F. Houghton, Peoria, Illinois; Mr. Walter Y. Morgan, Cleveland, Ohio; Mrs. Delia Browne, Louisville, Kentucky; Miss Florence Finch, Palestine, Texas.

Secretary—The Rev. W. L. Austin, New Albany, Ind.

“For light!” and “with light!” as the words we repeat,
Yet fuller and deeper the message they bring;
Still through every volume each line that we meet
In undertone earnest our motto shall ring.

“For light” do we ponder the history vast
Which spreads through the ages its sunshine and shade,
“With light” for the present, we come from the past,
With lessons whose impress we can not evade.

“For light” must we study the many-hued lines
Which Greece with her delicate pencil has traced;
While Rome with her pride and her grandeur combines
To deepen the picture no time can efface.

“For light” at the portals of Nature we wait—
Descend to her rocks and mount up to her stars—
Her atoms diffuse and her gases collate,
Yet learn, as her secrets she slowly unbars,

How, filling, pervading, encompassing all,
Still law—mighty law—through all systems doth reign;
The world and the atom respond to its call,
The dewdrop and ocean are bound by its chain.

“For light,” above all, when our vesper has chimed,
We bathe in the beams of an unsetting Sun;
When thus up the ladder of prayer we have climbed,
“With light” shall be blessed many thousands through one.

“For light!” and “with light!” ’tis for this we would live,
O fling our glad banner abroad to the sky!
Truths won for ourselves unto others we give,
Till light never-clouded shall greet us on high.

ALICE C. JENNINGS, Class of '86.

CLASS OF 1888.—“THE PLYMOUTH ROCKS.”

“Let us be seen by our deeds.”

CLASS ORGANIZATION.

President—The Rev. A. E. Dunning, D.D., Boston, Mass.

Vice Presidents—Prof. W. N. Ellis, 108 Gates Avenue, Brooklyn, N. Y.; the Rev. Wm. G. Roberts, Bellevue, Ohio.

Secretary—Miss M. E. Taylor, Cleveland, Ohio.

Treasurer—Miss M. E. Taylor, Cleveland, Ohio.

All items for this column should be sent, in condensed form, to the Rev. C. C. McLean, St. Augustine, Florida.

Badges for the Class of '88 sold only by Mrs. Rosie M. Baketel, Greenland, N. H. Price, 15 cents each.

All members who have interesting items of class news should send them promptly to the Rev. C. C. McLean.

The following circles of the Class of '88 have been formed, viz.: “Janes,” Brooklyn, N. Y., sixty-two members; “Vincent,” Wyoming, Iowa, over twenty members; “Washington Avenue,” Milwaukee, Wis., fourteen members. In Collamer, Ohio, there was organized a circle four years ago. It has enjoyed active vitality ever since, and is now doing most efficient work in astronomy. Aroused by the last Chautauqua Assembly, nearly thirty organized a new circle. All are of the Class of '88, except one of '82, one of '83, and five of '86. The latter includes an old lady in her 81st year, who is not only beautiful in character, but, seemingly, as bright in intellect as in the meridian of life. This circle favors a change in motto; one suggests “Perfect in principle, in practice pure.”

The Florida Chautauqua is now in session at Lake de Funiak, and closes March 9th. The program is varied and interesting. We hope to report a good increase in the Class of '88 at the close of the Assembly.

IOWA.—I am enrolled in the C. L. S. C. army, "Class of '88." Not until the middle of this month (December) was I able to commence my reading. The prescribed course I think grand, and I can but feel grateful for a plan so far reaching, and so full and beneficial in its results. Our class motto is excellent. I am a busy farmer, but I shall make known the advantages of the "Chautauqua University."

QUESTIONS AND ANSWERS.

BY A. M. MARTIN,
General Secretary C. L. S. C.

I.—QUESTIONS AND ANSWERS ON “COLLEGE GREEK COURSE IN ENGLISH”—FROM PAGE 187 TO END OF THE BOOK.

1. Q. Who was the third member of the great tragical triumvirate of Greece, Æschylus and Sophocles being the other two? A. Euripides.

2. Q. When was Euripides born, and what noted battle took place the year of his birth? A. 480 B. C., in the year of the battle of Salamis.

3. Q. Where were the closing days of Euripides spent? A. At the court of the king of Macedonia.

4. Q. Who are two of the translators of Euripides? A. R. Potter, who has made a metrical translation, and T. A. Buckley, who has produced a version in prose.

5. Q. From what play of Euripides are the most of the extracts presented by our author taken? A. From the “Alcestis.”

6. Q. Under what title has Robert Browning rendered a version of “Alcestis?” A. “Balaustion’s Adventure.”

7. Q. Who was Alcestis? A. The wife and queen to Admetus, king of Pheræ, in Thessaly.

8. Q. By grace from Apollo, on what condition was Admetus granted the privilege of not dying? A. On condition of his being able to find some one who would agree to die in his stead when his turn should come.

9. Q. Who became the required substitute? A. Alcestis, the wife of Admetus.

10. Q. After her death by whom was she brought back to life and restored to her husband? A. By Heracles.

11. Q. From what drama of Euripides does our author take a celebrated chorus, in part eulogistic of Athens? A. The "Medea."

12. Q. Who stands alone as representative to us of Greek comedy? A. Aristophanes.

13. Q. What two comedies of Aristophanes retain for us more interest than perhaps any other of his works? A. "The Frogs" and "The Clouds."

14. Q. Who were the especial targets of these two comedies respectively? A. Euripides of the "Frogs" and Socrates of the "Clouds."

15. Q. Who is first in fame among ancient lyric poets? A. Pindar.

16. Q. What does Sappho remain to this day in general estimation among those entitled to adjudge her just rank, from the various trustworthy indications that survive? A. The foremost woman of genius in the world.

17. Q. What is the only complete poem that has come down to us from Sappho? A. The "Hymn to Aphrodite."

18. Q. On what does the fame of Simonides chiefly rest? A. On his epigrams.

19. Q. What is the most celebrated, perhaps, of all the epigrams of Simonides? A. That on the Spartan Three Hundred who fell at Thermopylæ.

20. Q. What is the great name in Greek idyllic poetry? A. Theocritus.

21. Q. What two other pastoral poets are associated with Theocritus, in a kind of parasitic renown? A. Bion and Moschus.

22. Q. From what two idyls of Theocritus does our author give presentations? A. The "Death of Daphnis," and the "Festival of Adonis."

23. Q. Who is first among the masters of eloquence? A. Demosthenes.

24. Q. The name of what other orator is associated with that of Demosthenes? A. Æschines.

25. Q. What are the most celebrated of Demosthenes's public orations? A. The "Olynthiacs," the "Philippics," and the oration on the "Crown."

II.—QUESTIONS AND ANSWERS ON "CHEMISTRY," FROM PAGE 85 TO PAGE 156, INCLUSIVE.

26. Q. Why is chlorine a substance of very great commercial importance? A. On account of its extensive use as a bleaching agent.

27. Q. Of what common article is chlorine an important constituent? A. Salt.

28. Q. What are the three most striking properties of chlorine? A. Its noticeable weight—greater than that of the air—its greenish color, and its exceedingly irritating odor.

29. Q. In connection with what two principal properties does chlorine, as a chemical agent, manifest its activities? A. Its affinity for hydrogen and its affinity for the metals.

30. Q. Of what may the substance known as bleaching-powder be spoken in a general way as consisting? A. Of lime saturated with chlorine.

31. Q. When was bromine first recognized as an elementary substance, and by whom discovered? A. In the year 1826, by Balard, a French chemist.

32. Q. Where does the substance bromine occur? A. In the brine of the ocean, and in the water of mineral springs, united with certain metals in the form of bromides.

33. Q. To what does bromine show very decided resemblances, in its chemical relations? A. To chlorine, having affinities for the same substances, only less in intensity.

34. Q. In what processes is bromine an important substance? A. In the processes of photography.

35. Q. In what form has bromine had a very wide and beneficent use, as a remedial agent? A. In the form of potassic bromide.

36. Q. What is the other member of the chemical family to which it may be said chlorine and bromine belong? A. Iodine.

37. Q. Where are all these three elements found? A. In sea water.

38. Q. From what source is iodine obtained? A. From sea weeds.

39. Q. To what are the chemical characteristics of iodine throughout closely allied? A. To those of chlorine and of bromine, only in general, iodine may be said to have weaker chemical affinities than either of the other two.

40. Q. What are two of the principal uses of iodine? A. In photographic processes, and as a remedial agent.

41. Q. What remarkable statement is made of fluorine? A. That it has never been known to be produced isolated, that is, in a separate or uncombined form.

42. Q. What property above all others is characteristic of fluorine? A. Its striking affinity for silicon.

43. Q. With what substance is fluorine never known to form any compound? A. With oxygen, which can be said of no other element.

44. Q. What are three considerations upon which the importance of oxygen depends? A. The surpassing abundance of the substance itself, the great number of compounds into which it enters, and the activity of its chemical powers.

45. Q. To whom is the first discovery of oxygen usually attributed? A. Dr. Joseph Priestly, an English clergyman and student of natural science.

46. Q. What is the most prominent compound of oxygen? A. Water.

47. Q. What are some of the remarkable properties of sulphur? A. The ease with which it melts; the readiness with which it takes fire and burns in the air; the striking blue flame produced when it burns; the choking and

disagreeable odor attendant upon its combustion; and its burning when in the pure form without leaving any ashes.

48. Q. From what localities is the principal supply of sulphur for commerce obtained? A. From the volcanic districts of the island of Sicily.

49. Q. What is said as to the number of elements with which sulphur combines? A. It combines in simple form of union with a majority of the elements known.

50. Q. What are three important compounds of sulphur? A. Sulphuretted hydrogen, sulphur di-oxide, and sulphur tri-oxide.

THE TRUSTEES REORGANIZE CHAUTAUQUA.

On the thirteenth of January the Chautauqua Board of Trustees held its annual meeting in the elegant rooms of the Young Men's Christian Association of the city of Pittsburgh, to prepare the way for the next great Assembly. Mr. Lewis Miller, Dr. J. H. Vincent, Messrs. F. H. Root, Jacob Miller, E. A. Skinner, W. A. Duncan, Dr. J. T. Edwards, Rev. J. Lester, Rev. H. H. Moore, and most of the trustees were present, but as usual, of the twenty-four members, letters of apology were received from a few who were detained at home by sickness or urgent business matters. Those present, however, were fully prepared to go forward and meet the responsibilities of the hour. Dr. T. L. Flood, editor of *THE CHAUTAUQUAN*, and Judge Holt, attorney for the corporation, were present to look after their respective departments.

As they came together for deliberation the trustees felt the inspiration of a history of grand successes, of a present satisfactory, and of a future full of hope. Hence the boldness of their plans, and the energy with which they were carried into effect. Chautauqua has a constituency which is of inestimable value, in the prayers and sympathies of many thousands of people who have never seen those beautiful grounds.

Wherever the Board of Trustees hold their annual meetings a lively interest is created, especially among press reporters and in the C. L. S. C. part of the community. In this respect Pittsburgh surpassed any other place ever visited, Jamestown and Cleveland not excepted. On reaching the city it was found that a reception had been arranged by the alumni and members of the Chautauqua Circle, to be held in Christ Church on the evening of the 13th, and that an elaborate program of exercises had been provided. The Rev. Mr. Williams, of the Pittsburgh *Christian Advocate*, occupied the chair. Music was furnished by Hamilton's Junior Orchestra. Dr. Hirst, pastor of Christ Church, delivered, in chaste and eloquent language, an address of welcome. Prof. Holmes, Registrar of the Chautauqua University, in reply,

spoke at length, explaining its aims and method of operation. President Miller followed in his happiest vein, and made clear the point that the educational scheme of the Circle was well suited to meet the constant and progressive changes ever going forward in society. On being introduced, Dr. Vincent was received by the great audience with a storm of applause. In his own usual taking way he unfolded the principles embraced in the Chautauqua Idea. We deal mostly, he said, with the mature mind that is athirst for knowledge. We make use of practical methods to supply the great want of the day, which is a rational society.

Dr. Flood, editor of THE CHAUTAUQUAN, was presented and spoke for a few moments. The music was fine, the speaking the happiest, and after the formal exercises had closed a season of free social intercourse followed. The power Chautauqua had exerted upon the city of Pittsburgh appeared in the great number present, who rose to their feet as witnesses; and most of the cities of the nation could produce like evidence of its popularity and influence.

The lavish expenditures of money which have been made upon buildings at Chautauqua in the past have created such facilities for work of all kinds that at present nothing further is required in that direction. This was a satisfaction not only to those who have heroically carried heavy financial burdens, but to those who have regretted that they were able to give only their sympathies to the cause. The brief address made by President Miller to the Board of Trustees consisted of a brief and cheery review of the past and a hopeful glance into the future. There is, he said, much yet to be done, sacrifices to be made, for Chautauqua is yet in its infancy, and its enlarged work from year to year will demand increased attention. Secretary Duncan in his annual report informed the trustees that during the past year his receipts had exceeded his expenditures by nearly ten thousand dollars, and that this sum had been used as far as it would go to liquidate the floating debt.

The following written report was presented by Chancellor Vincent:

“For the first time in the history of the Chautauqua Assembly I present to the Board a formal report. This has hitherto seemed to me unnecessary, and you have generously accepted a verbal statement in lieu of a full, official

communication. I no longer thus tax your generosity, but under a keen conviction that an important crisis has arrived in our history, I beg leave to lay before you the following statements and suggestions:

“The Chautauqua movement is a marvel even to its projectors. However all-embracing may have been the original conception of our noble president, Mr. Miller, when he proposed a summer gathering in the grove at Chautauqua, the gradually unfolding scheme has been a source of surprise and delight to the world of curious and interested observers.

“Chautauqua in its various departments is a unit. However diverse the outward forms, the name which marks them all proves them one. The ‘Chautauqua Assembly,’ the ‘Chautauqua Literary and Scientific Circle,’ the ‘Chautauqua School of Languages,’ the ‘Chautauqua Teachers’ Retreat,’ the ‘Chautauqua Young Folks’ Reading Union,’ the ‘Chautauqua School of Theology,’ the ‘Chautauqua University’—all are but developments of the radical idea of Chautauqua, which is popular and symmetrical education; education for all people; education in all lines, according to varied tastes, needs and opportunities.

“Our constituency is as broad as are the aims of the institution: Sunday-school and other Christian workers, day school teachers, students of language, ministers of the gospel, citizens who mold the nation, mothers who mold citizens by making homes—these all, and all beside who seek knowledge, character and usefulness, are the people for whom Chautauqua was organized.

“With this wide reach of purpose it was necessary that Chautauqua should project the lines of its intention in plans and departments, that the world might see its magnitude, and that the full territory it proposes to occupy might be preëmpted. Until this projection was made, the Chautauqua Idea was irrepressible. And now Chautauqua with its variety of departments is not like a mere pile of buildings, with additions, lean-tos, unrelated edifices, and other after-thoughts, the results of unmanageable ingenuity. It is a growth and development, a provision according to the highest law, to meet the necessities which called it into existence.

“In this growth of twelve years there have been no unnecessary additions. To have omitted any of them would have made Chautauqua less

than it is; and to have made Chautauqua less than it is would have been a mistake—almost a disaster. Because of the broad and varied provisions now included in the Chautauqua movement, it will be greater and stronger for all time to come.

“It would not have been easy to organize these departments at first under a single charter. The separate schemes under separate constitutions came into being. Each is stronger to-day because of the relative independence of its origin. The time may have come, I think the time has come, for an external union of departments which have all along been practically one. No antagonism between them has ever seemed to me possible, but there is a way of preventing even the seeming or fear of such antagonism.

“At the first meeting of the Board of Trustees of Chautauqua University, I proposed the appointment of a committee whose business it should be to bring into complete external unity all departments of Chautauqua. This committee has never acted. I now renew the proposal, with some practical hints looking toward this result.”

Dr. Vincent then presented several suggestions designed to harmonize the various Chautauqua interests.

The report continues:

“The financial condition of Chautauqua is a subject to which I have heretofore given little attention. I trusted implicitly to the wisdom of the Board, whose large ideas of the Chautauqua work, whose enthusiasm in it, and whose generous courtesy toward me, have caused them to give me the largest liberty, and to treat with great gentleness what they have sometimes felt to be excessive expenditure.

“My dreams and aspirations concerning the development of Chautauqua have led me to plan largely, and to spend liberally, that the attention of great-hearted men might be attracted to our work, the sympathy of progressive educators secured, and the great centers of influence in pulpits, colleges and newspapers be commanded in the interest of Chautauqua. A careful analysis of these expenditures will show that there has been no extravagance, although a greater economy might have been exercised.”

The report of Dr. Vincent closed with the following words:

“Trusting that you will see your way clear to coöperate in the plans proposed, and commending our great institution to him who is the ‘Master of Assemblies,’ this report is respectfully submitted.”

The report of Dr. Vincent was submitted to a special committee, which presented the following report, which was adopted as below:

The special committee to which was referred the report of the Superintendent of Instruction, begs leave respectfully to report:

1. That we recommend to the Board to reorganize the union of the several associations, schools, and departments of the Chautauqua Assembly, the Chautauqua School of Theology, the Chautauqua University, and the Chautauqua Literary and Scientific Circle, under a single organization to be known as THE CHAUTAUQUA UNIVERSITY.

2. (This article calls for necessary legislative action.)

3. The work of the new organization shall be carried on under the following departments:

I.—THE CHAUTAUQUA ASSEMBLY, embracing:

- a.* The Summer meetings at Chautauqua;
- b.* The Sunday-School Normal Department;
- c.* The School of Languages;
- d.* The Chautauqua Teachers’ Retreat;

II.—THE CHAUTAUQUA LITERARY AND SCIENTIFIC CIRCLE;

III.—THE CHAUTAUQUA SCHOOL OF LIBERAL ARTS, now known as “The Chautauqua University,” and with powers as provided in its charter;

IV.—THE CHAUTAUQUA SCHOOL OF THEOLOGY, with purposes and powers as in its charter;

V.—THE CHAUTAUQUA PRESS.

4. There shall be three committees, of three persons each, appointed by the Board, to coöperate with the Chancellor in the management of the above departments. Persons may be eligible to appointment on these committees who are not members of the Board.

These committees shall be:

- a.* A Committee on Assembly;
- b.* A Committee on C. L. S. C., the C. S. L. A., and C. S. T.;
- c.* A Committee on the Chautauqua Press.

5. We approve of the recommendation of the Superintendent of Instruction, of the establishment of the “Chautauqua Press,” as a part of the “Chautauqua University.”

6. The income from the general membership fees in the Chautauqua Literary and Scientific Circle, in the Chautauqua School of Liberal Arts, and in the Chautauqua School of Theology, with such annual appropriations as may be made by the Board to these departments, shall constitute their fund respectively, out of which all expenses of each of these departments shall be paid annually; the surplus in the department treasury of the C. L. S. C., of the C. S. L. A., and of the C. S. T., from year to year, shall be paid to the treasurer of the Assembly Board. There shall be two assistant treasurers, one to have charge of the funds of the C. L. S. C., and the other of the funds of the C. S. L. A. and of the C. S. T., both of whom shall make an annual report to the Board.

7. We recommend that the Superintendent of Instruction in the Assembly Board shall hereafter be known as “The Chancellor of the University.”

8. It shall be the duty of the Chancellor of the University to arrange and conduct the program of the Chautauqua Assembly; to engage speakers, teachers, leaders of music, and such other assistants as the program may require; to conduct the affairs of the C. L. S. C., the C. S. L. A., and the C. S. T.; he shall submit a report to the meeting of the Board in January of each year, which shall contain a statement of his expenditures in the several

departments during the preceding year, and an estimate of the probable expenses for the year ensuing.

9. The duties of the other officers shall be those specified in the by-laws as already adopted by the Assembly, the Chautauqua School of Theology, and the University, or as may be hereafter adopted.

As it has been abundantly demonstrated that this section of the lake is rich in natural gas, it may be expected that as the season opens the work of development will commence. Chautauqua has also an inexhaustible mineral fountain, which many have found not only a pleasant beverage, but rich in health-giving qualities.

Appearances indicate that the next Assembly will be of the first importance. Many will probably meet J. B. Gough—the hero of a thousand platforms—for the last time. Dr. Deems is to come among us once more, and the original Fisk Jubilee Singers will be there, and they have no equals in reproducing the fast vanishing songs of the plantation.

EDITOR'S OUTLOOK.

THE GREAT GREEK DRAMATISTS.

The Greek drama, which is now before our C. L. S. C. students in its English rendering, presents many interesting aspects to the modern mind. We are well aware that this statement will surprise some readers; but let them consider a few facts. Is it not a remarkable thing that the Greek drama, which the world will not suffer to be forgotten, was all produced in the space of about half a century? And compare the fact with our own history, noting that English drama of the imperishable type is all gathered into a single brilliant period, of which Shakspeare is the central light in dramatic poetry. Æschylus was born in 525 B. C., Sophocles in 495, and Euripides in 480. The three are nearly on a level in merit, Æschylus having the more force, and his compeers in dramatic fame more of the refinements of art. They are not three Shaksperes in one constellation, but three orbs whose combined light is less than that of our English poet. No one has satisfactorily explained why the tragic drama should so isolate itself in the centuries of a people; but it may be said to be a rule that if a people produce a great drama at all, this choice fruit will appear only in a single period. But since Greece and England are the only nations having a great tragic stage—for we do not reckon the French drama as in the first rank—the rule has no well-defined value. It is remarkable, too, that the great epic poets are more numerous than the great masters of tragedy. Greece, indeed, is known to us for one epic and three tragic poets; but every great people before ours has had a respectable epic poem, whereas in most nations tragic poetry is rare or inferior. The great tragedies are so few that one may easily know them and prize them. No other form of literature presents us with so few masterpieces.

Another good aspect of the drama in Greece is that it came in the period of the full-flowering of Greek egotism, or if the phrase is happier, of Greek patriotism. Here, too, we may find an analogy in the England of Shakspeare.

The age of Elizabeth is easily fixed upon as that of self-satisfied British patriotism. It is also true that alike in the Athens of the dramatists and the London of Shakspeare there was the stir and bustle and heroic energy of national life. It is not to be overlooked that the dramatists of Greece, like the literary statesmen of England were in public life. They sought and held office; and, indeed, they, like Socrates, were soldiers besides. Æschylus fought at Marathon, Salamis, and Plataea, the three great battles of his country. Shakspeare did not do his work on the Avon, but in the din of London, and many a thing which surprises us in his plays may be explained by the close and uninterrupted contact of the dramatist with the active men of his time. He learned, for example, all his law phrases in convivial association with lawyers, much as he learned scripture by hearing the prayer-book read in the churches. (He always quotes the prayer-book, never the Bible.) So in Athens, our statesmen-dramatists lived in the full press of life, and their drama reflects the opinions and proverbs of their day. Men of the lamp could not have caught the spirit and attitude of the Athenian mind toward the problems of life which underlie the Greek plays. The just-enough and not too much or too little of philosophy—the mean between dogmatic theology and crude irreligion—the man of the world, and he alone, can hit. We may safely reason that while many forms of literature can be best wrought by men out of the world, some forms seem to require their producers to be in the world and of the world; and among these, the drama is especially reserved to men who combine practical experience with erudition, and also possess the indescribable mystery of genius. The student will be well repaid for his pains who struggles to understand the spirit—a strange one to us—which is peculiar to the Greek drama, the singular aspects and functions of religion, and the mode in which it is apparently held fast by the tragic poet, who is also a man of the world. There are also profitable studies to be made of those glimpses of unchanging human nature which the tragedies afford us. One theory is that we study old classics in order to know an older and extinct type of mankind: a truer view is that the virtues and vices of the elder man are simple and undisguised by social varnish. In any case, the student who understands, for example, the woman Medea, has a useful lesson in “the proper study of mankind.”

CHAUTAUQUA AT NEW ORLEANS.

Almost every interest of the country is represented at New Orleans this winter. Every prominent manufactory, all leading trades, the great branches of commerce, and particularly educational institutions have exhibits of more or less importance. The eye of the country is turned southward. Whatever is worthy our civilization has been collected there for study. In educational matters many departments have been given position, that they may be studied by the eager learner; for people are eager to know the world's work. You see it in their keen observation of the displays made throughout the long galleries, and their quick notice of the comparative merits of the exhibits. To them the work from the Indian schools, from the colored people, from the far away territories, and from foreign lands are studies in comparative civilization. Every sign of advancement is quickly seized upon; and in no department is more eagerness to know manifested by visitors than in the "Chautauqua Alcove."

Chautauqua and the C. L. S. C. have a very good representation in the south gallery of the government building, under the general supervision of Prof. E. A. Spring, a member of the faculty at Chautauqua. This exhibit is attracting a great deal of attention. All members and friends of the C. L. S. C. who may visit the Exposition are earnestly and cordially invited to visit the Chautauqua exhibit. An idea of Prof. Spring's work may be obtained from a few extracts from a letter received from him in the holidays:

CHAUTAUQUA ALCOVE, NEW ORLEANS, LA., Dec. 26, '84.

There is a large placard up in this exhibit, as follows:

U. S. BUREAU OF EDUCATION.

CHAUTAUQUA ALCOVE.

*Any one of the sixty thousand of the members of the C.
L. S. C. who may be here is requested to register.*

I give everybody one of a little handful of Spare Minute Course circulars that I brought with me. I have given out about a hundred—and had conversations, some of them with evident conviction—in German, French, English, and the language of signs. By help of my Italian, I have tried to talk to some of the

many Mexicans here, but did not get deep enough to broach the C. L. S. C.

I had yesterday and to-day considerable talk with the intelligent gentlemen representing the French Republic school system—M. Buisson and his assistant. General Eaton says that the brother of this Mons. Buisson is the genius of education in France. They expressed themselves as much interested in the scope of the noble Chautauqua plans.

The principal and seven lady teachers from Normal, Ill., just left me. They will come again, to learn more of Chautauqua. Some Texas gentlemen, one of whom, Prof. Hogg, I have long known as a fellow member of the National Educational Association, have been here again to-day, to say that they would be here to see me with about two hundred teachers from Texas at noon to-morrow.

Dr. Mitchell was here this afternoon.

On Saturday, the 27th, I was all ready for the two hundred teachers, who advanced in a body with a banner, and I gave them a regular lecture on Chautauqua and its out-reachings.

Then I said my friend, Prof. Hogg, had seen me model in Philadelphia, in 1879, at convention of the National Educational Association, where he read a paper on “The Education of the Hand, the Head, and the Heart,” and he had asked me to show this company of teachers some clay modeling, so I would occupy a few minutes in that, as it was one of the methods in the Chautauqua plan to train the hand by clay, and through that educate the head; and if the hand and the head were truly educated, as they ought to be, the heart should be developed too. So, laying out a colossal head in relief, I made a few remarks as to the value of a little easy practice of clay modeling in schools; and then, turning to the board with clay on it, I worked eighteen minutes, and made a head of “an American Teacher.”

General Eaton came to me after their vote of thanks, and as soon as they had gone—in the most congratulatory frame. “Chautauqua could afford to pay you two months’ work for that!” he said, shaking me by the hand. Two of the class of ’86 C. L. S. C., from Lockport, Dr. Mitchell, Mons. Buisson and a few others expressed themselves as much interested.

I was glad to find I had so many “Spare-minute” circulars—and I must have given out seventy-five or one hundred yesterday, beside one hundred to the Texans, generally accompanied with a conversation of more or less length.

All this in the midst of busy work and good progress in the mechanical embellishment of our alcove. It will be very attractive when completed, and I have so planned that I can work at it all along, adding new features from day to day.

It is very interesting, how near to people’s hearts and inner lives I sometimes get in these little talks. It is a plan that touches the aspirations and longings of many a true soul. I wish sometimes that words could be instantaneously photographed. It is impossible for me to write as fully as I should like.

Our only CHAUTAUQUAN (November) with one copy of the C. L. S. C. circular, with its cut of the Hall, has done good service.

I very much hoped that Dr. Vincent could manage to come here. Many people have asked, the first thing, if Dr. Vincent is to be here. Every state should have a Chautauqua headquarters—this alcove will get them all ready for it.

Monday morning.—Damp, muddy, discouraging to many people. The car drivers have struck and the hour’s ride, long and tiresome at best, is now cut off. Hundreds of teachers who have been pouring into New Orleans these last days of their holiday, are prevented from seeing and learning by this four or five miles of mud before they reach the Exposition. There have been great hindrances all along to the completion of the Exposition, and many grumblers. But I have never been discouraged! Everything

from the first start has been delightful. When the roof leaked, I moved some things away, told the roofer, and it was at once mended. When it came on a hard storm night before last, I laid down on the floor, rolled in my Kansas blanket, and liked it so well that I shall camp out here in the Chautauqua precincts; at any rate, till there is some comfortable conveyance away.

More anon.

Ever yours faithfully,

EDWARD A. SPRING.

WINTER SPORTS IN CANADA.

Winter undoubtedly has its hygienic value; and a part of this value we get without effort. It is not only a comfort to be freed from the annoyances of insect life, but it is also a gain for health that many of the atmospheric impurities are removed by frost. But to get the largest value from winter as a frost cure, we need to avail ourselves of the system of healthful and invigorating amusements which prevail in Canada, and have made that country famous. That portion of our population which is employed outdoors in winter is, *pro tanto*, undoubtedly the most healthy. For the rest of us the only possible equivalent is winter sports. It is unfortunately true that the variable character of our weather precludes us from exact imitation; but our inventive genius ought to be equal to the task of bridging over the soft places in our winters. In Canada, the long and comparatively equable winter makes it a simple thing to provide healthy and innocent amusements which may be enjoyed as regularly as any business is carried on. It is not to be forgotten, however, that the Canadians are the only people in the world who know how to keep warm out-doors as well as in-doors. They have learned to perfection this art, for lack of which our out-door employments are more or less dangerous. Our laborer does not keep warm in winter, and his "colds" become consumption. In Canada, young girls accomplish in this respect what stout men fail to do among us; they keep warm whether they are flying in sleighs or on toboggans. These forms of enjoyment are well organized; there are toboggan clubs, and "society" means some form of

winter sport. The miserable imitation called “roller-skating,” which is alarming thoughtful people in many of our villages, is only a craze, a temporary insanity; the winter sports of Canada are a national institution. The physical and moral wholesomeness of the roller-skating rink is more than doubtful. The moral and physical healthfulness of the sports by which Canadians make winter a season of joy, can not be questioned.

On the average, our winter in the United States is not a healthful and invigorating season to us. We lose the greater values and expose ourselves to special dangers. We live in-doors, with a temperature ten degrees too high. We shut in with us invisible plagues which breed diphtheria and other diseases. We are enfeebled by refraining from exercise and breathing unwholesome air in our houses. We come to the spring weaker than we were when winter began. We have moped by hot fires and breathed vitiated air, when we ought to have been out in the winter blast, using our muscles and filling our lungs with the clean winds. Two or three conditions seem wanting for a reform of these habits. One is the art of keeping warm in the cold air; another is a keener sense of the value of winter exercise, and a third is some devices by which the “soft spells” of weather shall not arrest our sports nearly every week.



THE RELATIVE PRONOUN “THAT.”

This word is a demonstrative pronoun and a conjunction; and in some idiomatic phrases it is also a relative pronoun. By idiomatic phrases, we mean that use has constructed certain forms of expression which are wholes, though consisting of several words. *All that we know* is an idiomatic phrase; use and habit have welded the words together. In the fifteenth and sixteenth centuries, there grew a habit of using *that* very freely as a relative pronoun. The Bible of 1611 is full of illustrations of this habit. During the present century this use of *that* has been by the best writers gradually restricted, and at present the rule for *that* as a relative pronoun, probably, is about as stated above—the word is used, as a relative, only in idiomatic expressions. The history of this word would make a very interesting chapter. We have in the foregoing statement merely suggested one line of change in its use, and we call attention to this change for a

particular reason. Among the excellent books published by Appleton & Co. is a reprint Cobbett's English Grammar, and in this reprint Mr. Ayres, the editor of it for this republication, lays down in his introduction, and illustrates by significant bracketing in Cobbett's text, a new rule for the use of the relative *that*. This new rule is, in substance, that the restrictive relative is *that*—*who* and *which* being coördinating relatives. This proposed reform is unfortunately timed. By a progress in use which has been unobtrusive, and unaided by dogmatism, the number of *thats* on a printed page has been reduced to tolerable proportion. If we accept the new rule we shall not only go back to excessive use of *that*, but we may even increase the evil of too much *thating*. The word fills two important functions in present good use; to add the office of expressing all the restrictive uses of the relative pronouns, would probably increase *thating* so as to render an English page unsightly. Take a sentence: "He said that that man that that boy said that he saw was not that man that that boy thought that he saw." Mr. Ayres tries to show that certain sentences which contain *who* and *which* as relatives are ambiguous in meaning, and that the substitution of *that* would make the meaning clear. As to such sentences, we may say that if they are really ambiguous in sense, the remedy is to reconstruct them. It is not necessary to use *that* to pull them out of their obscurity. It is easy, however, to show that a detached sentence might mean something which it does not mean. The meaning of a text is helped out by the context. Aphorisms usually have not context auxiliaries, and usually are ambiguous; but the ordinary use of language is to express our meaning by paragraphs rather than by single sentences. Every ellipsis furnishes an opening for the entrance of small criticism; and ellipsis is one of the large facts of English writing.

In short, the critic of ambiguous sentences will have abundant employment on the best writers, if he is allowed to break off any sentence from its yoke-fellows in the paragraph. We advise our readers not to make haste to adopt the rule of Mr. Ayres. The important question is: How do good writers employ the word *that* in their books? The answer is that good English writers employ the word as a conjunction and as a demonstrative; and as a relative only when phrase idiom compels such use. In this country, the practice is to use *that* a good deal as a relative; but there has been a great decline in this use of it, especially during the last thirty years. At the

present time our best writers seem to be following the English practice. We hope that Mr. Ayres will not succeed in turning reform backward. With *who* and *which* to employ as relative pronouns—and occasional help from *that*, *what*, and *as*, in idioms—the English language is not poor. We need not recall the restrictive *that* from its honorable retirement.

EDMUND ABOUT.

The death of this versatile French writer removes from modern literature another of the few French literary men who are known all over the world. About was born in 1828, and has enjoyed a cosmopolitan fame since 1860. His literary work had the charm of contemporary interest, and at the same time the merit of philosophical breadth and insight. He gained fame at home in a way our men of letters would not travel, by writing a Dictionary of Railroads. He was equally interesting and instructive whether he wrote a novel or a political pamphlet; for in both, Edmund About's personality was in the foreground. He was not an egotist, however, in his books, but his *I* was a modest one which rather relieved others of responsibility for his opinions, than obtrusively forced the author upon our admiration. He had a keen zest for current thought and fact; and, though our sensationalist newspaper men would not fellowship him, he was one of the best editors of his age. He was always an editor—even when he wrote the railroad dictionary—and his political pamphlets are among the best presentations of questions and situations. He saw the heart of a current issue, and with easy grace and perfect poise he described it from the point of view of a modern cosmopolitan gentleman. His "Roman Question" was, in its day, equally intelligible, interesting, amusing and illuminating in Paris, London, Vienna or New York. He described the situation under the Pope as King of Rome, setting out in full relief those peculiarities of the Roman situation which were picturesquely illogical for all men of the world. He had a marvelous power of suggestion. The first sentence of the "Roman Question" is like this—we do not attempt to recall the exact words or figures: "There are in the States of the Church 1,366,328 souls, *not counting the little Mortara boy.*" The last clause referred to a charge that priests at Rome had stolen a Jewish boy and were making a good Catholic of him against the will of his family. The incident made a great uproar at the time, and About recalled to the

mind of the reader the whole story, and, without expressing an opinion, attracted the sympathy of his Protestant readers by the mere allusion. Probably his books contain more examples of strong, suggestive allusions to recent or contemporary events, than those of any other writer; and it was always his special art to allude *only*, leaving his reader to his own opinion. The delicacy of his touch and the fine flavor of his criticism were remarkable, even in this age of keen and witty French writers. He became editor of the *XIX. Siecle* (century), after the war with Germany; but he had always been a journalist in some form, and more than one paper had its editions suppressed by Napoleon III. because they contained the fine but biting satire of About. Some years ago (in 1870, we believe) he was blackballed in the French Academy, but he was recently elected to that august body. He died before he was installed in his academic chair.

EDITOR'S NOTE-BOOK.

We regret exceedingly that a serious illness makes it impossible for Mr. Richard Grant White to furnish his usual paper to the present issue of THE CHAUTAUQUAN. Another month Mr. White will probably be able to continue his articles.

The glimpse we are getting, even at this early day, of the Chautauqua program for 1885, is very inviting. The regular School of Science will be under the charge of Prof. Edwards, president of Chamberlain Institute, Randolph, N. Y., and that of Pedagogy, under Dr. Dickinson, secretary of the board of education, Boston. Such people will be present as John B. Gough, Dr. Deems, Miss Willard, Mrs. Livermore, Bishop Foster, Dr. Boardman, of Philadelphia; Dr. G. P. Hays, of Denver, who will organize a school of Christian work; the Schubert Quartette, of Chicago; the original Fisk Jubilee Singers, for two weeks, and Miss Henninges, the noted singer of Cleveland, O. A very superior organist, Prof. Isaac V. Flagler, has been engaged for the entire Chautauqua season.

The alarm which the recent terrible earthquake in Spain has caused has led to the compilation of some interesting figures relative to the number of shocks which have occurred in late years. Between 1872 and 1883 no less than 364 earthquakes are recorded as occurring in Canada and the United States, not including Alaska. Of these the Pacific slope had 151, the Atlantic coast 147, and the Mississippi valley 66. Thus it appears that an earthquake occurs about once in every twelve days somewhere in the United States and Canada, and about once a month on the Atlantic coast. These are exclusive of the lighter tremors which do not make an impression on observers, but which would be recorded by a properly constructed seismometer, an instrument designed to detect the slighter shocks.

“Just about twenty years ago,” writes Dr. Felix Oswald in a recent letter to THE CHAUTAUQUAN, “when I was stationed at Sidi Belbez, in western

Algiers, I had a conversation with a half-civilized Sheik, who had visited our camp and seemed to take a good deal of interest in the portrait of a *mitrailleuse* ("Gatling gun") that had been photographed together with a group of Zouave artillerists. After scrutinizing the picture and comparing it with the original, he clutched his head, as if stunned by his emotions. 'Where do they teach such things?' he inquired, and then suddenly burst out: 'What a pity that education and Gatling guns can not be had at home!' For North America, at least, THE CHAUTAUQUAN seems to have solved one of those problems."

In a yellowish, time-worn volume bearing the title, *The Allegheny Magazine, or Repository of Useful Knowledge*, issued in Meadville, Pa., on July 4, 1816, we find in a paper on Chautauqua the following: "The tradition among the Seneca Indians is, that when their ancestors first came to the margin of this [Chautauqua] lake and had reclined their weary limbs for the night, they were roused by a tremendous wind which suddenly and unexpectedly brought the waves upon the shore to the jeopardy of their lives. The aboriginal history as handed down from father to son further represents that in the confusion of the scene a child was swept away by the surge beyond the possibility of recovery. Hence the name of the lake *Chaud-dauk-wa*; the radix from which this is formed signifying *a child, or something respecting a child*. The word is usually spelled *Chautauqua*; but, according to the pronunciation of the venerable Cornplanter, whose example is the best authority, it should be written *Chaud-dauk-wa*, the two first syllables of which are long, and the consonant at the end of each is to be distinctly sounded."

Mr. Francis Murphy, the apostle of temperance, who, by the way, is engaged to speak at Chautauqua next season, is a very useful and popular man in Pittsburgh, Pa. Mr. Murphy has recently been invited to become the pastor of a People's Church which leading citizens of Pittsburgh propose to establish. He is a powerful man with the masses, and his method of "Gospel-Temperance" is a wise one. By his efforts tens of thousands of drinkers, drunkards and saloon keepers have been led to become better men. We shall watch the new departure in Pittsburgh with a great deal of interest.

Bishop Hurst has discovered in Cairo, Egypt, the next largest university to Chautauqua in the world. His rich article on the "Mohammedan University," in this impression, fixes the number of students in attendance at about 15,000. The C. L. S. C. numbers more than 60,000, and the class of 1888, organized this school year, will reach nearly, if not quite, 20,000 members.

The recent terrible explosions in London have set us to counting up the similar outrages which have been perpetrated of late in England. In 1881, attempts were made to blow up the armory at Salford, the Mansion House, London, the Lord Mayor's private apartment, the barracks at Chester, the Central Police Station at Liverpool, and the Town Hall at Liverpool. The activity of the dynamiteurs was checked about this time by the vigilance of the police, and nothing further was done until March 15, 1883, when the Local Government Board offices in Westminster, near the House of Parliament, were nearly destroyed by an explosion of dynamite. In 1884, attempts were made to shatter three railway stations in London, explosions occurred in Scotland Yard and at St. James Square and under London Bridge. Already, in 1885, there have been an explosion on a London underground railway, and the outrages in Westminster Palace and the Tower.

What shall we do? How shall we treat these outrages? We can do nothing. To be sure it is a shameful list of cowardly, ineffectual deeds. Yet they deserve more pity than rage. It is a sad thought, that in rich, cultured, high-bred old England, there can exist a class so weak, cruel, and miserable that it tries to right its wrongs by methods more horrible than those of war.

A very suggestive scene took place recently in the Arkansas Assembly. Engrossing and enrolling clerks were to be elected. The members brought up the names of several ladies, discussing their ability, beauty, and claims to recognition, in most eloquent terms. After a long and amusing discussion, both positions were filled by ladies. This move gives to the self-supporting women of Arkansas a new outlook. The possibility of securing such positions will incite hundreds of women to prepare for clerkships, which if

not found in the legislature will surely be found elsewhere, as the peculiar ability of women for such work is recognized.

The legislature of Georgia, at its past session provided a similar opportunity for the women within its borders. Eight to ten clerks have been regularly employed each session to assist the clerk of the lower house of the legislature. Of its own accord the House directed that women be hereafter employed to fill these positions. This was done, and the bills engrossed by them are said to have been remarkably neat and accurate. This ready sympathy for the women who must earn their bread, and manly effort to make places for them, is very characteristic of the generous southern heart.

The Assembly at Lake de Funiak, Florida, will be in session when this number of THE CHAUTAUQUAN is on its way to our subscribers. The opening takes place on February 18. It is the first attempt at planting the Chautauqua Idea so far south, but after its fashion it is sure to take root. The preparations made by Mr. Gillet and his associates give promise of a good program. We expect an account of the meeting for the April number of THE CHAUTAUQUAN.

Two big schemes to attract patronage have of late come before the country. At the time the New Orleans Exposition seemed to stagger under its load of expense, and money was absolutely necessary, the Louisiana Lottery tried to get control of the Exposition. General Grant's embarrassment was seized upon by the incorrigible Barnum, who proposed buying the invaluable curiosities and relics of the General, to display in his summer pilgrimages through the country. It makes a person of taste blush to think of this impudence, to remember that there is a very large class of people who are willing to drag into advertising the most dignified and sacred institutions in the country.

The commercial side of Chautauqua Lake does not often reveal itself in the educational work which finds its center there. The beautiful country which forms the setting for the fair lake has, however, more than one most interesting industry. Just now ice cutting is at its height. There is a transit

company which packs dressed meats, eggs, butter, and other perishable articles, at Chicago. When these refrigerator cars start from that city, ice is placed in the cars, which is expected and found to keep the stores in fresh condition, as far as Salamanca; here the cars must be replenished, and it is to these storehouses that the ice which is now being cut from the lake is sent. The company employs men and teams near the lake to cut the ice, and the process is a very interesting one.

Mr. Edmund Yates, the editor of the *London World*, has been committed to prison for four months for allowing in his columns a bit of gossip connecting in an injurious statement the name of a young woman with that of a young nobleman. It is a refreshing sign of the times. Popular sentiment has tolerated an immense amount of personality, of curiosity, and of absolute impudence in the social columns of newspapers. Mr. Yates's punishment will emphasize the fact that the public is not so depraved as editors often consider it. By the way, how like is this affair to that earlier one of Mr. Yates's, when he was turned out of the Garrick Club for publishing a disrespectful paragraph about Thackeray, a fellow-member. It is to be hoped that Mr. Yates will soon learn that it is a mean thing to make one's bread by selling a friend's peculiarities or a neighbor's mistakes and sins.

The Christian revolt of the Jews of Bessarabia, and the establishment of the "National Jewish Society of the New Testament," was discussed by Bishop Hurst in the January issue of this magazine. The founder of this new sect, Rabinowitz, has been since found dead at his home in Kishenev. It is believed that he was murdered. The Christian authorities believe that it is the work of the orthodox Hebrews, and it is not improbable that such is the case. Apostasy in religion very rarely receives from men Gamaliel's advised treatment, and unless the law can secure safety for these reformers, there is but little chance that they will escape the fate which all the history of the past teaches us that religious fanaticism believes to be the just and only treatment.

It is gratifying to know that in all probability the \$250,000 required for the pedestal of the Bartholdi statue will soon be in the hands of the

committee. The difficulty in raising the money has revealed a new side of American generosity. The financial agent of the pedestal committee probably explained it, when he said recently: "The American people are peculiar about these matters. You touch their sympathies and sensibilities, and money flows like water. For flood or fire sufferers you can raise a million dollars in forty-eight hours and have a million more advanced for emergencies by bankers who know that it will be promptly replaced by willing givers. But we haven't got along to the appreciation of art—of great masterpieces like the Bartholdi statue—and so it was hard to raise money for it. In France, under similar conditions, the fund would have been raised in a week."

Apropos of the above a step that is being taken in many cities and towns of late, will undoubtedly do much to cultivate among us the lamented lack of "appreciation of art." It is the establishment of city and village art museums. Worcester, Mass., has had \$25,000 left to her, recently, to invest in an art museum. Smaller sums have been raised in several other towns. A good opportunity to study art thoroughly may be secured to any village by a donation of \$1,000. Casts, photographs, engravings, and a few standard works are sufficient to cultivate correct ideas, and lay the foundation of knowledge. It is the only way in which to raise the standard of taste in the villages remote from the few cities of America which boast art museums.

The question of the date of the birth of Elizabeth Barrett Browning interested the readers of the C. L. S. C. some time ago. The year alone was ascertained. If any one was troubled that we were unable to answer the query exactly, the answer of Mr. Robert Browning to a lady asking for the date of Mrs. Browning's birth may be of some consolation: "I know neither the day, month, nor year of Elizabeth Barrett Browning's birth. It is a subject upon which I have never had the slightest curiosity."

One of the most active public men of the last generation has been Schuyler Colfax. He has been prominent, both as a political leader and as a public speaker. Mr. Colfax's life of a little more than sixty years was immensely busy. He made his career, beginning as errand boy and clerk in his grandfather's store. After he was eighteen years of age he took up the

study of law, then launching out as a journalist, and finally, at twenty-five, entering the world of politics as secretary of the Whig National Convention, to which he had been sent as a delegate. When the new Republican party was started, Mr. Colfax was sent as a representative to Congress, and from that time he was closely identified with his party, serving particularly as Speaker and Vice President. He was of that large class of industrious, quick witted men who make themselves indispensable in whatever relations they are placed.

The “Imperial Dictionary” promises to be the rival of all the old standard dictionaries among scholarly people. Its form is four good sized volumes, which signifies that the English language grows and grows, and that words need fuller explanation. Mr. Gilder, editor of *The Century*, explained to us, when on a recent pleasant visit to the *Century* offices, that the “Imperial Dictionary” was built on “Webster’s Dictionary” in England, and that scholarly men had devoted ten years to the task. Now the Century Company have more than two hundred scholars engaged in making improvements on the English edition. It will be seven years before the new American edition will be ready for the market.

A timely and practical department of the Chautauqua University is the School of Journalism. This school is under the able direction of H. W. Mabie, one of the editors of the *Christian Union*. The demand for such schools is great, and the fact that all the work between teacher and pupil in this new undertaking will be conducted by correspondence, is an additional argument in its favor. The plan is briefly this: Three courses of study, with supplementary readings for those who have time for them, have been prepared; theses are expected on subjects assigned, and these will be criticised with special reference to vigorous style; constant correspondence will furnish needed help and hints. The plan is a wise one, its director is able, and there is no doubt but there are numerous young men and women to whom it will open the long desired way out of the woods.

One of the most romantic spots of American history is that of the Florida Chautauqua. Ponce de Leon’s famous quest for the Fountain of Youth lay through this region, and Lake de Funiak itself is fabled to be one of the

springs by which the old knight encamped. Perhaps here he plunged into the clear waters and vainly waited to see himself changed to vigorous youth again. However that may be, the road he laid out is a thoroughfare for Florida travelers to-day, and about the clear lake still hangs the tradition that it is the fabled Fountain of Youth. Ten miles from Lake de Funiak is a second spring which still bears the gallant Spaniard's name. It will be a rare opportunity for dreaming over those early adventures that visitors to Lake de Funiak will have.

The proposed new word, "Thon," which was suggested in the program in THE CHAUTAUQUAN for January as a suitable subject for an essay, seems to have caused our readers some trouble. A word of explanation may help them. We have no pronoun of the singular number and common gender in English. The absence of such a word leads to many awkward circumlocutions. To obviate this trouble Mr. C. C. Converse, a lawyer, has compounded the word *thon*, from *that* and *one*—declined: nominative *thon*, possessive *thons*, objective *thon*. Its use is evident. In this sentence is an example: If George or Anna will meet me I will go with *thon*. The word has been much discussed and much amusement is caused by using it—a practice which, however, demonstrates the need we have for such a word. Prof. March, of Lafayette College, writes: "I do not know that any other vocable would have so good a chance for this vacancy." Prof. Norton, of Harvard, says: "Such a pronoun would undoubtedly be a convenience, did it exist. The difficulty lies in it being *yours*. All forms of speech have grown, and I do not recall an instance of the use by a civilized race of any word, not a noun or a verb, deliberately invented by a philologist, however ingenious."

C. L. S. C. NOTES ON REQUIRED READINGS FOR MARCH.

COLLEGE GREEK COURSE IN ENGLISH.

Articles on Euripides may be found in the following works: Mahaffy's "Classical Greek Literature;" Blackwell's "Introduction to the Classics;" "Studies of the Greek Poets," by J. A. Symonds; *Encyclopædia Britannica*; "Phæton," *Fraser's Magazine*, vol. xlv, p. 488; "Sea Studies" (J. A. Froude), *Fraser's Magazine*, vol. xci, p. 541; "Vindication of Euripides," *National Quarterly*, vol. xix, p. 1.

P. 188.—"The Raging Hercules." One of the most precious remains of Euripides, full of tragic pathos. While Hercules is absent from home, Lycos, tyrant of Thebes, persecutes his father, wife and children. As they are about to be put to death, Hercules returns and a scene of vengeance follows, and Lycos is the one to suffer death.

"Balaustion," ba-la-us'ti-on.

"Sicilian Expedition." See "Brief History of Greece," page 31.

P. 191.—"Mistress." Artemis.

P. 192.—"I-ol'cos." An ancient town in Thessaly, the place from which the Argonauts set sail.

"Stygian barge." The Greek's view of the world entered immediately after death is given in the following quotation from Seemann's "Classical Mythology:" "It was supposed to be a region in the center of the earth, with several passages to and from the upper world. Through it flowed several rivers—Co-cy'tus, Py,ri-phleg'e-thon, Ach'e-ron and Styx. The last of these encompassed the lower world several times, and could only be crossed by the aid of Charon, the ferryman, who was depicted as a sullen old man with a bristling beard. The Greeks, therefore, used to place an

obolus (small copper coin) in the mouths of their dead, in order that the soul might not be turned back by Charon for lack of money. On the farther side of the river the portals were watched by the dreadful hell-hound Cerberus, a three-headed monster, who refused no one entrance, but allowed none to leave the house of Pluto. All souls on reaching the lower world had to appear before the tribunal of Minos, Rhadamanthus, and Æacus. Those whose lives had been upright were then permitted to enter Elysium, where they led a life of uninterrupted bliss; while those who on earth had been criminal and wicked were consigned to Tartarus, where they were tormented by the Furies and other evil spirits. Those whose lives had not been distinctly good or bad remained in the Asphodel Meadow, where, as dim shadows, they passed a dull, joyless existence.”

P. 194.—“Koré,” kō'rā. Persephone or Proserpine, the wife of Pluto.

P. 195.—“Moirai,” moy'rī.

P. 197.—“Strophe.” In Greek tragedy, in its highest development, there was a group of persons, composed of both sexes, who constituted the chorus. When the actors paused the chorus sung or spoke, accompanied by solemn music, moving from one side of the stage to the other. The time of this movement was adapted to the stanzas, so that one, called the strophe, was given as they passed in one direction, and the next, the antistrophe, as they passed back.

“Daughter of Pelias.” Alcestis.

“Seven-chorded shell.” Tradition tells that the first lyre was made by Mercury, out of the shell of a tortoise, which he caught a few hours after his birth. Lyres were employed in recitations of epic poetry, and consisted of a tortoise shell sounding bottom, from which arose two horns, joined near the top by a transverse piece of wood, to which the upper ends of the strings, usually seven in number, which were stretched perpendicularly from the bottom, were fastened.

“Carnean feast.” One of the great national festivals of Sparta, held in honor of Apollo, who had for a surname Carneus, which was derived by some from Carnus, a son of Jupiter and Latona, and by others from Carnus, a soothsayer.

P. 199.—“Lustral bath.” In their early history the only rite of purification observed by the Greeks was that of ablution in water, but afterward sacrifices and other ceremonies were added. These were used to purify individuals, armies and states, and to secure the blessing of the gods. The word lustral is derived from the Latin verb *lusto* and signifies to purify by means of propitiatory offerings.

P. 200.—“Othrys.” A range of mountains in Thessaly.

“Pythian’s sake.” Apollo’s sake.

P. 205.—The lines at the top of the page, spoken by Hercules, contain the same sentiment that runs all through “Rubáiyát,” the poem written by Omar Khayyám. Compare the extracts from this book given in the “Talk About Books,” in THE CHAUTAUQUAN for February, 1885, with these stanzas. To further show the similarity in thought, we select one stanza from the poem:

“Waste not your hour, nor in the vain pursuit
Of this and that, endeavor and dispute;
Better be jocund with the fruitful grape
Than sadden after none, or bitter fruit.”

P. 206.—“Asclepian train,” as-cle'pi-an. Train of physicians, who are often called the descendants of Æsculapius, the god of the medical art.

P. 210.—“Gorgon.” A terrible winged woman, who dwelt with her two sisters on the borders of Oceanus, the river that flowed around the ancient world. She was beheaded by Perseus, who accomplished the perilous task by the help of Hermes and Athena.

P. 211.—“Son of Sthenelus,” sthen'e-lus. Euristheus, who assigned to Hercules his twelve labors.

P. 212.—“Electra.” Daughter of Agamemnon and Clytemnestra. On the return of Agamemnon from the Trojan war, Clytemnestra and her paramour murdered him. When her young brother, Orestes, had grown to manhood, Electra excited him to avenge the death of Agamemnon, and assisted him in slaying their mother.

P. 213.—“Medea.” The wife of Jason, the Argonautic hero.

“Pieria,” pi-e'ri-a. A narrow strip of country along the southeastern coast of Macedonia.

“Harmonia.” Daughter of Mars and Venus, and wife of Cadmus.

P. 216.—In connection with the chapter on Ar-is-toph'an-es, the following works may be read: Mahaffy's “Classical Greek Literature” (readings will be found in this book on all the characters mentioned in “College Greek Course”), “Aristophanes,” *National Quarterly*, vol. iii, p. 70: *Fraser's Magazine*, vol. xii, p. 222.

P. 219.—“Creon.” Cleon is meant, the “leather-seller” who for six years was the most influential man in Athens. He took command of the forces at Sphac-te'ri-a, during the Peloponnesian war, and fulfilled the promise he

had boastingly made, that he would capture the Spartans within twenty days if the Athenians would send him against them.

P. 220.—“Tableaux vivants,” tä-blō vĕ-vă^{ng}. Living representations, in which persons are grouped as in pictures. We frequently use only the first of these French words.

“Soph'ist.” The Sophists were the leading public teachers in Greece during the fifth and fourth centuries B. C. In its original sense, the word meant a wise man, and as such could properly be applied to Socrates. But in his day, as a class, they were “ostentatious imposters, flattering and duping the rich for the sake of personal gain.”

P. 225.—“Rhea.” The wife of Saturn, and the great goddess of the world.

“Hebrus.” The principal river in Thrace.

P. 226.—Readings on Pindar will be found in Talfourd's “History of Greek Literature,” *National Quarterly*, vol. xxxii, p. 203; *London Magazine*, vol. ii, p. 60.

Readings on Sappho, *The Atlantic* (T. W. Higginson), vol. xxviii, p. 83; *Harper's Magazine*, vol. lvi, p. 177; *Appleton's Magazine*, vol. vi, p. 158.

Readings on Simonides, *Westminster Review*, vol. xxxii, p. 99; *Fraser's Magazine*, vol. ii, p. 52.

P. 228.—“Dithyrambics,” dith-y-ram'bics. Originally songs in honor of Bacchus; later, any poems written in a wild and enthusiastic manner.

“The Ivy-clad Boy.” Bacchus.

“Bromius.” One of the surnames of Bacchus, signifying the shouter.

“Eriboas.” See index of “College Greek Course.”

P. 229.—“Prophet of Nemea's strand.” Jupiter.

“Orchomenus,” or-kom'e-nus. An ancient and powerful city of Bœotia.

“Min'yans.” An ancient Greek race, said to have migrated from Thessaly. Their ancestral hero, Minyas, is said to have been a son of

Neptune.

P. 230.—“A-glai'a,” “Eu-phros'y-ne,” “Tha-li'a.” The names of the Graces.

“A-so'pi-chus.” See index to “College Greek Course.”

“Cle-o-dā'mus.” Usually written Cleodæus. A descendant of Hercules, who made an unsuccessful attempt to lead the Heraclidæ back into their own land, the Peloponnesus. Temenus, his grandson, succeeded in the attempt.

“Bellerophon.” A Corinthian, who obtained possession of the winged horse, Pegasus, who rose with him into the air, whence by means of arrows he killed the Chimæra, a fire-breathing monster which had three heads, one that of a lion, one of a dragon, and one of a goat. It had made great havoc in Lycia and the surrounding countries. Afterward he conquered the Solymi, a warlike race inhabiting the mountains of Lycia, and the Amazons, a mythical, warlike race of females.

P. 232.—“Typhon.” A monster who wished to acquire the sovereignty of gods and men, but who was subdued, after a fearful struggle, by Jupiter, and confined in a Sicilian cave. He begot the winds.

P. 233.—“Phal'a-ris,” B. C. 570. A cruel and inhuman tyrant of Agrigentum, who was put to death in a sudden outbreak of popular fury. He is said to have burned alive the victims of his cruelty, in a large brazen bull.

P. 240.—“A-don'is.” A beautiful youth beloved by Venus. He died from a wound which he received from a wild boar. The grief of the goddess was so great that the gods of the lower world allowed Adonis to return to the earth for six months every year. In this myth the death of the youth every year probably represents winter, and his return, summer.

“Cypris” and “Cyth-e-re'a.” Venus.

P. 241.—“Arethusa.” The nymph of the famous fountain of Arethusa, on the island of Ortygia.

P. 242.—“Meles.” A small stream in Ionia, on the bank of which Homer is said to have been born.

“Pegassean fountain.” The inspiring well of the muses on Mt. Helicon, said to have been formed from a kick given by Pegasus. It is sometimes called the Hippocrene.

“Daughter of Tyndarus.” Helen of Troy.

“Son of Thetis.” Achilles.

“Eros.” Cupid.

“Al-ci'des.” Hercules.

“Orpheus.” See C. L. S. C. Notes in THE CHAUTAUQUAN for November, 1884. Eurydice is the wife of Orpheus, instead of Proserpine, as there stated.

P. 244.—“Daphnis.” A Sicilian hero, son of Mercury, and a nymph. A Naiad fell in love with him and made him swear he would never love another. But he met and loved a princess, and the Naiad smote him with blindness. He besought his father for help, and the latter removed him to the abode of the gods, and caused a fountain to gush forth on the spot whence he was taken up.

“Thirsis.” A herdsman who laments the death of Daphnis.

“Priapus.” Son of Bacchus. One of the divinities presiding over agricultural pursuits.

P. 245.—“Gäl'in-gale.” A rush-like, or grass-like plant, often called sedge.

“Ly-ca'on's son.” Pandarus. One of the commanders in the Trojan war.

P. 246.—“Cicala,” si-cā'lä. Usually written cicada. The locust.

P. 247.—“Dilettanteism,” dil-et-tan'te-ism. Admiration of the fine arts.

P. 251.—“Gol'gi.” A Sicyonian colony, inhabiting a town of the same name in Cypris.

“Idalium.” A town of Cypris.

P. 253.—For supplementary reading on Demosthenes see Talfourd's "History of Greek Literature;" *The North American Review*, vol. xxii, p. 34; *New York Review*, vol. ix, p. 1; *National Review*, vol. xii, p. 99.

P. 255.—"Ignatius Loyola," ig-na'sheus loi-o'la. (1491-1556.) A Spaniard; the founder of the Society of Jesus. He served as page in the court of Ferdinand and Isabella, and later engaged in the wars against the French and the Moors. He was severely wounded in battle, and was made lame. His thoughts were then turned toward a religious life. Long fasts and scourgings often brought him near to death. He attended the University of Paris, where he took the master's degree at the age of forty-three. Afterward he gathered a few followers about him as the nucleus for his society, which in a short time became so famous.

P. 270.—"Margites." A poem ascribed to Homer, which holds up to ridicule a man who pretended to know many things, and knew nothing well.

P. 275.—"Milo." A Roman of daring and unscrupulous character. He was impeached for bribery and for interfering with the freedom of elections, and Cicero undertook his defense.

P. 278.—"Cyrtilus." The stoning of this man and his family occurred when the Athenians, under Themistocles, retreated from their city to Salamis, after learning that Thermopylæ was in the possession of the Persians.

P. 281.—"Laocoön." While the Trojans were debating whether they should receive the wooden horse into the city, Laocoön, a priest, rushed forward and warned them not to do it, and struck his spear into its side. As a punishment, Minerva sent two monstrous serpents, which crushed him and his two sons to death.

P. 282.—"Bema." A raised place, from which an orator addressed public assemblies.

CHEMISTRY.

P. 77.—“Champs de Mars,” Shäng duh Mars. Field of Mars. The name given to the place devoted to military exercises in France. It is an extensive parade ground, about 3,000 feet long and 1,500 feet wide, lying on the left bank of the Seine. There are four rows of trees on each side, and it is entered by five gates. It was finished in 1790, and in their eagerness to have it ready for the first great feast of the French Revolution, on July 14th, of that year, 60,000 volunteers, men and women, worked night and day for two weeks, and completed it in time. At this feast the king swore allegiance to the constitution. The Champs de Mars has been the scene of many great historic events. The World’s Fair of 1867 was held there.

P. 78.—“Academy of Science.” This was organized in France in 1666. In 1795 it, with four other academies, viz.: the French Academy, Academy of Painting and Sculpture, Academy of Belles Lettres, and the Academy of Moral and Political Science, was revived in a new form, under the name of the *Institut National*. This institution is the most important of its kind in the world. These academies now have the same relation to the *Institut* that colleges bear to a university. In the Academy of Science at present there are sixty-three members and one hundred corresponding members. It bestows an annual prize of about \$2,000, for the most important astronomical observation, a prize of nearly \$600 for productions on natural science, and other rewards for inventions, discoveries, and improvements. Its sessions are all held in public, and are much frequented.

P. 80.—A free translation of the note at the bottom of the page: Having attained an altitude of 22,960 feet, he still wished to go higher, and so disburdened himself of all the objects which he could in any way do without. Among these objects was a chair of white wood, which chanced to light in a thicket, very near a young girl who was tending some sheep. Great, indeed, was the astonishment of the shepherdess! The sky was clear, the balloon invisible. What else could she think of the chair than that it had come from Paradise? The only objection that could be raised against the conjecture was the rudeness of its construction. The workmen in the higher world, said the incredulous, could not be so unskillful. The discussion was still going on, when the papers, in publishing all the particulars of the aerial voyage of Gay Lussac, announced, among the natural results of the ascent, this which up to this time had seemed a miracle.

P. 85.—“Scheele,” shā'leh.

P. 91.—“Litmus paper.” Paper that has been prepared for use as a test for acids and alkalies. Litmus is a blue coloring matter, extracted from lichens which are found along the rocky coasts of the Mediterranean, and other tropical lands. They are largely used for dyeing purposes, and when prepared with potash or soda, they produce litmus. A strong infusion of litmus is made with boiling water, and a little sulphuric acid is added. Unsized paper is dipped into this infusion, which gives it a blue color. The application of any acid will change the blue to red, and then the blue color may be immediately restored by immersing the paper in an alkali. So delicate a test is it, that the paper has to be preserved in closely stoppered bottles, to prevent the access of acid fumes.

P. 94.—“Berthollet,” ber-to-lā.

P. 100.—“Balard,” bā-lār.

P. 101.—“Liebig,” lee'big.

P. 107.—“Varech,” vār'ek; “Barilla,” ba-ril'la.

P. 108.—“Courtois,” koor-twä.

P. 114.—“Nicklès,” nē-klā.

P. 115.—“Puy Maurin,” pwe-mō-ra^{ng}; “Haüy,” ä-we.

NOTES ON REQUIRED READINGS IN “THE CHAUTAUQUAN.”

TEMPERANCE TEACHINGS OF SCIENCE.

[1.](#) “Boerhave,” bōr'häv, Hermann. (1668-1738.) A Dutch physician. He gave much attention to the distinction between mind and matter, and condemned the doctrines of Epicurus, Hobbes and Spinoza. He published several works on the study and practice of medicine, and held the chair of chemistry, botany, and medicine in Leyden University.

[2.](#) “Saracens.” The Mohammedan people who, coming from Mauritania, invaded Europe in the early part of the eighth century. In Spain they took the name of Moors. They applied to all unbelievers in Mohammedanism the name Giaours (jour) as a term of reproach.

[3.](#) “Lorenz Oken.” (1779-1851.) A German naturalist, and the author of several works. He was professor of medical science for a time at Jena, and editor of the celebrated periodical, *The Isis*, devoted to natural science. At the time of his death, he held the position of professor of natural science in Zurich, Switzerland. A statue has been erected to his honor in Jena, Germany.

[4.](#) “Bentham,” Jeremy. (1748-1832.) An English writer on politics and jurisprudence. In opposition to Blackstone’s views, he wrote “Fragments on Government.” His numerous literary works were more kindly received in France than in England. One of his latest works was the “Art of Packing,” that is, of arranging juries so as to obtain any verdict desired. He wrote a book on the “Defense of Usury,” showing the impolicy of placing restraints upon dealings in money.

[5.](#) “Benjamin Rush.” (1745-1813.) A celebrated American physician, one of the signers of the Declaration of Independence. During the ravages of the

yellow fever, in 1793, he distinguished himself by rendering extraordinary services, and his history of that epidemic is a valuable work.

6. “Nostrum Mongers.” Sellers of quack medicines.

7. “Circenses,” sir-sen'sēs. A Latin term, meaning race-courses. Here it can be translated recreations.

8. “Panes,” pa'nēs. Bread, means of subsistence.

9. “Languedoc,” lang'gue-dock. A name applied during the middle ages to a province in the south of France, which is now divided into several departments, among which are Aude, Hérault, and Upper Garonne.

10. “Bunsen,” Christian Karl, Baron von, generally known as Chevalier Bunsen. (1791-1860.) One of the most distinguished statesmen and scholars of Germany. Through the favor of Niebuhr, who was Prussian minister at Rome, he was appointed secretary to the Prussian embassy at that court, where he remained twenty years, and then succeeded Niebuhr as minister. Later he was sent as Prussian ambassador to England. He was highly esteemed by Frederick William III. and Frederick William IV., both of whom frequently took him into their counsel. He was one of the most zealous workers in bringing about the union of the German states. His widow has published the “Memoirs of Bunsen.”

11. “Turnerhalls.” Gymnasia which were established throughout Germany through the enterprise of Friedrich Ludwig Jahn, in the latter part of the eighteenth century, for the purpose of fitting young men to endure the fatigues of war.

12. “Jean Jacques Rousseau.” (1712-1778.) One of the most eloquent French writers and singular characters of his age. He was denounced on account of his subversive theories and the immoralities of his life. His erratic social and political teachings are redeemed in part by the strong desire he had to increase the happiness of the laboring classes.

13. “Goldwin Smith.” (1823- .) An English author, and a warm friend to the federal government during the civil war. Coming to the United States in 1868, he became professor of English history in Cornell University.

SUNDAY READINGS.

The selection given in *THE CHAUTAUQUAN* as a Sunday Reading for October 5, 1884, was from Gotthold's "Emblems." The note on Gotthold was crowded out of the C. L. S. C. Notes. Many inquiries have been made concerning him; for this reason we insert the following:

"Christian Scriver, a Lutheran clergyman and writer of devotional works in the seventeenth century, the contemporary and friend of Spener, was born at Rendsburg, in Holstein, January 2, 1629. His childhood was spent under the care of a widowed mother in the trying period of the Thirty Years' War; but a wealthy merchant—a brother of Scriver's grandmother—finally made provision for his needs. After suitable preparatory studies, Scriver became a private tutor, and in 1647 entered the University of Rostock. In 1653 he was archdeacon at Stendal, and in 1667 pastor at Magdeburg, with which position he combined other offices, *e. g.*, that of a scolarch, and finally of a senior in the government of the church. He refused to leave Magdeburg in answer to repeated calls to Halberstadt, to Berlin, and to the court of Stockholm, but in advanced age was induced to accept the post of court preacher at Que Dinburg. In 1692 he suffered an apoplectic stroke, and on April 5, 1693, died. He had been married four times, and had had fourteen children born to him, but he outlived all his wives and children except one son and one daughter.

"The name of Scriver has lived among the common people through the publication of his 'Seelenschatz' (Magd. and Leipsic, 1737, Schaffhausen, 1738, sq., five parts in two vols., folio), a manual of devotion which he dedicated to 'the Triune God,' and which deserves high commendation. Another work deserving of mention is Gotthold's 'Zufällige Andachten' (first edition 1671, and often), a sort of Christian parables, 400 in number, which are based on objects in nature and ordinary occurrences in life. The 'Siech. u. Siegesbette' describes a sickness through which he passed, and the aids and comforts derived from God's goodness in that time. Prittius has published a work of consolation entitled 'Wittwentrost,' from Scriver's literary remains."

For Scriver's life see Prittius's preface to the "Seelenschatz;" Christmann's "Biographie" (Nuremburg, 1829); Hagenbach's "Wesen u.

Gesch. d. Reformat.," vol. iv; "Evangelisch Protestantismus," vol. ii, 177 sq.; Herzog's "Real-Encyklop," s. v.

1. "Renan," rŭh-näng. (1823- .) A French philosopher, who has published several treatises on comparative philology, and translations of scriptural books with critical introductions, and has written much for periodicals. He was sent at the head of a scientific commission to explore Tyre and Sidon, Lebanon and other localities, and made many interesting discoveries.

2. "Whitefield," George. (1714-1770.) The founder of Calvinistic Methodism. He set the example of preaching in the open air, and at one time is said to have addressed 60,000 persons at Moorfields. He quarreled with Wesley on the subject of predestination, but afterward was reconciled to him, although he never agreed with him in doctrine. He made several visits to the United States.

3. "President Edwards," Jonathan. (1745-1801.) Son of Jonathan Edwards, the divine. He was president of Union College, Schenectady, N. Y. His complete works were published in two volumes.

4. "Tholuck," tō'look. (1799-1877.) A German divine. In 1826 he was called to the University of Halle, as professor of theology, where he spent the remaining years of his life.

STUDIES IN KITCHEN SCIENCE.

1. "Brassica oleraceæ," bras'si-ca ō-ler-a'se-ē.

2. "Bore-cole." A variety of cabbage, not having its leaves packed into a firm head, but loose and curled.

3. "Daucas carota," dau'cus ca-ro'ta.

4. "Beta vulgaris," bē'ta vul-gā'ris.

5. "Mangold-Wurzel." Commonly written mangel wurzel.

6. “Allium Cepa,” al'li-um sē' pa.

THE PREPARATION OF VEGETABLES.

1. “Entreés,” o^{ng}' trā. The first course of dishes served on the table.

2. “Mayonnaise,” mā' yon-naise.

THE CIRCLE OF THE SCIENCES.

1. “Tufa.” A kind of volcanic sandstone, composed of pulverized volcanic rocks. It is formed whenever a shower of rain accompanies the fall of cinders, during the eruption of a volcano.

2. “Drift period.” The name applied to the time in which that remarkable bed of earth, gravel, and stones of all dimensions, was deposited. It has puzzled all geologists to account for this formation, which is the lowest of the three groups of the superficial covering of the earth, and no completely satisfactory theory has yet been advanced.

3. The large New Zealand bird described was called the moa.

4. “Carboniferous period.” Coal age. By careful study it has been found that in the progress of the earth's development a number of great ages have existed—each distinguished from the others by some marked change. That of coal plants is placed by geologists as the fourth age, counting upward from the lowest formation. It was remarkable for the alternate low elevation of the land above the sea level, and its submergences; and also for the luxuriant growth of vegetation, which, under the great pressure and heat to which it was subjected while the surface was submerged, was changed into coal.

5. “Spectroscope.” The name given to the apparatus used for the study of the spectrum. “When a ray of sunlight admitted through an aperture in a dark room is concentrated upon a prism of rock salt”—or glass—“by means of a lens of the same material, and then after emerging from the prism is received on a screen, it will be found to present a band of colors, in the

following order: red, orange, yellow, green, blue, indigo, and violet.”—*Ganot*. This band is called the spectrum. That there are other but invisible lines than those mentioned in the spectrum, is proven by the use of the thermopile, oftener called thermomultiplier, mentioned in the article on “Home Studies in Chemistry.” This is a complicated instrument used for detecting minute differences in the degrees of heat; its description without an accompanying illustration would be of no benefit to any one. So delicate is it that the heat of the hand held at a distance of three feet is sufficient to deflect the needle. The spectroscope is composed of three telescopes, mounted on a common foot, whose axes converge toward a glass prism. One of the telescopes is movable, and can be adjusted so as to give the observer the clearest view of the spectrum. The ray of light is admitted through the telescope and falls upon the prism, which decomposes it, and the spectrum is formed on the opposite side of the prism. In the telescope which the observer uses a powerful magnifying glass is placed. The third telescope is used for measuring the relative distances between the lines.

HOME STUDIES IN CHEMISTRY.

[1.](#) “James Dwight Dana.” (1813- .) An American geologist and mineralogist; the author of several works on these and other sciences. He went out as mineralogist and geologist appointed by the United States Government with the exploring expedition sent to the Southern and Pacific Oceans in 1838, and returned in 1842. For the next fourteen years he was busily engaged in preparing for publication the reports of this exploration. These were published by the government and formed valuable records. For many years he was one of the editors of the *American Journal of Science and Arts*. He has been elected to membership in several learned European societies and royal academies.

[2.](#) “Eisenlohr,” i'zen-lore. (1799-1872.) A German physicist. He was a Heidelberg student, and in 1819 removed to Mannheim, where he became a teacher of mathematical and physical science in the lyceum. He was afterward a professor at Carlsruhe. A Manual of Physics is his chief work.

[3.](#) “Geissler’s tubes,” ghīce'ler. The spectrum of any gas can be best obtained by placing the gas in these tubes, and then passing the electric

current through. If the gas is hydrogen, the spectrum will consist of a bright red, a green, and a blue line. Each gas casts its own spectrum. In this way the spectroscope aids in the analysis of substances. The different spectra formed reveal the elements.

4. “Sir John Herschel.” (1792-1871.) An English astronomer. His great enterprise was his expedition to the Cape of Good Hope to take observations of the heavens in the southern hemisphere. He remained there four years. His published results of his observations furnish one of the most valuable works on astronomy. He did not confine his studies to astronomy alone, but gave great attention to the subject of the atmosphere. He held that from eighty to ninety miles above the earth a perfect vacuum exists, and that three fourths of all the atmospheric air lies within four miles from the earth’s surface. His studies in meteorology were also very valuable, as well as his important discoveries in photography. Among his published works are: “Essays, from the *Edinburgh* and *Quarterly Reviews*, with Addresses and Other Pieces,” “Physical Geography,” and “Familiar Letters on Scientific Subjects.” Herschel held various positions of honor in his lifetime, being at one time president of the Royal Astronomical Society, and afterward Master of the Mint for five years. He was one of the eight foreign associates of the French Academy of Sciences.

5. “Fire Worshipers.” A Persian sect which worships fire as an emanation of the divine being. “Fire worshipers” is the English name for the Guē'bers (also called Ghē'ber or Giaours—jours). They call themselves *Beh Din*, “those of excellent belief.” The Arabs completed the conquest of Persia in the seventh century, and the great mass of the nation adopted the faith of the conquerors. Those who refused to do so were subjected to persecution. Some of them took refuge in the wilderness of Khorasan, and others in Kohistan. The latter in the ninth century emigrated to India and settled in the neighborhood of Surat. Their descendants still inhabit the same region, and are called Parsees. The descendants of those who remained in Persia have gradually decreased in numbers and sunk into ignorance and poverty, though still preserving a reputation for honesty, chastity, industry, and obedience to law superior to that of the other Persians. They are estimated to number about 7,000.

[6.](#) “Bunsen Burner.” In this burner, at the lower end of the hollow stem through which the gas passes, there is a lateral orifice which admits the air necessary for combustion. This orifice can be made larger or smaller by means of a diaphragm which is used as a regulator. If a moderate amount of air enters, the gas burns with a luminous flame, but if a strong and steady current is admitted, the carbon is rapidly oxidized, the flame loses its brightness, and burns with a pale blue light, scarcely perceptible, and with intense heat.

[7.](#) “Voltaic Arc.” A most beautiful effect, obtained from the electric light. At the terminals of a battery, pieces of charcoal are connected and placed in contact until the current causes them to become incandescent. Then they are separated about the tenth of an inch, and it is found that a luminous, exceedingly brilliant arc connects the two points.

[8.](#) “Hell Gate.” The name of a narrow channel between Long Island and Manhattan Island. Until recently the numerous reefs made it impassable for large ships and dangerous for small ones. In 1851 the first efforts were made to open the channel, by submarine blasting. In 1876, after many vain attempts, the work was carried to a successful issue. The total amount of money expended by Congress for this work since 1868 was \$1,940,000.

[9.](#) “Æolus.” A descendant of the founder of the Æolian race. He became the ruler of certain islands in the Tyrrhenian Sea, which from him were called the Æolian Islands. He is said to have taught his subjects to use sails on their ships, and to have foretold the nature of the winds that were to rise. Homer said of him that Jupiter had given him rule over the winds. This led to his being regarded as the god of the winds, which he was supposed to keep shut up in a mountain.

TALK ABOUT BOOKS.

It would be difficult for a biography of Sydney Smith, that man who always took short views of life, hoped for the best, and put his trust in God, to be other than interesting. Mr. Reid's biography[L] is so interesting that the reader quite forgets to criticise. It is a many-sided sketch of the brave hearted dominie. It tells his history, to be sure, but one gets a very good idea of many of his associates as well; it tells his route through life, and as a happy idea adds descriptions and illustrations of the various localities in which he lived, as they are to-day. There is just enough quotation from the reverend Sydney to give pith to the sober, clear narrative of the writer, and just enough of the "Times" to keep one in sympathy with his age. Several letters and essays never before printed appear in the volume. Mr. Reid, we are pleased to see, presents the courage, the unfailing hope, and the abundant common sense of his subject as characteristics of more importance than his wit.

It is moderate praise of the book[M] produced by Mrs. Mitchell to say that all lovers of art and its history will find it a valuable acquisition to their libraries. The author has chosen the historical method of presenting her subject, and begins with Egyptian sculpture, passes on to Chaldean, Assyrian, and Persian; then to that of Phœnicia, Asia Minor, and Greece, and ends with works of the Italian masters. Feeling that "description can not by any possibility supersede the sight of the artistic creations," she has freely illustrated the book with accurate representations of many of the great masterpieces. There can be no work better suited for the use of those who desire to acquire a knowledge of this branch of art.

M. Gaillard has added one more to the many books already issued for the purpose of teaching "French Conversation." [N] The system he has adopted differs from all the others in this respect: questions alone are given, to which the scholar is to frame his own answers. A clue to the words needed in the replies, and to the construction of the sentences and idioms will be found in the questions. Thus the memorizing of set sentences which

never will fit in anywhere save in the recitation room, is avoided, and the pupil is obliged to think for himself instead of merely observing how the words are used by others. Theoretically the plan is a good one. As a text-book for common use in schools and elsewhere, we doubt, somewhat, its feasibility.

No tourist to the White Mountains can afford to do without Mr. Drake's book.[\[Q\]](#) The last edition of it is prepared expressly for their use, and contains in the form of an appendix a complete guide-book. One of the covers is provided with a pocket, within which is placed a map of the White Mountains, and one of Vermont and New Hampshire. This pocket will also prove convenient for carrying memoranda. The book contains many fine illustrations, is printed from large, clear type, and is handsomely bound. And as one sees in word pictures the scenery of the mountains, and is delighted with racy little incidents of travel, and with anecdote, or is thrilled with some perilous adventure, he can not help saying that author, artist, and publisher have all done their part toward making an attractive book.

It was a good idea to publish a dictionary of the "Women of the Day."[\[P\]](#) Miss Hays has undoubtedly put an immense amount of labor into the neat little volume which she has just sent out to the world. However, the publication has been too soon. More labor is needed to make the book as useful as it ought to be. More than once her biographies of the best known women are incorrect, as when she located Marion (which name, by the way, she spelled Marian) Harland's present home at Newark, N. J., a place she left years ago. Again, in some of the sketches the work is poorly arranged. Why should Miss Willard's whereabouts in 1878 be tacked on at the end of the article, after it had been brought up to 1882, instead of being inserted in its proper order? For all that, it is a very useful work. It will be of great help to the general reader interested in eminent women.

A valuable series of "Outlines" of the Philosophy of Hermann Lotze has been undertaken by Prof. Ladd, of Yale College. A leading philosopher of Germany, Lotze's works have been sealed to all English readers, save those who were able to overcome philosophical German. This series will furnish an opportunity long desired by those interested in German thought to make themselves familiar with Lotze's ideas. "Outlines of Metaphysics"[\[Q\]](#) is the first work issued.

Mrs. Jackson's "Ramona"[\[R\]](#) takes rank at once in the highest class of fiction. The fascination in its pages holds one from beginning to end, and he closes the book with much the same impression as if he had just returned from a day's exquisite enjoyment of wild and rugged mountain scenery. The characters possess an individuality such as is found in those drawn by Dickens, and the fine shaping of plot and incident recalls George Eliot's "Romola." The story of "Ramona" has to do with Indian life in Southern California and Mexico, and is of historical interest. As one reads of the wrongs cruelly inflicted upon the noble *Alessandro* and the heroic Christian spirit with which he endured them all to the bitter end, there comes a sense of shame that under American laws, base, unprincipled men could commit such deeds of plunder and violence with impunity. The character of *Ramona* is unique. Her devoted love for *Alessandro*, the gladness with which she accepted the life of deprivation and danger at his side, and the development, through heavy sorrows, of her deep, true, womanly nature, give the book a richness of color and a depth of pathos seldom met.

In "Dorcas,"[\[S\]](#) a story of anti-Christ, the lives and sufferings of the early Christians in Rome are depicted. Dorcas and her friends hid themselves away for many long months in the Catacombs, to escape persecution. In two instances while there, the miracle of bringing the dead back to life occurred, one of those restored being Marcellus, the affianced husband of Dorcas, a young Roman nobleman who was put to death for accepting the Christian religion. The accession of Constantine gave them their freedom. The book affords a good study in the high style of its diction and the purity of its language. It is valuable, too, for its record of the customs of those days, and for its historical incidents.

Students of English who enjoy theories about words and expressions will find in "Elements of English Speech"[\[T\]](#) a full measure of them, most ingeniously supported. The book is in no way suitable for readers who are unacquainted with Latin, Greek, French, and German, but for those who have dabbled a little in each it will furnish interesting reading, and some ideas of real value.

The house of D. Appleton & Co. is publishing some excellent textbooks. Among these is "Elements of Geometry,"[\[U\]](#) a work on plane and solid geometry. The arrangement of the book, its admirable fitness to the

needs of the pupils just beginning the sciences, and its abundant exercises make it a very satisfactory work for teachers.—In their series of “Science Text-Books,” “Elements of Zoölogy,”[\[Y\]](#) by C. F. & J. B. Holder, is one of the most entertaining, practical, and, beside, thorough, elementary works on animal biology we have ever seen. The illustrations are excellent.—A capital “Second Reader” is “Friends in Feathers and Fur, and Other Neighbors.”[\[W\]](#) We like the idea of giving the young folks good, clear type.—But best of all is “Appleton’s Chart Primer,”[\[X\]](#) a pretty little book with numbers of beautifully colored pictures for color lessons, and a cover so brilliant that it will make it a pleasure for little ones to learn their lessons.

A new edition of “The Water Babies,”[\[Y\]](#) abridged by J. H. Stickney has been issued. It is a delightful fairy story for land babies. Little Tom, a poor chimney sweep who belonged to a very cruel master, went one day to work in a grand house. Coming down the wrong chimney, he found himself standing opposite a large mirror in a very beautiful room in which a little, sick girl was lying. The sight of himself in the glass, black and impish, and the screams of the little girl frightened him so that he jumped from the window, caught the branches of a tree, slid to the ground and ran for his life, pursued by different members of the family, who supposed him to be a thief. They could not catch him, however, and soon gave up the attempt. Two or three days after his body was found in a stream of water, and all the people thought him dead. But they were mistaken; that body was only the old covering of Tom; he had been changed into a beautiful water baby, whose life in that fairy land is told in a very fascinating manner, showing that there, also, little folks ought to work for the good of others.

The “Water Babies” is one of a series of “Classics for Children,” a series arranged on the sensible idea that children can be taught to enjoy good literature, as they are taught to read. Among the other works which have appeared in this course are a “Primer and First Reader,”[\[Z\]](#) Scott’s “Lady of the Lake,”[\[AA\]](#) and Kingsley’s “Greek Heroes.”[\[AB\]](#) Others are in preparation.

“Which: Right or Wrong?”[\[AC\]](#) is an interesting story centering about the Framingham Assembly. It gives some bright pictures of life there, and teaches some excellent lessons.

“The Mentor”[\[AD\]](#) is a very neat little book written for the use of men and boys who wish to appear to good advantage in cultivated society. It treats of personal appearance, manners at the dinner table and in public, conversation, odds and ends, calls and cards, and closes with a chapter answering the question, “What is a Gentleman?” It contains a number of quotations from eminent authors.

A beautiful device is that of “The Guest Book,”[\[AE\]](#) in which the hostess may record the coming and the going of her guests. It contains short, beautifully illustrated selections concerning hospitality, from prominent writers, with blank pages left between for autographs, incidents, and sketches relating to pleasant calls and visits. In the hands of every woman who loves to entertain her friends it will prove a treasure-house of pleasant memories.

Not often are our social foibles “taken off” more pointedly than in “The Buntling Ball.”[\[AF\]](#) It is a really clever, and withal sprightly, satire on some of the vulnerable points of New York society. *Mrs. Buntling*, wife of a “potentate in pork,” returning from Europe, issues invitations for a ball. She has obtained a list of “all the names considered of decisive note,” and, regardless of the fact that she knows none of them issues a general invitation. The fact that everybody comes is one of the sharpest points in the play. Choruses are introduced in true Greek drama style, and the “Knickerbocker young men,” “maneuvering mammas,” “wall-flowers,” “gossips,” “Anglo-maniacs,” etc., carry on dialogues with the principal characters, in which they give the whole philosophy of New York society, in the frankest manner and in all sorts of happy, sprightly verse. The mystery of its authorship has been turned to good account by the publishers, who offer a prize of \$1,000 to the successful guesser.

Marion Harland, in writing “Eve’s Daughters,”[\[AG\]](#) has done a noble work for women. The book must exert a good influence wherever it goes, and do much toward breaking down the barrier of false modesty and ignorance in regard to herself, that woman, too often, has taken pride in rearing. It begins with the life of the baby girl and follows her as the representative of her sex, through all the years down to old age. Strong, plain, helpful things are said, and said only as a brave, womanly woman can

say them, in regard to the physical life of women. Every mother ought to read the book, and read it with her daughters.

“Memories of the Manse”[\[AH\]](#) is a quiet little picture of the life, home, family, and parish of a Scotch minister who lived, a number of years ago, in Glenarran. The rugged outlines of the stern character belonging to that northern people are well drawn, and dashes of color, showing the tender and loving side of human nature, appear here and there, brightening up the scene. The experience of the eldest son, who was “a clever lad, and had just returned after working his way through college, wearing a wonderfully clerical dress and air, an eye-glass, and a highly comfortable opinion of himself,” only to find that he was ridiculed instead of admired by his former associates, and his honest surprise at his unpopularity furnish a touch of humor to the whole work.

The books which Samuel Smiles has put upon the market are eminently valuable to boys and men who are in trades. He has done much to dignify labor and to show how essential is brain and thrift and education to manual labor. In his late volume, “Men of Invention and Industry,”[\[AI\]](#) the material is particularly good. It is fresh, and the stories of successful men give a grip to the book which is very effective. The lack of literary finish of which some complain in Mr. Smiles’s work is but a minor matter when we think of the serious purpose, the earnest desire to show how handicrafts may be developed, and how great opportunities lie in the way of mechanics to benefit society and to attain distinction. Among his men of invention and industry are Phineas Pett, the English ship builder; John Harrison, the inventor of the marine chronometer, and Frederick Koenig, inventor of the steam printing machine. A digression from the main object of the book is the chapter on “Industry in Ireland,” but it is a pleasing digression. The abundant resources which Mr. Smiles shows to exist in Ireland, will be surprising to many readers. Her fisheries, her iron, coal and clay beds, her linen industries, and her ship building are well described. The development of these resources he justly concludes to be the solution of the “Irish trouble.”

Mr. Harrison, in giving to the public the life and literary works[\[AJ\]](#) of the author of “Home Sweet Home,” has met a want that many persons have felt, to know something more of this author. No trouble has been spared in

gathering the data for the biography, and much valuable information has been given to the world which, but for his efforts, might have been lost. He has, however, entered so fully into details as frequently to detract from the interest of the work. The circumstances under which “Home Sweet Home” was written, are given.

FOOTNOTES

[L] A Sketch of the Life and Times of the Rev. Sydney Smith. By Stuart J. Reid. New York: Harper & Brothers. 1885.

[M] A History of Ancient Sculpture. By Lucy Mitchell. New York: Dodd, Mead & Company. Price, according to binding, \$12.50, \$18.00, or \$25.00.

[N] French Conversation. By J. D. Gaillard. New York: D. Appleton & Co.

[O] The Heart of the White Mountains. By Samuel Adams Drake. Illustrated by W. Hamilton Gibson. New York: Harper & Brothers, Franklin Square.

[P] Women of the Day. A Biographical Dictionary of Notable Contemporaries. By Frances Hays. Philadelphia: J. B. Lippincott & Co. 1885.

[Q] Outlines of Metaphysics. Dictated Portions of the Lectures of Hermann Lotze. Translated and edited by George T. Ladd. Boston: Ginn, Heath & Co. 1884.

[R] Ramona. By Helen Jackson. (H. H.) Boston: Roberts Brothers. Price, \$1.50.

[S] Dorcas, the Daughter of Faustina. By Nathan C. Kouns. Author of “Arius the Libyan.” New York: Fords, Howard and Hurlbert. 1884.

[T] Elements of English Speech. By Isaac Bassett Choate. New York: D. Appleton & Co. 1884.

[U] Elements of Geometry. By Eli T. Tappan, LL.D. New York: D. Appleton & Co. 1885.

[V] Elements of Zoölogy. By C. F. & J. B. Holder, M.D. New York: D. Appleton & Co. 1884.

[W] Friends in Feathers and Fur, and Other Neighbors. By James Johonnot. New York: D. Appleton & Co. 1885.

[X] Appleton’s Chart Primer: Exercises in Reading at Sight, and Language and Color Lessons For Beginners. By Rebecca D. Rickoff.

[Y] The Water Babies. By Charles Kingsley. Edited and abridged by J. H. Stickney. Boston: Ginn, Heath & Co. 1884. Mailing price, 40 cents. Introduction, 35 cents.

[Z] Primer and First Reader. By E. A. Turner. Boston: Ginn, Heath & Co. 1885.

[AA] The Lady of the Lake. By Sir Walter Scott. Edited by Edwin Ginn. Boston: Ginn, Heath & Co. 1885.

[AB] The Heroes; or, Greek Fairy Tales for My Children. By Charles Kingsley. Edited by John Tetlow. Boston: Ginn, Heath & Co. 1885.

[AC] Which: Right or Wrong? By M. L. Moreland. Boston: Lee and Shepard, Publishers. 1883.

[AD] The Mentor. By Alfred Ayers. New York: Funk & Wagnalls. 1884.

[AE] The Guest Book. Designed and illustrated by Annie F. Cox. Boston: Lee and Shepard. New York: C. S. Dillingham, 618 Broadway. 1885.

[AF] The Buntling Ball. A Græco-American Play. New York: Funk & Wagnalls. 1884.

[AG] Eve's Daughters. By Marion Harland. New York: Charles Scribner's Sons. 1885.

[AH] Memories of the Manse. By Anne Breadalbane. Troy, N.Y.: H. B. Nims & Co. 1885.

[AI] Men of Invention and Industry. By Samuel Smiles, LL.D. New York: Harper & Brothers. 1885.

[AJ] John Howard Payne. By Gabriel Harrison. Illustrated. Philadelphia: J. B. Lippincott & Co.

BOOKS RECEIVED.

Daddy Darwin's Dovecote. A Country Tale. By Juliana Horatio Ewing. Boston: Roberts Brothers. 1885. Price, 35 cents.

Flatland. A Romance of Many Dimensions. By A Square. Boston: Roberts Brothers. 1885. Price, 75 cents.

Memoirs of the Rev. David Brainard. Based on the Life of Brainard prepared by Jonathan Edwards, D.D. Edited by J. M. Sherwood. New York: Funk and Wagnalls. 1885.

PARAGRAPHS FROM NEW BOOKS.

A FRAGMENT ON THE CULTIVATION AND IMPROVEMENT OF THE ANIMAL SPIRITS.
—It is surprising to see for what foolish causes men hang themselves. The most silly repulse, the most trifling ruffle of temper, or derangement of stomach, anything seems to justify an appeal to the razor or the cord. I have a contempt for persons who destroy themselves. Live on, and look evil in the face; walk up to it, and you will find it less than you imagined, and often you will not find it at all; for it will recede as you advance. Any fool may be a suicide. When you are in a melancholy fit first suspect the body, appeal to rhubarb and calomel, and send for the apothecary; a little bit of gristle sticking in the wrong place, an untimely consumption of custard, excessive gooseberries, often cover the mind with clouds and bring on the most distressing views of human life.... The greatest happiness which can happen to any one is to cultivate a love of reading. Study is often dull because it is improperly managed. I make no apology for speaking of myself, for as I write anonymously, nobody knows who I am, and if I did not, very few would be the wiser—but every man speaks more firmly when he speaks from his own experience. I read four books at a time; some classical book, perhaps, on Monday, Wednesday, and Friday mornings. The “History of France,” we will say, on the evenings of the same days. On Tuesday, Thursday, and Saturday, Mosheim or Lardner, and on the evenings of those days, Reynolds’s Lectures or Burns’s Travels. Then I have always a standard book of poetry, and a novel to read when I am in the humor to read nothing else. Then I translate some French into English one day and retranslate it the next; so that I have seven or eight pursuits going on at the same time, and this produces the cheerfulness of diversity, and avoids that gloom which proceeds from hanging a long while over a single book. I do not recommend this as a receipt for becoming a learned man, but for becoming a cheerful one.—*From Reid’s “Life and Times of the Rev. Sydney Smith.”*

SCENES ON A STAGE COACH.—The views of the mountains as the afternoon wore away, grew more and more interesting. The ravines darkened, the summits brightened. Cloud-shadows chased each other up and down the steeps, or, flitting slowly across the valley, spread thick mantles of black that seemed to deaden the sound of our wheels as we passed over them. On one side all was light, on the other all gloom. But the landscape is not all that may be seen to advantage from the top of a stage coach.

From time to time, as something provoked an exclamation of surprise or pleasure, certain of the inside occupants manifested open discontent. They were losing something where they had expected to see everything.

While the horses were being changed, one of the insiders, I need not say it was a woman, thrust her head outside of the window, and addressed the young person perched like a bird upon the highest seat. Her voice was soft and persuasive. “Miss!” “Madam!” “I’m so afraid you find it too cold up there. Sha’n’t I change places with you?” The little one gave her voice a droll inflection as she briskly replied: “Oh, dear, no, thank you; I’m very comfortable indeed.” “But,” urged the other, “you don’t look strong; indeed, dear, you don’t. Aren’t you very, very tired, sitting so long without any support to your back?” “Thanks, no; my spine is the strongest part of me.” “But,” still persisted the inside, changing her voice to a loud whisper, “to be sitting alone with all those men!” “They mind their business, and I mind mine,” said the little one reddening; “besides,” she quickly added, “you proposed changing places, I believe!” “Oh!” returned the other, with an accent impossible to convey in words, “if you like it.” “I tell you what, ma’am,” snapped the one in possession, “I’ve been all over Europe alone, and was never once insulted except by persons of my own sex.”—*From Drake’s “Heart of the White Mountains.”*

EVERY MAN HAS HIS PRICE.—It is a curious trait in human nature, that each individual places the highest value on himself; treats the world as if it were only in existence on his account, looks upon himself as if he were the central point round which all things turn—and that yet, in spite of this universal self-appreciation, so many persons make themselves the slaves of others, or of some insignificant desire of their own. This contradiction in the

human mind, this inordinate pride of men in combination with ignorance of their own true value, this insatiable self-seeking in connection with so contemptible a depreciation of themselves, is so common that we are only astonished that thoughtful persons, perceiving it in others, are not thereby led to discover it in themselves.... Every man has a price at which he sells himself. What is thy price? Hast thou ever weighed what thou art really worth? Go into thy chamber and devote some moments of earnest thought to an examination of thyself, and try to discover for what earthly good thou wouldst be likely to give thyself away. Look no farther back than the last year; pass in review thy secret thoughts and silent wishes even of the last few weeks only! Ah! a short while will no doubt suffice to show thee thy weak points, which, had they been assailed by any tempter, would soon have revealed to thee at what price thou wouldst have sold thy goodness, thy Christian principles, thy heaven on earth, thy eternal prospects. Thou shudderest? Thou wouldst rather not look into thyself? But if thou valuest thy goodness, thy Christian principles, thy heaven on earth, thy eternal prospects, ah, shrink not from this self-investigation?—*From Zschokke's "Meditations on Life, Death, and Eternity."*

ADVICE TO AN INEXPERIENCED TEACHER OF HISTORY.—But the method of teaching history must be determined in the main by the object aimed at. If the object is to deposit in the mind the greatest number possible of historical facts, there is perhaps no better way than to confine the instruction to drill upon the contents of a manual by question and answer, with frequent examinations in writing. Such a method would probably be effective in two ways; it would give learners positive knowledge, or the semblance of it, and it would pretty certainly make them hate history. I do not hesitate to say that the ultimate purpose of school instruction should be to incite an interest in history, and to create a love for historical reading.

A word may be here most conveniently said on the subject of chronology. A few dates should be well fixed in the memory; they should be carefully selected by the teacher, and some explanation given of their significance. But “a few,” you will say, is a little indefinite. Of course, opinions will differ as to the number of indispensable dates in any history, though there might be a general assent to the principle of requiring the pupil to commit as few as possible. Of the two hundred and fifty dates given in

“Smith’s *Smaller History of Greece*,” I insist on fifteen, and I think the number might be reduced to ten. But if learners are properly taught, they will, of course, be able to determine a great many dates approximately.

Remembering that you must make history interesting, to that end use all available means to produce vivid impressions. This is a trite remark, but it will bear repeating. Casts, models, coins, photographs, relief maps, may not be at your command, but maps of some sort you must have. Historical instruction, without the constant accompaniment of geography, has no solid foundation—“is all in the air.”—*From “Methods of Teaching History.”* [\[AK\]](#)

THE COMING OF LUTHER.—The events of the sixteenth century have been too often regarded as constituting a break in history. But to the eye of thought reviewing the course of history, the continuity remains unbroken. Luther was but the child of the ages preceding; the Protestant revolution was the natural and orderly sequence of a long course of preparation. It was indispensable indeed for a time that men should regard the Reformation as breaking with the past, in order that they might estimate more deeply the meaning of the truth which had been revealed to them, and secure its firmer establishment. In the turmoil of an age of transition it is not always given to the leaders to discern the route by which they have been led. Luther entered upon the inheritance of Wycliffe and of Huss, and still further was he indebted to the spirit of German mysticism. But his greatness was also peculiarly his own. He was not so much a theologian as a man who afforded in his own rich nature, unveiled so completely before his age, the materials for theology. His life was a type of humanity for his own and succeeding ages. He lived through the religious experience of the Mediæval dispensation before he came to his knowledge of a higher birthright. Viewed from the standpoint of a formal theology, he is full of inconsistencies and contradictions, and even dangerous errors. But regarded simply as a man, with his rich endowment of human instincts and yearnings, to which he gave the freest, most unguarded expression, he was in himself a revelation of the human consciousness in its freshness and simplicity, with which a complete theology must come to terms. It is because the explosive utterances of his vigorous, tumultuous nature have been weighed as if they were carefully formed, dogmatic statements, that

Luther has been so often misunderstood by Protestant as well as by Roman Catholic writers.—*From Allen's "Continuity of Christian Thought."* [\[AL\]](#)

NATURAL RESOURCES OF IRELAND.—Ireland is a much richer country by nature than is generally supposed. In fact, she has not yet been properly explored. There is copper ore in Wicklow, Waterford, and Cork. The Leitrim iron ores are famous for their riches; and there is good ironstone in Kilkenny, as well as in Ulster. The Connaught ores are mixed with coal beds. Kaolin, porcelain clay, and coarser clay abound; but it is only at Belluk that it has been employed in the pottery manufacture. But the sea about Ireland is still less explored than the land. All around the Atlantic's seaboard of the Irish coast are shoals of herring and mackerel, which might be food for man, but at present are only consumed by the multitudes of sea birds which follow them.—*From Smiles's "Men of Invention and Industry."*

FOOTNOTES

[\[AK\]](#) The Pedagogical Library. Edited by G. Stanley Hall. Vol I. Methods of Teaching History. Second Edition. Boston: Ginn, Heath & Co. 1885.

[\[AL\]](#) The Continuity of Christian Thought. By Alexander V. G. Allen. Boston: Houghton, Mifflin & Co. 1884. Price \$2.00.

SPECIAL NOTES.

We have been asked the meaning of the term the “geography of the heavens.” Professor Hiram Matteson, in his excellent little treatise entitled “The geography of the heavens,” makes in his preface the following explanatory remarks: “I have endeavored to teach the geography of the heavens in nearly the same manner as we teach the geography of the earth. What that does in regard to the history, situation, extent, population, and principal cities of the several kingdoms of the earth, I have done in regard to the constellations; and I am persuaded that a knowledge of the one may be as easily obtained as of the other. The systems are similar. It is only necessary to change the terms in one to render them applicable to the other. For this reason I have yielded to the preference of the publisher in calling this work ‘Geography of the Heavens,’ instead of *Uranography*, or some other name more etymologically apposite.”

It will be noticed from Chancellor Vincent’s article on “The Chautauqua School of Liberal Arts,” found in this impression of THE CHAUTAUQUAN, that the Sunday-school Normal department of Chautauqua will hereafter be known as “The American Church Sunday-school Normal Course.”

Messrs. L. Prang & Co. have begun to send out valentines of as much beauty and artistic merit as their Christmas and Easter cards. Those of the present season have been of rare beauty—the coloring of many of them is exquisite.

The following, clipped from the text-book of the Chautauqua Musical Reading Union noticed in THE CHAUTAUQUAN for February, will be of interest to many of our music-loving readers:

The aim is not so much to give technical instruction in the science, as to invite the wider outlook which is so important in real musical culture. No

person receives any pecuniary benefit from this organization, but the labor is freely given in the hope of benefiting others. The books required will be furnished from the Boston office at a discount from the retail prices, or they may be ordered through any local bookseller. Local circles may be formed in cities, towns, or small villages, greatly to the advantage of all who thus associate themselves. Scarcely anything can be conceived that will yield more delightful entertainment, together with improvement of mind and heart, than such a local circle as may be formed in connection with the C. M. R. C. All who are really in earnest about the improvement of the musical taste of the community in which they live, should exert every effort to bring about such an organization. For plans and information as to how these circles may be made successful, address the director, who will gladly furnish suggestions, and will send list of prices at which the required books will be furnished. *Please enclose stamp for reply.* A fee of fifty cents will be required to defray the expense of registration, correspondence, etc., which amount, with the name and postoffice address *plainly written* (including county and state), should be forwarded at once, directed to W. F. Sherwin, Director C. M. R. C., New England Conservatory, Boston, Mass. Certificates will be given for each course, and a diploma upon the completion of the four. A "round-table" will be held (*à la C. L. S. C.*) each year during the Chautauqua Assembly. For price list of books and any other information, address as above.

The following special course in physiology is announced:

Wonders of the Human Body. A. Le Pileur. \$1.25.
Physiology for Practical Use. James Kinton. \$2.15.
Mental Physiology. W. B. Carpenter. \$3.00
The Foundation of Death. Axel Gustafson. \$2.00.

ERRATA

In list of C. L. S. C. graduates which appeared in THE CHAUTAUQUAN for February:

Pettit, Harriet L., California, instead of *Pennsylvania*.
Arann, the Rev. J. M., not *Araun*.
Hon, George V., not *How*.
Hoerner, George P., not *Hoemer*.

NAMES TO BE ADDED

To the list of graduates in the class of 1884:

Black, Jennie L.	Pennsylvania.
Burgess, Miss Anna E.	Ohio.
Carter, Anna B.	California.
Carter, Emily B.	California.
Chamberlin, Lydia L.	Massachusetts.
Clark, Miss Annie	Rhode Island.
Coleman, William H.	Ohio.
Horsman, Mrs. George	Wisconsin.
Holden, Mrs. Sarah K.	Canada.
Jones, Mrs. E. J.	Ohio.
Marsh, Miss Susanna	Dakota.
Millar, Mrs. Lizzie L. S.	Minnesota.
Safley, Agnes E.	Minnesota.
Scott, Mrs. Lucie M.	New York.
Walker, Ezra L.	Ohio.
Weaver, the Rev. Wm. C.	Pennsylvania.

The following persons passed a creditable examination in the Advanced Normal Course of 1884 at Chautauqua:

Miss Fannie L. Armstrong, Hempstead, Texas.
Mrs. A. W. Briggs, Elma, Erie Co., N. Y.
Mr. O. W. Bowers, McLallen's Corners, Pa.

At Lakeside, Ohio, Assembly, the following passed an Advanced Normal examination:

Mrs. Abby A. Parish, Brooklyn Village, Ohio.

Transcriber's Notes:

Obvious punctuation errors repaired.

Page 316, "made" changed to "make" (all diligence to make their calling)

Page 316, "lotty" changed to "lofty" (such a lofty understanding)

Page 317, repeated "der" removed (the under side of the cabbage leaves)

Page 319, "entreés" changed to "entrées" (toothsome entrées)

Page 331, repeated "mon" removed (for common school purposes)

Page 341, "What is true of the processes of the is equally true of almost every other manual industry" changed to "What is true of the processes is equally true of almost every other manual industry". The revised sentence at least makes some sort of sense, but it's possible that words are in fact *missing* from the original.

Page 348, repeated "of" removed (Yale College class of 1853)

Page 350, “invarible” changed to “invariable” (an invariable part of each evening’s work)

Page 367, “Calvanistic” changed to “Calvinistic” (The founder of Calvinistic Methodism.)

Page 368, “cuurse” changed to “course” (The first course of dishes)

Page 368, “on” changed to “in” to match article title (Home Studies in Chemistry)

*** END OF THE PROJECT GUTENBERG EBOOK THE
CHAUTAUQUAN, VOL. 05, MARCH 1885 ***

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